

Lidl 1517 Norcross, GA - Ground Up

4990 Jimmy Carter Blvd. Norcorss, GA 30093

Report generated by Tucker Teague (Place Services, Inc.)



Submittal #PSI-0003 - Plumbing Package

|             |                                      |                  |                                            |
|-------------|--------------------------------------|------------------|--------------------------------------------|
| Creator     | Tucker Teague (Place Services, Inc.) | Submission due   | -                                          |
| Assignee    | Tucker Teague (Place Services, Inc.) | Response due     | 09/17/2026                                 |
| Reviewers   | -                                    | Closed on        | -                                          |
| Status      | Pending approval                     | Reference number | -                                          |
| Type        | Product data                         | Specification    | 22 11 19 DOMESTIC WATER PIPING SPECIALTIES |
| Description | -                                    |                  |                                            |

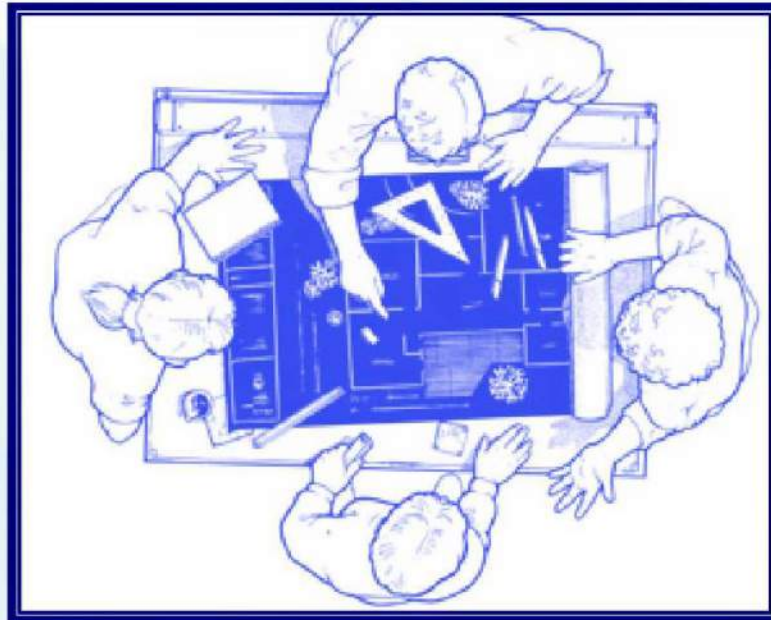
Approval (rev. 0)

|              |   |              |   |
|--------------|---|--------------|---|
| Responded by | - | Responded on | - |
| Response     | - |              |   |
| Notes        | - |              |   |
| Attachments  | - |              |   |

Submission (rev. 0)

|              |                                                                                                                                 |              |            |
|--------------|---------------------------------------------------------------------------------------------------------------------------------|--------------|------------|
| Submitted by | Tucker Teague (Place Services, Inc.)                                                                                            | Submitted on | 09/03/2026 |
| Notes        | -                                                                                                                               |              |            |
| Attachments  | PDF attachments are included in this export.<br><a href="#">↓ #1517 Lidl Norcross, GA - Plumbing Submittals.pdf</a>   (Page 2 ) |              |            |

# Plumbing Submittal



## LIDL NORCROSS

Submitted by

**GOLDEN HANDS PLUMBING**

**8/26/2026**

### **COWAN SUPPLY**

3032A, BANKERS IND DR, ST A  
ATLANTA, GA 30360-2715  
Phone: 404 351-6351  
Fax: 404 351-1259

### **SHOP DRAWING / SUBMITTAL REVIEW**

- ☒ NO EXCEPTION TAKEN
- ☐ REVISE AND RESUBMIT
- ☐ MAKE CORRECTIONS NOTED
- ☐ REJECTED

SUBMITTAL WAS REVIEWED FOR DESIGN CONFORMITY AND GENERAL CONFORMANCE TO CONTRACT DOCUMENTS ONLY. THE CONTRACTOR IS RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AT JOBSITE FOR TOLERANCE, CLEARANCE, QUANTITIES, FABRICATION PROCESSES AND TECHNIQUES OF CONSTRUCTION, COORDINATION OF HIS WORK WITH OTHER TRADES AND FULL COMPLIANCE WITH CONTRACT DOCUMENTS.

By: Tucker Teague  
Date: 9/3/2026

Place Services, Inc  
Canton, GA 30115

**Spec#****Table of Contents**

|    |         |                                                                                                                  |
|----|---------|------------------------------------------------------------------------------------------------------------------|
| 1  | .....   | <b>ALL MODELS ARE SUBJECT TO APPROVAL</b>                                                                        |
| 2  | .....   | <b>DF-1</b>                                                                                                      |
| 3  | 630619  | ELKAY VRCTLDDSC SS ADA WM BI-LEVEL VP<br>PUSHBUTTON VP BUBBLER DRINKING FTN                                      |
| 4  | .....   | <b>HB-1</b>                                                                                                      |
| 5  | 65231   | WOODFORD B65C-4 3/4X4 SWT CHROME AUTO DRAIN<br>FRZLS WALL HYDRANT W/ BOX & ANTI-SIPHON VAC<br>BREAKER            |
| 6  | .....   | <b>HB-2</b>                                                                                                      |
| 7  | 1624299 | TSBRASS MV-0771-12N-BVT BRASS 49.93 GPM 4H 6 IN.<br>CENTERSET WM DBL LEVER HDL MANUAL FCT                        |
| 8  | .....   | <b>L-1</b>                                                                                                       |
| 9  | 342134  | KOHLER 2084-0 SOHO WHT 20 X 18 X 7IN VC 1H WM<br>COM LAV SINK                                                    |
| 10 | 834804  | KOHLER 13460-CP SCULPTED POL CHROME 0.5 GPM<br>1H DM TOUCHLESS SENSOR FCT                                        |
| 11 | 258659  | ZURN Z8743-PC 11/4X5 CP 17 GA BRASS COMM LAV<br>GRID DRAIN W/ PO PLUG & OVERFLOW HOLES                           |
| 12 | 974288  | MAINLINE ML27001 11/4 CP 22GA BRASS P-TRAP L/<br>CLEANOUT                                                        |
| 13 | 1166059 | ZURN Z8804-XL-LR-PC 1/2X3/8 COMPCOMP CP BRASS<br>XL ANGLE STOP & SUPPLY KIT W/ HDL EXT LEAD FREE                 |
| 14 | 1017492 | MAINLINE ML102EZ WHT LAV-GUARD TRAP & SUPPLY<br>COVER F/ TUBULAR P-TRAP 2-ANGLE STOP & 2-<br>SUPPLY ADA APPROVED |
| 15 | 815216  | JRS 0700-M32 LAV CARRIER                                                                                         |
| 16 | .....   | <b>RD-1</b>                                                                                                      |
| 17 | 1380508 | JRSMITH 1015Y04                                                                                                  |
| 18 | .....   | <b>RH-1</b>                                                                                                      |

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**Disclaimer:**

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|    |         |                                                                                                                                |
|----|---------|--------------------------------------------------------------------------------------------------------------------------------|
| 19 | 2693001 | PRIER P-RH4 SANITARY ROOF HYDRANT ASSEMBLY-<br>ROOF MOUNT BRACKET & SANITARY HYDRANT                                           |
| 20 | .....   | <b>SK-1</b>                                                                                                                    |
| 21 | 1822786 | KROWNE HS-5 HAND SINK SS 12 1/2 X 9 3/4 X 5 5/8IN 20<br>GA SS 2H 4 IN. CENTERSET 1 STATION WM HAND<br>WASH SINK INCL WALL BRKT |
| 22 | 1225542 | KROWNE 10-400L POL CHROME 0.5 GPM 2H 4 IN.<br>CENTERSET WM DBL LEVER HDL BAR FCT                                               |
| 23 | .....   | <b>SK-2</b>                                                                                                                    |
| 24 | 970812  | ELKAY ELUHAD281655 LUSTERTONE CLASSIC<br>LUSTROUS SATIN 30 1/2 X 18 1/2IN SGL BASIN UM SS<br>KITCHEN SINK                      |
| 25 | 2102151 | ELKAY LKAV3032CR ELKAY AVADO 1 HOLE BAR<br>FAUCET PULL-DOWN SPRAY LEVER HANDLE CHROME                                          |
| 26 | 2715468 | ISE 80926 BADGER 5 1/2 HP CONT DISPOSER (OLD<br>PART# 79883-ISE)                                                               |
| 27 | .....   | <b>SK-3</b>                                                                                                                    |
| 28 | 2081398 | ELKAY SS8354LR4 STURDIBILT BUFFED SATIN 102 X 27<br>1/2 X 44IN 14 GA SS 4H FM SCULLERY SINK INCL LEFT &<br>RIGHT DRAINBOARD    |
| 29 | 1130255 | KROWNE 17-109WL POL CHROME 1.15 GPM 2H 8 IN. WS<br>WM TRIPLE LEVER HDL PRE-RINSE FCT W/ PULL-DOWN<br>SPRAY                     |
| 30 | .....   | <b>UB-1</b>                                                                                                                    |
| 31 | 1245652 | GUYGRAY 88531 MIB1HAAB 1/2 SWT ICE MAKER BOX<br>W/ ARRESTER LF                                                                 |
| 32 | .....   | <b>WC-1</b>                                                                                                                    |
| 33 | 2002188 | KOHLER K-84325-0 WHITE KINGSTON ULTRA 11/2 TOP<br>SPUD TOILET                                                                  |
| 34 | 1208094 | ZURN ZTR-6200-WS1 1.6GPF FLUSHVALVE                                                                                            |
| 35 | 793988  | BEMIS 1655SSCT 000 WHT ELONG OF L/ COVER SSTA<br>PLAS TOILET SEAT                                                              |

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|    |         |                                                                                                          |
|----|---------|----------------------------------------------------------------------------------------------------------|
| 36 | .....   | <b>WC-1 CARRIER OPTIONS</b>                                                                              |
| 37 | 1886448 | JRS 0211LY SGL LH CLOSET CARRIER                                                                         |
| 38 | 1466233 | JR SMITH 0211DY BACK TO BACK DOUBLE CARRIER                                                              |
| 39 | 1101626 | JR SMITH 0230Y-M58/54 VERTICAL SINGLE CLOSET CARRIER                                                     |
| 40 | .....   | <b>PP-1</b>                                                                                              |
| 41 | 997733  | B&G 103401LF - NBF-36 3-SPEED WET ROTOR CIRC 115V/60HZ AB1953 LEAD FREE, 10GPM @ 15'TDH                  |
| 42 | 532078  | BG 113210 TC-1 TIMER                                                                                     |
| 43 | 2174063 | B&G 101501LF 3/4 SS FLG SET                                                                              |
| 44 | 684757  | BG AQS 3/4 AQUASTAT CONTROL 113224                                                                       |
| 45 | .....   | <b>FD-1</b>                                                                                              |
| 46 | 2421064 | MIFAB TS5-1 TOP SET STRAINER FOR 5" NB ROUND STRAINER                                                    |
| 47 | 2726031 | MIFAB A4-3-30-4 3" DRAIN BODY ONLY                                                                       |
| 48 | .....   | <b>FD-2</b>                                                                                              |
| 49 | 2514481 | MIFAB TSS5-1 5" TOP SET SQUARE HEELPROOF STRAINER                                                        |
| 50 | 2726031 | MIFAB A4-3-30-4 3" DRAIN BODY ONLY                                                                       |
| 51 | .....   | <b>FD-3</b>                                                                                              |
| 52 | 2102082 | ZURN ZN511-3NL-S-Y FLOOR DRAIN W/ SEDIMENT BUCKET                                                        |
| 53 | .....   | <b>FS-1</b>                                                                                              |
| 54 | 1401756 | ZURN Z1900-3NL-2 3X12 NEO-LOC SANI-FLOR ACID RESISTANT ENAMEL COAT CI RECEPT W/ 6 SUMP DEPTH & 1/2 GRATE |
| 55 | .....   | <b>FS-2</b>                                                                                              |

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|    |         |                                                                                                                                                                          |
|----|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 56 | 64495   | ELMUST 63M DURASTONE WHT 24 X 24 X 10IN MOP<br>SVC BASIN                                                                                                                 |
| 57 | .....   | <b>PRV</b>                                                                                                                                                               |
| 58 | 1240786 | WATTS LFN223M1-S-B 0298617 3 TXT HIGH CAP IRON<br>BODY 25-75PSI WTR PRESS RED VLV W/ STR SET AT<br>50PSI LEAD FREE                                                       |
| 59 | .....   | <b>GREASE INTERCEPTOR</b>                                                                                                                                                |
| 60 | 1460432 | MISC NON-STOCK ITEM ***DO NOT USE TO BUY<br>PRODUCT***                                                                                                                   |
| 61 | .....   | <b>EWB-1</b>                                                                                                                                                             |
| 62 | 1918115 | RHEEM ELD52-TB 683744 480V 6KW/6KW 50 GAL COM<br>LT DUTY ELEC WTR HTR W/ TERM BLOCK 58-5/8H<br>20-1/4D                                                                   |
| 63 | .....   | <b>GAS METER</b>                                                                                                                                                         |
| 64 | 2496658 | ON F-5500-3114-1150 INSERTION THERMAL MASS FLOW<br>METER W/ 1.0 NPT ADAPTER, 24 VDC LOOP POWERED<br>4-20MA AND RS485 SERIAL COMM, INTEGRAL<br>ENCLOSURE WITH LCD DISPLAY |
| 65 | 2052116 | ONICON INSTL0094-FMH INSTALLATION KIT                                                                                                                                    |

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ALL MODELS ARE SUBJECT TO APPROVAL



DF-1



## PRODUCT SPECIFICATIONS

Elkay Wall Mount Vandal Resistant Bi-level ADA Cooler Non-filtered Non-refrigerated Stainless. Features shall include Vandal Resistant. Furnished with Vandal Resistant bubbler. Mechanical Front Bubbler Button activation. Product shall be Wall Mount (On Wall), for Indoor + Outdoor applications, serving 2 station(s). Unit shall be lead-free design which is certified to NSF/ANSI 61 & 372 (lead free) and meets Federal and State low-lead requirements.

|                                 |                                 |
|---------------------------------|---------------------------------|
| <b>Special Features:</b>        | Vandal Resistant                |
| <b>Finish:</b>                  | Stainless Steel                 |
| <b>Power:</b>                   | No Electrical Required          |
| <b>Bubbler Style:</b>           | Vandal Resistant                |
| <b>Activation by:</b>           | Mechanical Front Bubbler Button |
| <b>Mounting Type:</b>           | Wall Mount (On Wall)            |
| <b>Chilling Capacity:</b>       | Non-refrigerated                |
| <b>Dimensions (L x W x H):</b>  | 36-5/16" x 18-5/8" x 27-1/8"    |
| <b>Approx. Shipping Weight:</b> | 71 lbs.                         |
| <b>Installation Location:</b>   | Indoor + Outdoor                |
| <b>No. of Stations Served:</b>  | 2                               |

**\*\*When used in non-temperature controlled environments, unit(s) must be adequately winterized and/or protected from extreme heat to prevent damage where climates dictate.**

- Mechanically-Activated bubbler continues to supply water in event of service disruptions.



**Included with Product:** Water Cooler  
**Ships in one box.**

A Century of Tradition and Quality.  
For more than 100 years, Elkay has been making innovative products and providing exceptional customer care. We take pride in offering plumbing products that make life easier, inspire change and leave the world a better place.



## PRODUCT COMPLIANCE

ADA & ICC A117.1

ASME A112.19.3/CSA B45.4

Buy American Act

NSF/ANSI 61 & 372 (lead free)



Complies with ADA & ICC A117.1 accessibility requirements when installed according to the requirements outlined in these standards. Installation may require additional components and/or construction features to be fully compliant. Consult the local Authority Having Jurisdiction if necessary.

[Installation Instructions \(PDF\) - 98958C](#)

Water Systems are warranted for 12 months from date of installation.

**Warranty pertains to drinking water applications only. Non-drinking water applications are not covered under warranty.**

[Warranty \(PDF\)](#)

PART: \_\_\_\_\_ QTY: \_\_\_\_\_

PROJECT: \_\_\_\_\_

CONTACT: \_\_\_\_\_

DATE: \_\_\_\_\_

NOTES: \_\_\_\_\_

APPROVAL: \_\_\_\_\_

*In keeping with our policy of continuing product improvement, Elkay reserves the right to change product specifications without notice. Please visit elkay the most current version of Elkay product specification sheets. This specification describes an Elkay product with design, quality, and functional benefits to When making a comparison of other producers' offerings, be certain these features are not overlooked.*



630619

**IMPORTANT! INSTALLER PLEASE NOTE:**

This water cooler has been designed and built to provide water to the user which has not been altered by materials in the cooler waterways.

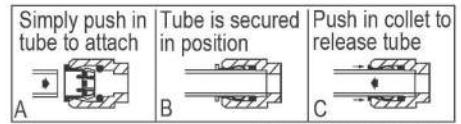
The grounding of electrical equipment such as telephone, computers, etc to water lines is a common procedure. This grounding may be in the building or occur away from the building. This grounding can cause electrical feedback into a water cooler creating an electrolysis which creates a metallic taste or causes an increase in the metal content of the water. This condition is avoidable by installing the cooler using the proper materials as shown below.

**NOTICE**

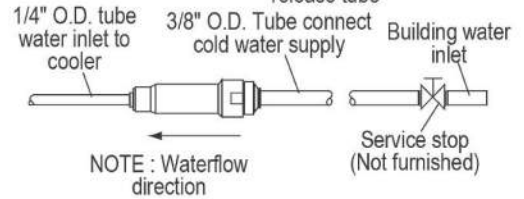
The water cooler must be connected to the water supply using a dielectric coupling. The cooler is furnished with a non-metallic strainer which meets this requirement. The drain trap which is provided by the installer should also be plastic to completely isolate the cooler from the building plumbing system.

These products are designed to operate on 20 psi to 105 psi supply line pressure. Simultaneous operation of both bubblers on a bi-level unit may not be possible depending on water supply pressure. If simultaneous operation is desired, please ensure a minimum of 50 psi supply.

**OPERATION OF QUICK CONNECT FITTINGS**

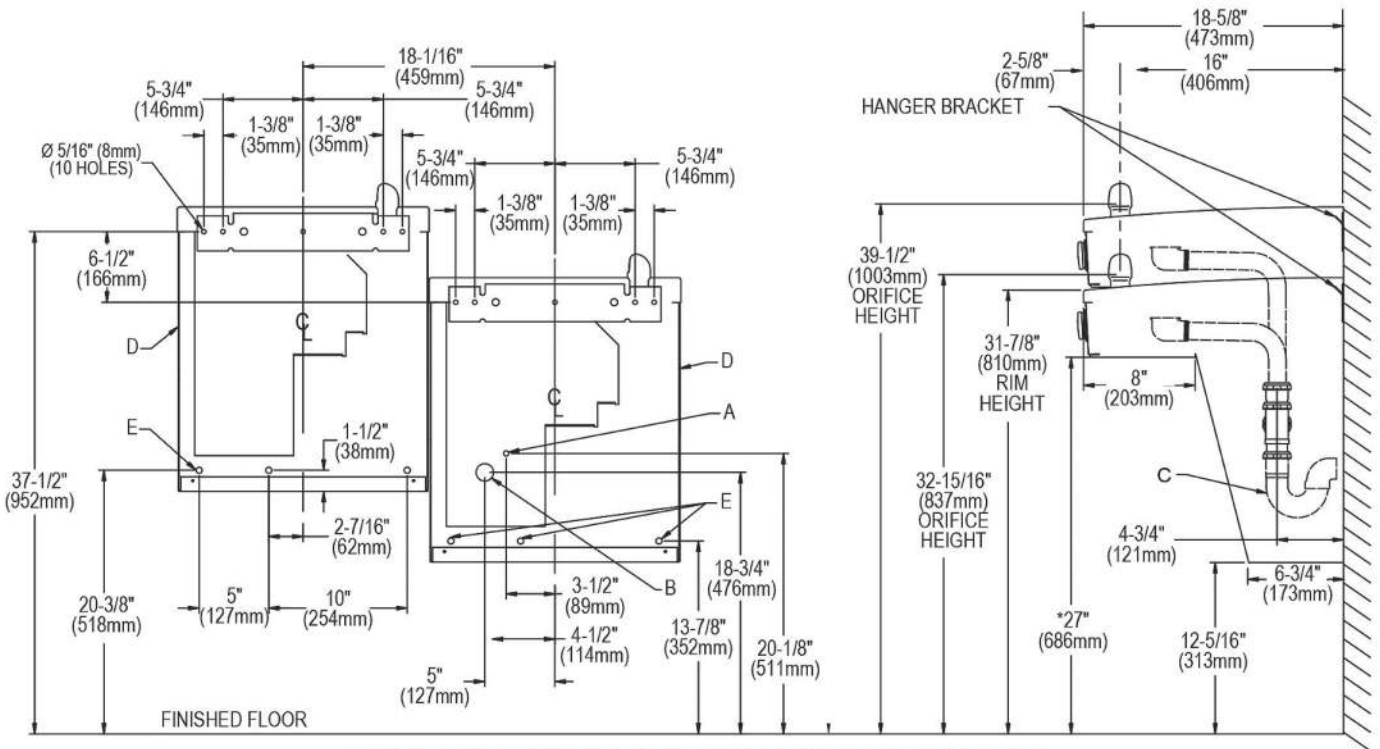


Pushing tube in before pulling it out helps to release tube



NOTE : Waterflow direction

630619



REDUCE HEIGHT BY 3" FOR INSTALLATION OF CHILDRENS ADA COOLER

**LEGEND**

- A = Recommended water supply location. Shut off valve (not furnished) to accept 3/8" O.D. unplated copper tube. 3" (76mm) maximum out from wall.
- B = Recommended location for waste outlet 1-1/2" O.D. drain. Drain stub 2" (51mm) out from wall.
- C = 1-1/2" Trap not furnished.
- D = Insure proper ventilation by maintaining 6" (152mm) (MIN.) clearance from cabinet louvers to wall.
- E = 7/16" Bolt holes for fastening unit to wall.

\*ADA REQUIREMENT

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HB-1



The Model 65, B65, RB65 are automatic draining, freezeless wall hydrants with single check hose connection anti-siphon vacuum breakers. Hydrants drain as handle is shut off, *even if hose is attached*. All models are designed for irrigation purposes to blend in with modern architecture for installation on restaurants, schools, office buildings, churches, apartments, motels, stores, shopping centers and industrial buildings.

### SPECIFICATIONS:

#### MODEL 65/B65/RB65

- Approved under ASSE Standard 1019-B
- Listed by IAPMO®
- Meets Government Specification WW-P-541b

#### VACUUM BREAKER - ANTI SIPHON -

- NIDEL® Model 34HA with 3/4 inch male hose thread
- Approved under ASSE 1011
- Listed by IAPMO®

**VALVE SEAT** – Permanent type brass valve body with hemispherical seating surface.

**VALVE** – One piece valve plunger accurately controls both flow and drainage with a minimum number of turns and without need for adjustments.

**DRAIN** – Under nozzle away from hands of operator and with a lip to divert water away from building.

**CASING** – Copper tubes.

**NO LEAD SOLDER** - All solder joints.

**STEM** – Hardened stainless steel stem resists damage.

**TEE KEY** – Loose key operates hydrant.

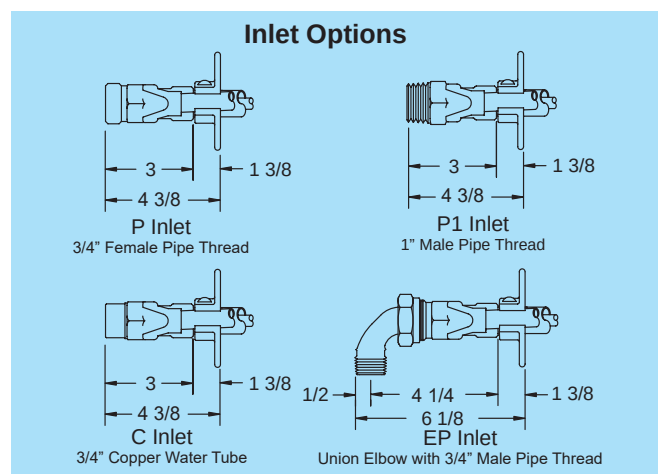
**OPERATING ROD** – 3/8" solid brass operating rod.

**INLETS** – As shown.

**WALL CLAMP** – Furnished on all 60 series except close coupled.

**MAX PRESSURE** – 125 p.s.i.

**MAX TEMPERATURE** - 120° F



### Specify as follows:

Wall hydrant shall be Woodford Model 65 (exposed type), B65, RB65 or MB65 (concealed box type), automatic draining with anti-siphon vacuum breaker. ASSE Standard 1019-B approved. 3/4" inlet and outlet (specify type of inlet). Hardened stainless steel operating stem and one-piece valve plunger to control both flow and drain functions. Exterior finish to be Chrome Plated (options: Polished Brass or Rough Brass). Loose tee key to be furnished with each hydrant. Wall thickness to be \_\_\_\_\_ inches.

**When ordering, specify model, inlet and wall thickness**

## Anti-Siphon Freezeless Wall Hydrants

## Model 65/B65/RB65



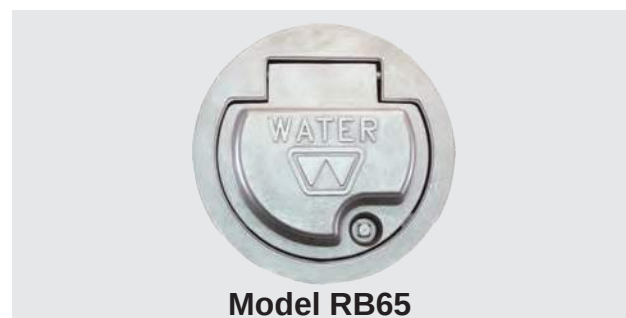
**Model 65**

Standard - Chrome (CH)  
Optional - Brass (BR) Polished Brass (PB)  
Fits one standard modular brick course.



**Model B65**

Exterior Finish: Box & Door  
Standard - Chrome (CH)  
Optional - Brass (BR) Polished Brass (PB)  
Anodized Aluminum Box (AL)  
Fits two std. modular brick course or one course of facing tile.

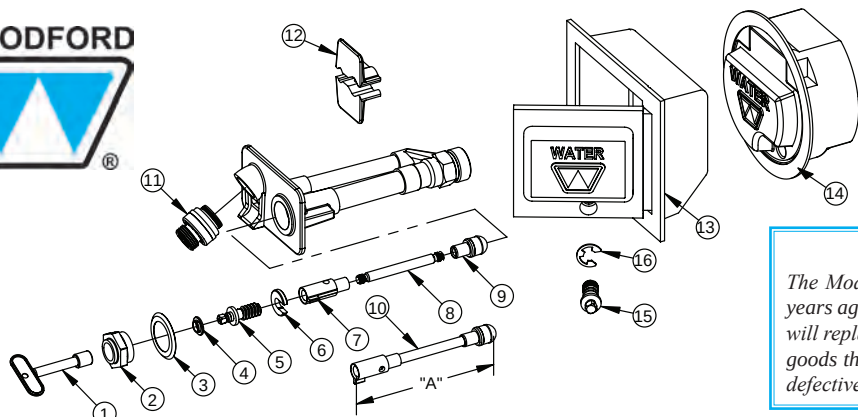


**Model RB65**

Exterior Finish: Box & Door - Chrome (CH) Only  
Designed especially for tilt-up wall construction.  
Install through 6" diameter hole.  
Two 3/8"-16 tapped holes for all-thread anchors in back of box.  
Lift and latch door stays open when hydrant is in use.

**SEE WOODFORDMFG.COM FOR INSTRUCTIONS  
AND DETAILS OF APPLICABLE PATENTS AND  
PATENTS PENDING OR CALL 1-800-621-6032**





### LIMITED WARRANTY

The Model 65 is sold with a limited warranty for five years against defects in material and workmanship. We will replace or issue credit (at our option) for defective goods that are returned for inspection and found to be defective within five years of purchase from Woodford.

### MODEL 65/B65/RB65 PARTS LIST

| ITEM | PART #                                 | DESCRIPTION                                                                                          |
|------|----------------------------------------|------------------------------------------------------------------------------------------------------|
| 1    | 50009                                  | Tee Key                                                                                              |
|      | 50010                                  | Long Tee Key (Box Models)                                                                            |
| 2    | 50264                                  | Head Nut – chrome                                                                                    |
|      | 50265                                  | Head Nut – brass                                                                                     |
| 3    | 50583                                  | I.D. RING                                                                                            |
| 4    | 50252                                  | Stem Washer                                                                                          |
| 5    | 55062                                  | Stem Screw                                                                                           |
| 6    | 50251                                  | "C" washer                                                                                           |
| 7    | 51014                                  | Yoke Nut                                                                                             |
| 8    | Operating Rod (Sold With Item 10 Only) |                                                                                                      |
| 9    | 50250                                  | Plunger                                                                                              |
| 10   | 554XX                                  | Operating Rod Assembly (Includes Items 7,8,9)<br>(Select required assembled length from table below) |
| 11   | 34HA-CH                                | 34HA Vacuum Breaker – chrome                                                                         |
|      | 34HA-BR                                | 34HA Vacuum Breaker – brass                                                                          |
| 12   | 55063                                  | Wall Clamp Assembly                                                                                  |
| 13   | 67BX                                   | Box/Door Assembly – Chrome                                                                           |
|      | 65BXF                                  | Box/Flat Door Assembly – Chrome                                                                      |
|      | 67BX-BR                                | Box/Door Assembly - Brass                                                                            |
|      | 65BXF-BR                               | Box/Flat Door Assembly - Brass                                                                       |
|      | 67BX-PB                                | Box/Door Assembly – Polished Brass                                                                   |
|      | 65BXF-PB                               | Box/Flat Door Assembly – Polished Brass                                                              |
|      | 67BX-AL                                | Box/Door Assembly – Aluminum                                                                         |
| 14   | RB67BX                                 | Round Box/Door Assembly - Chrome                                                                     |
|      | RK-65                                  | Chrome Repair Kit (Includes Items 1, 2, 4-7 & 9)                                                     |
| 15   | 51117                                  | Door Lock Screw - Chrome                                                                             |
|      | 51116                                  | Door Lock Screw - Brass                                                                              |
| 16   | 51120                                  | E-Ring                                                                                               |

#### Operating Rod Assy. "A"

| Wall Thickness | Length   | Part No. |
|----------------|----------|----------|
| 65             | B65/RB65 |          |
| CC             | CC       | 3 1/2"   |
| N/A            | 4        | 4 1/2"   |
| 4              | 6        | 6 1/2"   |
| 6              | 8        | 8 1/2"   |
| 8              | 10       | 10 1/2"  |
| 10             | 12       | 12 1/2"  |
| 12             | 14       | 14 1/2"  |
| 14             | 16       | 16 1/2"  |
| 16             | 18       | 18 1/2"  |
| 18             | 20       | 20 1/2"  |
| 20             | 22       | 22 1/2"  |
| 22             | 24       | 24 1/2"  |
| 24             | N/A      | 26 1/2"  |

Manufactured under one or more of the following patents:  
 U.S. Patents:  
 3,414,001  
 4,178,956  
 4,316,481  
 4,532,954  
 D216,790  
 D216,791  
 D277,365  
 D277,366  
 Canada Patents:  
 852,529  
 865,995  
 1,146,438

| Wall Thickness (Inch)       | CC    | 4     | 6      | 8      | 10     | 12     | 14     | 16     | 18     | 20     | 22     | 24     |
|-----------------------------|-------|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Overall Length - Hydrant    | 5     | 8     | 10     | 12     | 14     | 16     | 18     | 20     | 22     | 24     | 26     | 28     |
| Overall Length - Box Models | 7 3/8 | 8 3/8 | 10 3/8 | 12 3/8 | 14 3/8 | 16 3/8 | 18 3/8 | 20 3/8 | 22 3/8 | 24 3/8 | 26 3/8 | 28 3/8 |
| Shipping Wt. (Lbs)**        | 3.8   | 4.4   | 4.8    | 5.2    | 5.6    | 6      | 6.4    | 6.8    | 7.2    | 7.6    | 8      | 8.4    |

\*\* Add 8.3 Lbs. for brass box models.

\*\* Add 2.5 Lbs. for aluminum box models.

NOTE: Close Coupled Models are not recommended for use in freezing climates and cannot use wall clamp.

For more information contact...

## WOODFORD MANUFACTURING COMPANY, LLC.

2121 Waynoka Road, Colorado Springs, Colorado 80915 • Phone: (800) 621-6032 • Fax: (800) 765-4115

To view our complete product line visit: [www.woodfordmfg.com](http://www.woodfordmfg.com) or email: [sales@woodfordmfg.com](mailto:sales@woodfordmfg.com)



HB-2




**T&S BRASS AND BRONZE WORKS, INC.**

 2 Saddleback Cove / P.O. Box 1088  
 Travelers Rest, SC 29690

Model No.

**MV-0771-12N-BVT**

Item No.

Travelers Rest, SC: 800-476-4103 • Simi Valley, CA: 800-423-0150 • Fax: 864-834-3518 • www.tsbrass.com

This Space for Architect/Engineer Approval

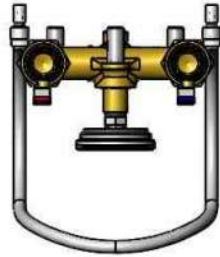
Job Name \_\_\_\_\_ Date \_\_\_\_\_

Model Specified \_\_\_\_\_ Quantity \_\_\_\_\_

Customer/Wholesaler \_\_\_\_\_

Contractor \_\_\_\_\_

Architect/Engineer \_\_\_\_\_


 3/4" Ball Valves  
 w/ Stainless Steel,  
 Color Coded  
 Handles & Nuts

 3/4" Check  
 Valves

3/4" Unions

 Bi-Metallic  
 Thermometer  
 w/ Stainless  
 Steel Case

 5"  
 [127mm]

 9 1/4"  
 [235mm]

 6"  
 [152mm]

Open

Open

Closed

Closed

 7 15/16"  
 [202mm]

 2 3/4"  
 [70mm]

 $\phi$  7/16" (3)  
 [11mm]

 $\phi$  5/16" (2)  
 [7mm]

Hose Swivel

 Stainless Steel  
 Hose Rack w/  
 Mounting Hardware

 8"  
 [203mm]

 3 3/16"  
 [81mm]

 3/4" NPT  
 Female  
 Thread Inlets

 1 1/2"  
 [38mm]

 14 3/16"  
 [361mm]

Wall

 9 3/8"  
 [239mm]

## Product Specifications:

 Natural Finish Washdown Station w/ 3/4" Mixing Valve, Ball Valves, Check  
 Valves, Thermometer, Hose Swivel & Stainless Steel Hose Rack

## Product Compliance:

NSF 61 Exempt (Non-Potable)

Drawn: DMH | Checked: JRM | Approved: JHB | Date: 07/17/20 | Scale: 1:8 | Sheet: 1

1624299




**T&S BRASS AND BRONZE WORKS, INC.**

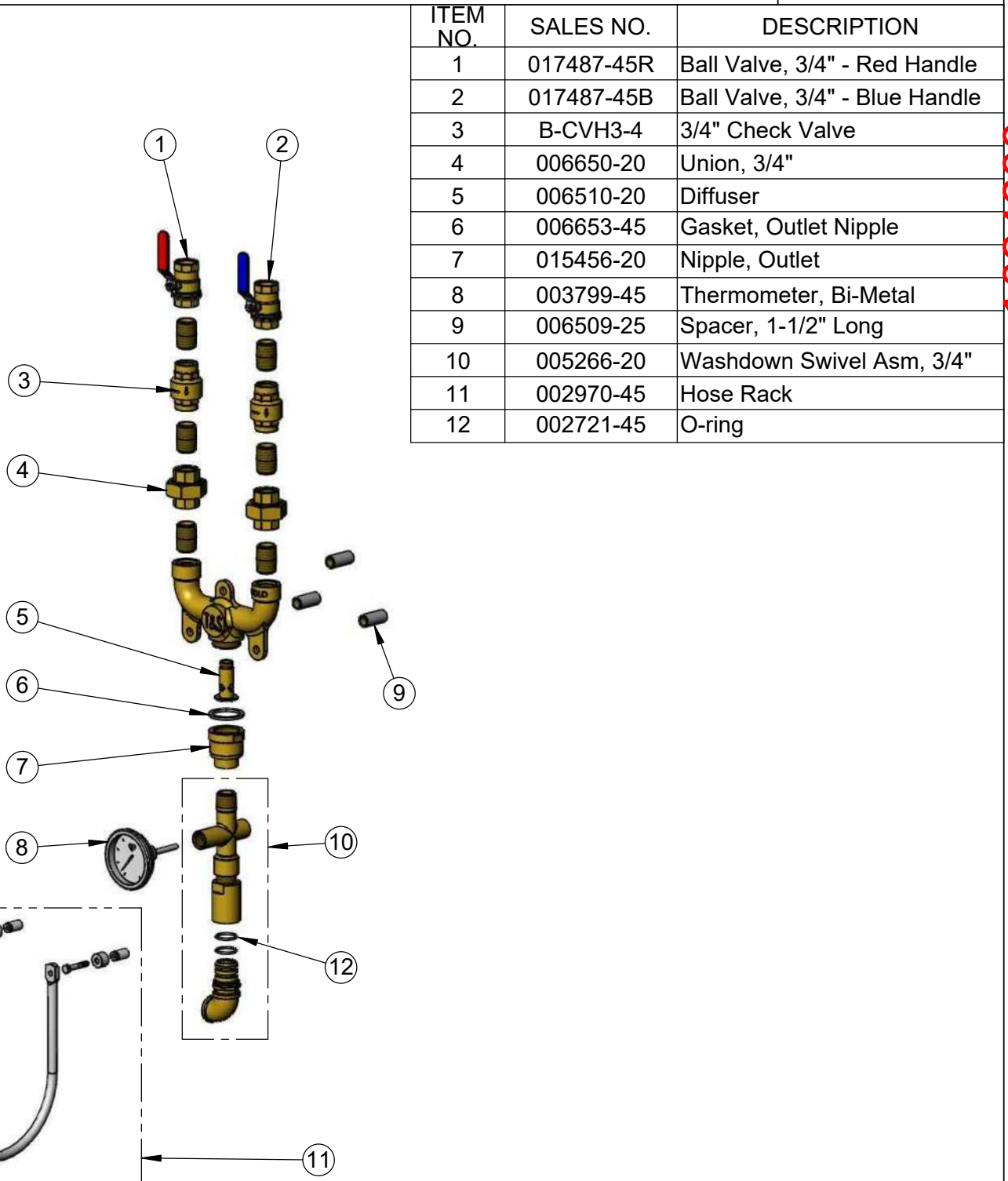
 2 Saddleback Cove / P.O. Box 1088  
 Travelers Rest, SC 29690

Model No.

**MV-0771-12N-BVT**

Item No.

Travelers Rest, SC: 800-476-4103 • Simi Valley, CA: 800-423-0150 • Fax: 864-834-3518 • www.tsbrass.com



1624299

**Product Specifications:**

Natural Finish Washdown Station w/ 3/4" Mixing Valve, Ball Valves, Check Valves, Thermometer, Hose Swivel &amp; Stainless Steel Hose Rack

**Product Compliance:**

NSF 61 Exempt (Non-Potable)



L-1





**Soho®**  
Wall-Mount Bathroom Sink  
**K-2084**

**Features**

- Made from premium materials that withstand high-volume usage
- 20" (508 mm) x 18" (457 mm)
- Faucet sold separately

**Material**

- Vitreous china

**Installation**

- Wall-mount
- Drilled for concealed arm carrier installation

**Recommended Products/Accessories**

- K-8998 P-Trap
- K-23726 Drain treatment
- K-23725 Cast iron cleaner



342134

|     |          |     |
|-----|----------|-----|
| ADA | CSA B651 | OBC |
|-----|----------|-----|

**Codes/Standards**

ASME A112.19.2/CSA B45.1  
ADA  
ICC/ANSI A117.1  
CSA B651  
OBC

**KOHLER® Vitreous China Bathroom Sink Lifetime Limited Warranty**

See website for detailed warranty information.

**Available Colors/Finishes**

*Color tiles intended for reference only.*

| Color                                                                               | Code | Description  |
|-------------------------------------------------------------------------------------|------|--------------|
|  | 0    | White        |
|  | 96   | Biscuit      |
|  | 7    | Black Black™ |



**KOHLER** Faucets

# Sculpted

Touchless single-hole lavatory faucet with Insight™ sensor technology, DC-powered, 0.5 gpm  
**K-13460**

**Features**

- 0.5 gpm (1.9 lpm) maximum flow rate
- Factory-set 30-second maximum continual run cycle
- Vandal-resistant aerator
- 24" (610 mm) flexible supply hose for easy installation
- Less drain
- Includes two AAA batteries

**Material**

- Brass

**Technology**

- Electronic Insight™ sensor technology for accurate and consistent activation

**Installation**

- Single-hole mounting
- Above-counter valve for easy installation and maintenance

**Recommended Products/Accessories**

K-23726 Drain treatment

**Optional Products/Accessories**

1160595 .35 gpm Aerator, Medium  
 1075455 Medium Spray 0.5 gpm (1.9 l/min)  
 1139284 Medium Aerated 1.0 gpm (3.8 l/min)  
 1139285 Medium Laminar 1.0 gpm (3.8 l/min)  
 1097180 Medium Aerated 1.5 gpm (5.7 l/min)  
 1104546 Medium Laminar 1.5 gpm (5.7 l/min)  
 79441 Sealing Washer

**ADA****CSA B651****OBC****Codes/Standards**

ASME A112.18.1/CSA B125.1  
 NSF/ANSI/CAN 372  
 DOE - Energy Policy Act 1992  
 ADA  
 ICC/ANSI A117.1  
 CSA B651  
 OBC  
 NOM Standards  
 NOM 012-CONAGUA

**KOHLER® One-Year Limited  
 Warranty**

See website for detailed warranty information.

834804

1-800-4KOHLER (1-800-456-4537)

Kohler Co. reserves the right to make revisions without notice to product specifications.

For the most current Specification Sheet, go to [www.kohler.com](http://www.kohler.com) USA or [www.kohler.ca](http://www.kohler.ca) Canada

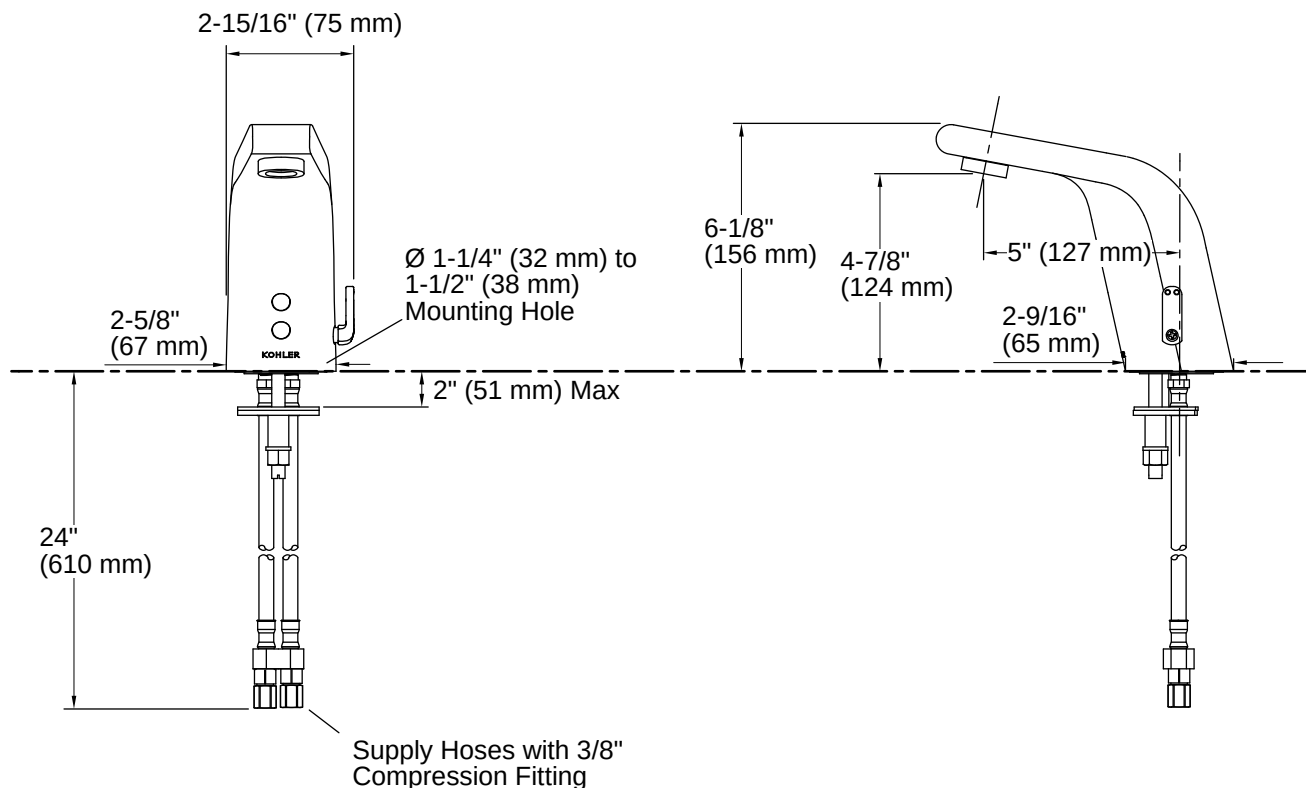
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THE BOLD LOOK  
 OF **KOHLER**



**KOHLER** Faucets

**Sculpted**  
 Touchless single-hole lavatory faucet with Insight™  
 sensor technology, DC-powered, 0.5 gpm  
**K-13460**

**Technical Information**

All product dimensions are nominal.

Power source: Battery - DC, included

Faucet flow rate: 0.5 gal/min (1.9 l/min)

**Spout:**

Spout reach: 5" (127 mm)

**Notes**

Install this product according to the installation guide.

A 1-1/4" (32 mm) minimum to 1-1/2" (38 mm) maximum diameter mounting hole is required.

The minimum distance between the back of the faucet spout and the wall must be 1-3/4" (44 mm). This will allow proper clearance for setscrew access.

The maximum distance between the bathroom sink basin edge to the base of the faucet spout must be 1-3/4" (44 mm).

The maximum height of the lip basin should be 3/4" (19 mm).

Product includes rosebud spray aerator. Other aerator styles are available.

ADA, OBC, CSA B651 compliant when installed to the specific requirements of these regulations.

1-800-4KOHLER (1-800-456-4537)

Kohler Co. reserves the right to make revisions without notice to product specifications.

For the most current Specification Sheet, go to [www.kohler.com](http://www.kohler.com) USA or [www.kohler.ca](http://www.kohler.ca) Canada

3-7-2026 19:31 - US/CA

THE BOLD LOOK  
 OF **KOHLER**



834804

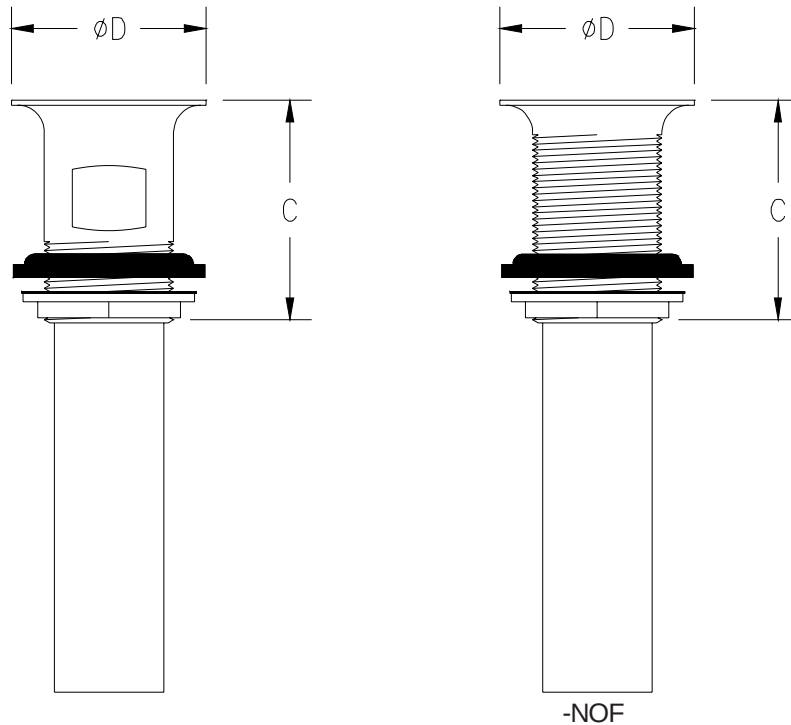


# SOLID TOP, OPEN-GRID P.O. PLUGS

## Z8743-PC

### TAG \_\_\_\_\_

**Engineering Specification:** Zurn Z8743-PC – Chrome plated cast brass solid top, open grid, P.O. plug. Furnished with 5"[127mm] long, chrome plated brass 17 gauge tailpiece and chrome plated cast brass locknut for sink depth up to 2 1/4"[57mm]. 6"-, 8"-, and 12"- tailpieces available.



258659

Note: All dimensions are for reference only. Do not use for pre-plumbing

| PRODUCT<br>NUMBER | TAILPIECE<br>in[mm] | DIMENSIONS |           | MASTER<br>CARTON | WEIGHT<br>CARTON |
|-------------------|---------------------|------------|-----------|------------------|------------------|
|                   |                     | Cin[mm]    | ØDin[mm]  |                  |                  |
| Z8743-PC          | 1 1/4[32]           | 2 9/16[65] | 2 1/4[57] | 50               | 50               |
| Z8743-1-PC        | 1 1/2[38]           | 2 9/16[65] | 2 1/4[57] | 30               | 30               |

### OPTIONAL ACCESSORIES

| Suffix     | Description            |
|------------|------------------------|
| _____ -NOF | No Overflow            |
| _____ -6   | w/6"[152mm] Tailpiece  |
| _____ -8   | w/8"[203mm] Tailpiece  |
| _____ -12  | w/12"[305mm] Tailpiece |

ZURN INDUSTRIES, LLC ♦ COMMERCIAL BRASS OPERATION ♦ 2640 SOUTH WORK STREET ♦ FALCONER NY 14733

Phone: 1-716-665-1132 ♦ Fax: 1-716-665-1135 ♦ World Wide Web: [www.zurn.com](http://www.zurn.com)

In Canada: ZURN INDUSTRIES LIMITED ♦ 3544 Nashua Drive ♦ Mississauga, Ontario L4V1L2 ♦ Phone: 905/405-8272 Fax: 905/405-





## 1-1/4" TUBULAR P-TRAP

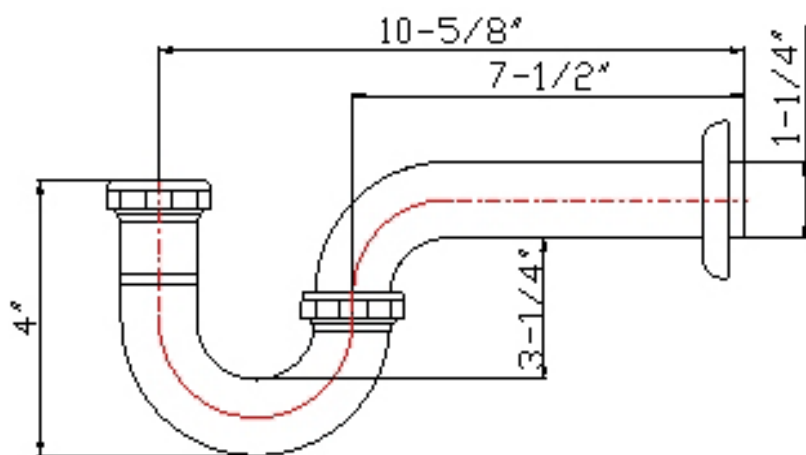
## SPECIFICATIONS

## 1-1/4" TUBULAR LESS CLEANOUT

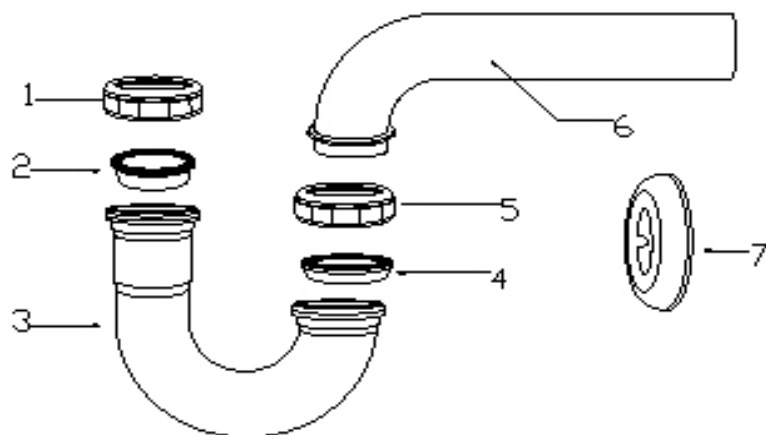
- 7-1/2" Brass Wall Tube
- 1-1/4" x 10-5/8"
- Zinc Nut Kit
- Shallow Escutcheon
- Chrome Plated



974288



| Model # | ML7001 | ML27001 |
|---------|--------|---------|
| Gauge   | 20     | 22      |



| No. | Part Name  | Material |
|-----|------------|----------|
| 1   | Nut        | Zinc     |
| 2   | Washer     | Rubber   |
| 3   | J-Bend     | Brass    |
| 4   | Washer     | Rubber   |
| 5   | Nut        | Zinc     |
| 6   | Wall Tube  | Brass    |
| 7   | Escutcheon | Steel    |

## CERTIFICATIONS

- cUPC



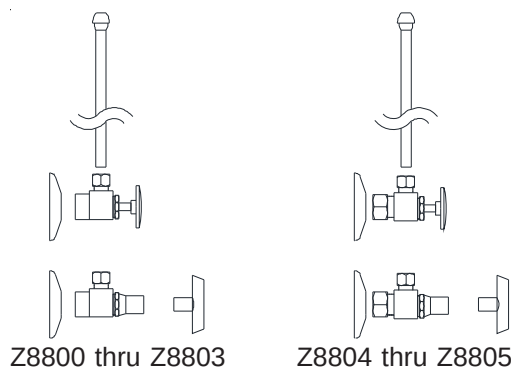


# STANDARD STOP WITH FLEXIBLE LAVATORY SUPPLY

## Z8800-XL-LR-PC TO Z8809-XL-LRLK-PC TAG \_\_\_\_\_

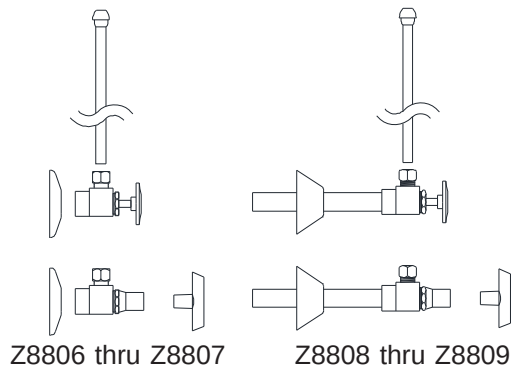
**Engineering Specification:** Zurn Z8800-XL-LR-PC to Z8809-XL-LRLK-PC Two Zurn chrome plated, solid brass angle stops with round wheel handles or loose key as specified, two 12"[305mm] flexible chrome plated copper lavatory risers complete with two chrome plated steel flanges. Z8808-XL-LR-PC to Z8809-XL-LRLK-PC include 5"[127mm] chrome plated copper extension tubes and deep bell steel flanges.

Zurn Lead Compliant\* "XL" products are a line of durable, high quality brass faucets and fixtures that are designed and manufactured to comply with state laws and local codes that mandate lead content levels. Zurn "XL" products are manufactured with \*not more than 0.25% lead content when used with respect to wetted surfaces of pipes and pipe fittings, plumbing fitting and fixtures: California Health & Safety Code § 116875; Vermont 9 VCA § 2470h.



Z8800 thru Z8803

Z8804 thru Z8805



Z8806 thru Z8807

Z8808 thru Z8809

| PRODUCT NUMBER   | CONNECTIONS                | EXT. | MASTER CARTON | WEIGHT CARTON |      |
|------------------|----------------------------|------|---------------|---------------|------|
|                  |                            |      |               | LR            | LRLK |
| Z8800-XL-LR-PC   | 3/8 [10] IPS x 3/8 [10] OD | -    | 25            | 24            |      |
| Z8800-XL-LRLK-PC | 3/8 [10] IPS x 3/8 [10] OD | -    | 25            |               | 25   |
| Z8802-XL-LR-PC   | 1/2 [13] IPS x 3/8 [10] OD | -    | 25            | 24            |      |
| Z8802-XL-LRLK-PC | 1/2 [13] IPS x 3/8 [10] OD | -    | 25            |               | 25   |
| Z8803-XL-LR-PC   | 1/2 [13] IPS x 1/2 [13] OD | -    | 25            | 24            |      |
| Z8803-XL-LRLK-PC | 1/2 [13] IPS x 1/2 [13] OD | -    | 25            |               | 25   |
| Z8804-XL-LR-PC   | 1/2 [13] NOM x 3/8 [10] OD | -    | 25            | 24            |      |
| Z8804-XL-LRLK-PC | 1/2 [13] NOM x 3/8 [10] OD | -    | 25            |               | 25   |
| Z8805-XL-LR-PC   | 1/2 [13] NOM x 1/2 [13] OD | -    | 25            | 24            |      |
| Z8805-XL-LRLK-PC | 1/2 [13] NOM x 1/2 [13] OD | -    | 25            |               | 25   |
| Z8806-XL-LR-PC   | 1/2 [13] SWT x 3/8 [10] OD | -    | 25            | 24            |      |
| Z8806-XL-LRLK-PC | 1/2 [13] SWT x 3/8 [10] OD | -    | 25            |               | 25   |
| Z8807-XL-LR-PC   | 1/2 [13] SWT x 1/2 [13] OD | -    | 25            | 24            |      |
| Z8807-XL-LRLK-PC | 1/2 [13] SWT x 1/2 [13] OD | -    | 25            |               | 25   |
| Z8808-XL-LR-PC   | 1/2 [13] SWT x 3/8 [10] OD | 5    | 25            | 28            |      |
| Z8808-XL-LRLK-PC | 1/2 [13] SWT x 3/8 [10] OD | 5    | 25            |               | 33   |
| Z8809-XL-LR-PC   | 1/2[13] SWT x1/2[13] OD    | 5    | 25            | 28            |      |
| Z8809-XL-LRLK-PC | 1/2 [13] SWT x 1/2 [13] OD | 5    | 25            |               | 33   |

### SUFFIXES

- ☐ -A Single supply option
- ☐ -C1 Standard Line Lock Shield Cap
- ☐ -CE Chrome Plated, Cast Brass, Flange with setscrew
- ☐ -D Chrome Plated Steel, Deep Bell Flange (Bell Type)
- ☐ -XP Cross Handle

\*REGULARLY FURNISHED UNLESS OTHERWISE SPECIFIED

- ☐ - 3 3[76] Nipple
- ☐ - 5 5[127] Nipple
- ☐ -15 15[381] Supply Tube
- ☐ -20 20[508] Supply Tube
- ☐ -30 30[762] Supply Tube
- ☐ -36 36[914] Supply Tube

ZURN INDUSTRIES, LLC. ♦ COMMERCIAL BRASS OPERATION ♦ 5900 ELWIN BUCHANAN DRIVE ♦ SANFORD NC 27330

Phone: 1-800-997-3876 ♦ Fax: 919-775-3541 ♦ World Wide Web: [www.zurn.com](http://www.zurn.com)

In Canada: ZURN INDUSTRIES LIMITED ♦ 3544 Nashua Drive ♦ Mississauga, Ontario L4V1L2 ♦ Phone: 905-405-8272 Fax: 905-405-1292



1166059



# Molded Trap Covers

## Product Specifications

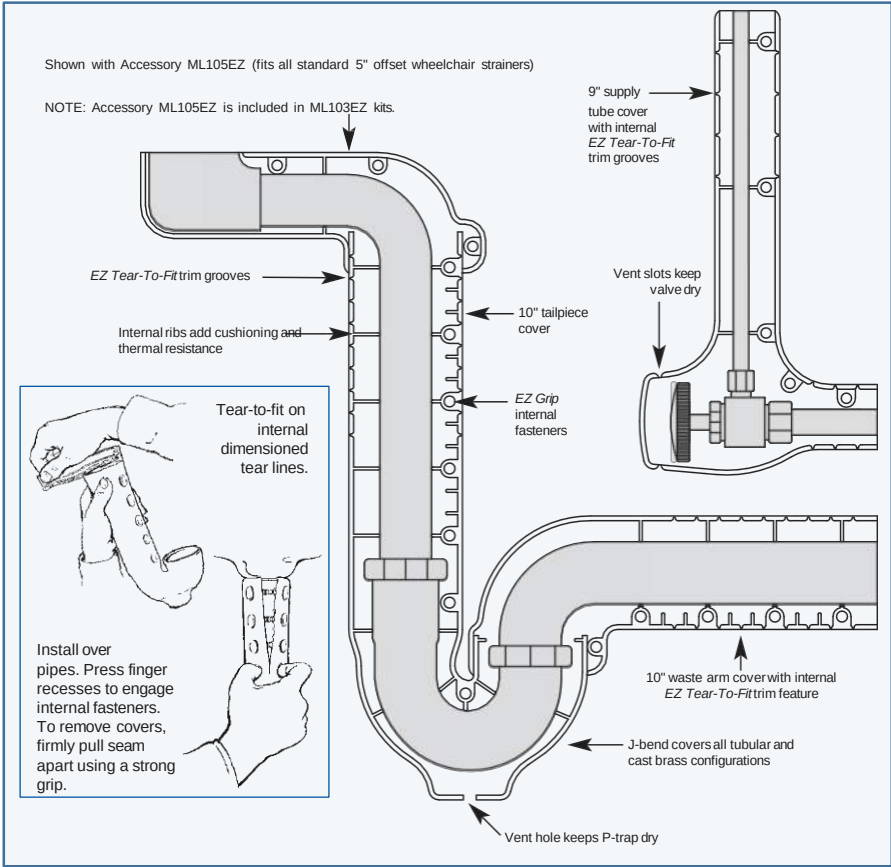
### FEATURES

EZ Series waste and supply piping covers satisfy ADA compliance requirements. Built-in EZ Grip internal fasteners and internal EZ Tear-To-Fit trimming feature allow for fast installation without tools. Cover kits are designed to fit on all tubular and cast brass 1-1/4" or 1-1/2" P-trap assemblies. Valve cover completely encloses 3/8" or 1/2" angle valve and supply tube for both handled and keyed type water stops.

| ✓ To Submit | Part #  | Description                                                                                  |
|-------------|---------|----------------------------------------------------------------------------------------------|
|             | ML101EZ | One P-trap cover, one angle valve and supply covers                                          |
|             | ML102EZ | One P-trap cover, two angle valve and supply covers                                          |
|             | ML103EZ | One P-trap cover, two angle valve and supply covers, one 5" offset wheelchair strainer cover |
|             | ML105EZ | One 5" offset wheelchair strainer cover                                                      |



1017492



| SPECIFICATIONS                     |                                               |
|------------------------------------|-----------------------------------------------|
| Material                           | Rigid high-impact, stain-resistant PVC        |
| Nominal Wall                       | 1/8" constant with internal ribs              |
| Durometer                          | 70-80 - Shore A                               |
| UV Protection                      | Will not fade or discolor                     |
| Durability                         | Virtually indestructible                      |
| Trimming (EZ Series)               | EZ Tear-To-Fit trim feature (no tools needed) |
| Fasteners (EZ Series)              | Internal EZ Grip fasteners, reusable          |
| Color                              | China white                                   |
| Paintability                       | Apply latex paint                             |
| Burning Characteristics ASTM D-635 | Self extinguished 0 sec (ATB) 0 mm (AEB)      |
| Bacteria/Fungus Resistance         | ASTM G21 and G22 — Result: 0 growth           |
| Maintenance                        | Wipe clean using common detergents            |

### Warranty

See warranty information for more details.

All dimensions listed are nominal. MAINLINE® reserves the right to make product and material changes at any time without notice.

Contact your sales representative for more information or visit our website at [mainlinecollection.com](http://mainlinecollection.com)





RD-1





H

DRAWING NUMBER  
S1015BS

SIZE  
A

SCALE:  
NONE

DATE:  
6-29-88

APPROVED BY:  
TD

CHECKED BY:  
MCD

DRAWN BY:  
EMB

FIGURE  
NUMBER

SMITH®

JAY R. SMITH MFG. CO.®

MEMBER OF MORRIS GROUP INTERNATIONAL  
POST OFFICE BOX 3237  
MONTGOMERY, ALABAMA 36109-0237 (USA)  
TEL: 334-277-8520 FAX: 334-272-7396 www.jrsmith.com

ASPE

MEMBER OF:

PDI

MEMBER OF:

LOCATION

ADJUSTABLE ROOF DRAIN

1

Position flat gasket on drain body, align holes in gasket with holes in body.

2

Determine if reversible collar is to be installed in the low (1 1/4" to 2 1/2") or high (2 1/4" to 4") position.

2a

Place reversible collar on body and align holes in collar with holes in gasket and body.

3

Secure collar to body with 1 1/2" long bolts insuring neoprene backed washers are installed with neoprene side facing down at proper locations between top of reversible collar and bolt head. Tighten bolts in the following manner: 1, 3, 2 & 4. Tighten No. 1 bolt approximately one full turn then No. 3, No. 2 and No. 4 (same manner as for No. 1). Continue this method until reversible ring has been secured evenly all around.

4

Wipe "O" Ring recess in reversible collar clean. Insert dry "O" Ring in groove of threads at inside diameter of reversible collar. Lubricate the face of "O" Ring in it's entirety with (1) tube of the lubricant provided and remaining threads.

5

Lubricate liberally the entire lead-in chamfer and first 2 threads of the adjustable sleeve. Thread adjustable sleeve into reversible collar and adjust to required height. Do not exceed minimum/maximum adjustment dimensions.

6

Check bolts a final time to insure tightness.

7

Run waterproofing membrane over top of adjustable sleeve to inside of bolt circle.

8

Install flashing clamp and secure with (4) bolts, use pipe sealant on threads. Again tighten bolts in the 1, 3, 2, 4 manner. (See 3)

9

Install Dome

**NOTE:** Stated flow rates for roof drains are based on particular test piping configurations and dimensions. Installed roof drain flow rates cannot be stated unless the attached piping configuration is known and the effects of the drainage calculated.

**NOTE:** Dimensions shown in parentheses are in millimeters.

1015BS

| REV. | DATE | DESCRIPTION | BY | CKD. BY | WEIGHT<br>POUNDS | VOLUME<br>CUBIC FEET | FIGURE NUMBER |
|------|------|-------------|----|---------|------------------|----------------------|---------------|
|      |      |             |    |         |                  |                      | 1015BS        |

**WARNING:** Cancer and Reproductive Harm - www.P65Warnings.ca.gov

1380508

RH-1





Build With Purpose

# P-RH4 Sanitary Roof Hydrant



full view



Find us in

**MasterSpec®**

a product of The American Institute of Architects



NSF 372

PRIER's P-RH4 Sanitary Roof Hydrant stores water in a lead free canister instead of requiring a drain line like traditional roof hydrants. While opening, the stored water re-enters the flow path and is first to exit the hydrant. The P-RH4 uses a piston system, rather than venturi, so it does not require winterization.

The Roof Mounting Bracket enables the hydrant to be attached to any structural member with rigid support, instead of to the skin of the building.

The bracket requires a single penetration point in the roof of 3 5/8", and can be attached to concrete, wood or steel, I-beams, trusses, perkins or other structures with standard construction materials (Unistrut, threaded rods, or other).

- Freezeless
- Lead Free
- No Drain Line Needed
- No Winterization Needed

## FEATURES

- Single hole penetration in roof of just 3 5/8"
- No winterization required, drain water contained within hydrant
- Standard 3/4" Garden Hose Outlet
- 1" NPT inlet
- ASSE 1057 approved, pressure rated to 125 PSI, temperature to 140°
- Component maintenance can be completed at roof level
- Hydrant head marked for High "H" or Low "L" desired flow rates
- Heavy-duty ductile cast iron head and lockable lever handle
- Easy to open and close O-ring seals utilizing pressure balance design
- High flow rate of 10 gallons per minute at 25 PSI differential
- Bucket catch on spout
- Included roof mounting bracket attaches to any roof structural member (truss, I-beam, concrete roof, or other)
- Lead-Free NSF 372 listed & labeled

## ORDERING

# P-RH4

Type

H4- Sanitary Roof Hydrant

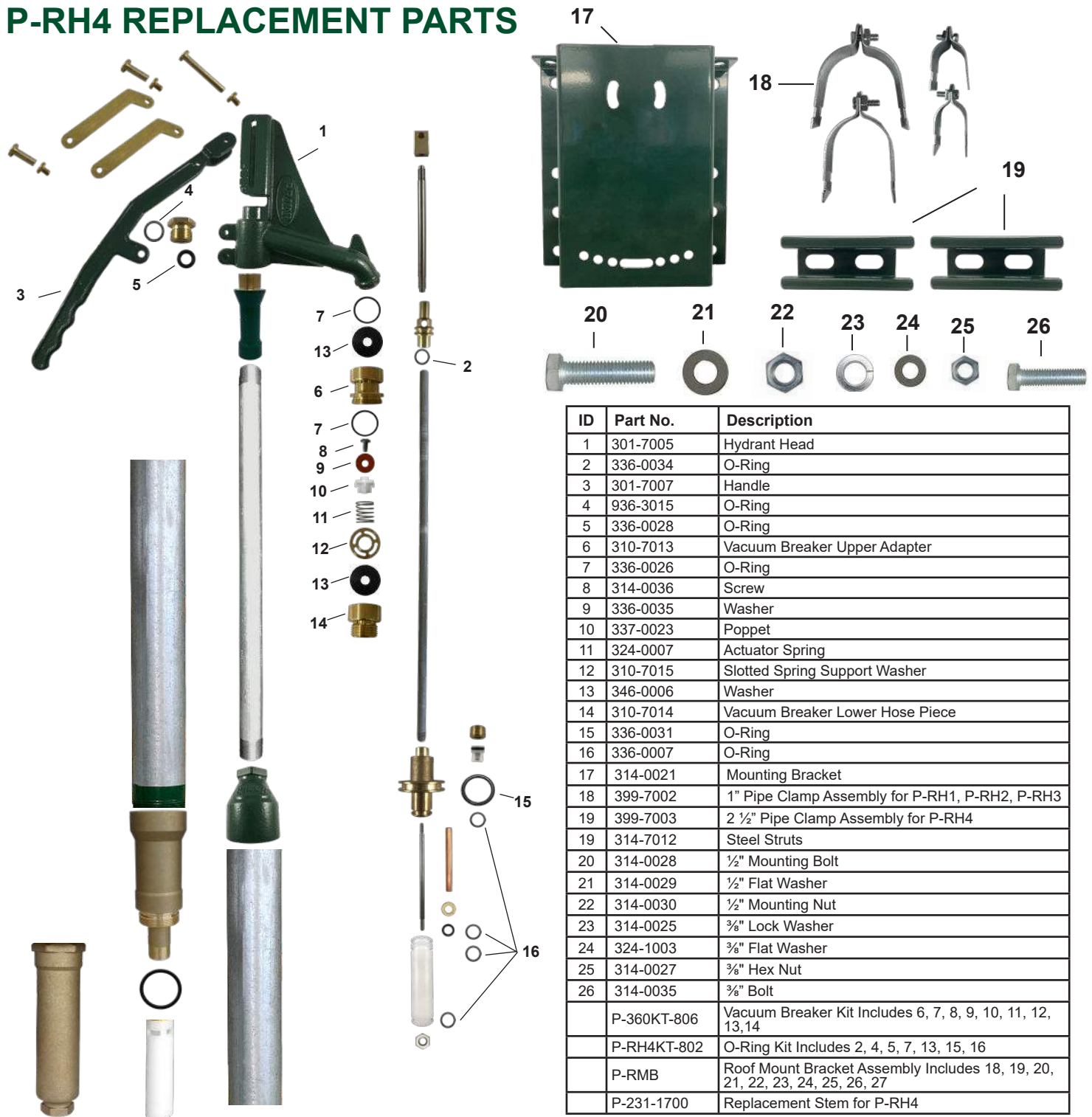


2693001



Build With Purpose

## P-RH4 REPLACEMENT PARTS



| ID | Part No.    | Description                                                                 |
|----|-------------|-----------------------------------------------------------------------------|
| 1  | 301-7005    | Hydrant Head                                                                |
| 2  | 336-0034    | O-Ring                                                                      |
| 3  | 301-7007    | Handle                                                                      |
| 4  | 936-3015    | O-Ring                                                                      |
| 5  | 336-0028    | O-Ring                                                                      |
| 6  | 310-7013    | Vacuum Breaker Upper Adapter                                                |
| 7  | 336-0026    | O-Ring                                                                      |
| 8  | 314-0036    | Screw                                                                       |
| 9  | 336-0035    | Washer                                                                      |
| 10 | 337-0023    | Poppet                                                                      |
| 11 | 324-0007    | Actuator Spring                                                             |
| 12 | 310-7015    | Slotted Spring Support Washer                                               |
| 13 | 346-0006    | Washer                                                                      |
| 14 | 310-7014    | Vacuum Breaker Lower Hose Piece                                             |
| 15 | 336-0031    | O-Ring                                                                      |
| 16 | 336-0007    | O-Ring                                                                      |
| 17 | 314-0021    | Mounting Bracket                                                            |
| 18 | 399-7002    | 1" Pipe Clamp Assembly for P-RH1, P-RH2, P-RH3                              |
| 19 | 399-7003    | 2 1/2" Pipe Clamp Assembly for P-RH4                                        |
| 19 | 314-7012    | Steel Struts                                                                |
| 20 | 314-0028    | 1/2" Mounting Bolt                                                          |
| 21 | 314-0029    | 1/2" Flat Washer                                                            |
| 22 | 314-0030    | 1/2" Mounting Nut                                                           |
| 23 | 314-0025    | 3/8" Lock Washer                                                            |
| 24 | 324-1003    | 3/8" Flat Washer                                                            |
| 25 | 314-0027    | 3/8" Hex Nut                                                                |
| 26 | 314-0035    | 3/8" Bolt                                                                   |
|    | P-360KT-806 | Vacuum Breaker Kit Includes 6, 7, 8, 9, 10, 11, 12, 13, 14                  |
|    | P-RH4KT-802 | O-Ring Kit Includes 2, 4, 5, 7, 13, 15, 16                                  |
|    | P-RMB       | Roof Mount Bracket Assembly Includes 18, 19, 20, 21, 22, 23, 24, 25, 26, 27 |
|    | P-231-1700  | Replacement Stem for P-RH4                                                  |

## SPECS

Furnish and install Model P-RH4 Sanitary Roof Hydrant manufactured by PRIER Products, Inc. It must be listed to ASSE 1057 and NSF 372 standards. This hydrant shall have freeze protection and a vacuum breaker/backflow preventer. Hydrant head shall be durable cast iron with freezable water flow and self-contained draining functions. Hydrant must be multi-piston design and completely evacuate drain canister upon use. The hydrant shall have a 1" NPT inlet and does not require winterization or a drain line. The roof penetration is to be sealed by the roofing contractor in accordance with the roofing codes and condition in their municipality. Hydrant shall be installed through a 3 5/8" single penetration in the roof and mounted to the interior structure with the Roof Hydrant Mounting Bracket (P-RMB) provided bracket must be manufactured of heavy gauge powder-coated steel and must allow for adjustment to meet roof angles up to 2/12 pitch. The mounting bracket be attached to concrete, a roofing truss, steel I-beam or interior wall with standard construction materials (threaded rods, Unistrut or other).

4515 E. 139th Street Grandview, MO 64030 | (800) 362-9055 | [PRIER.com](http://PRIER.com)

P-RH



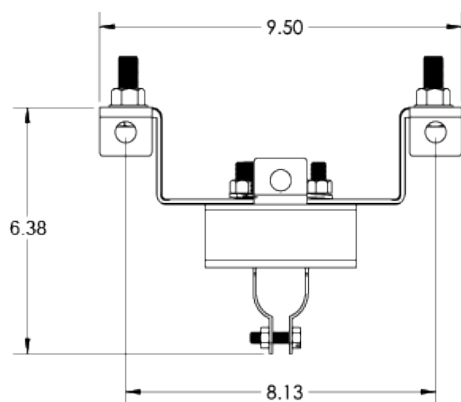
2693001



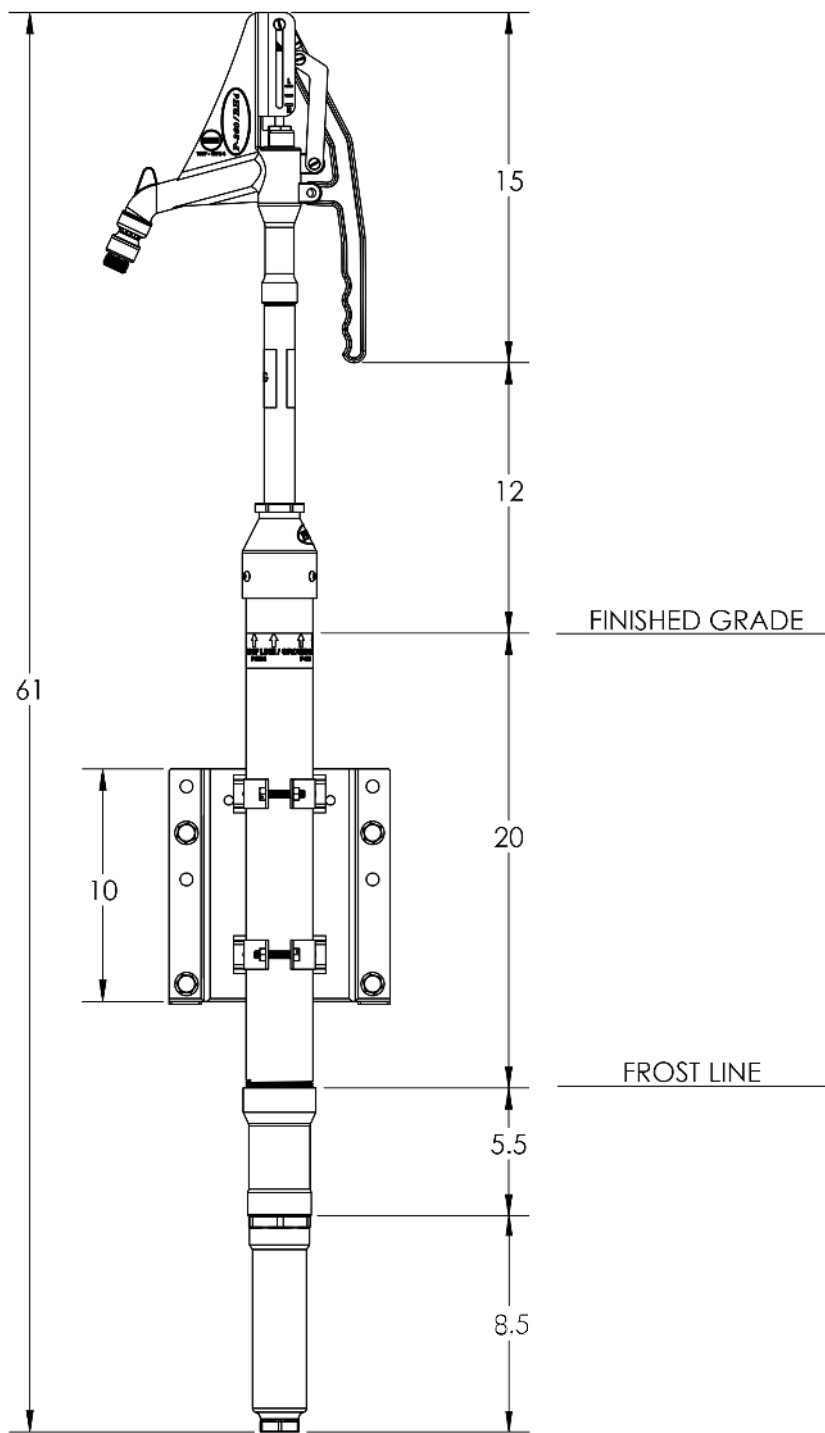
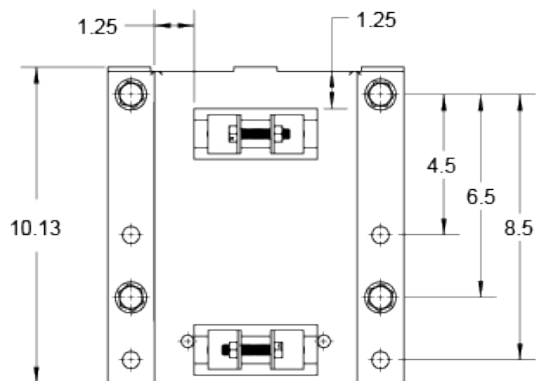
Build With Purpose

# DIMENSIONS

**Bottom Bracket View**



**Front Bracket View**



2693001



SK-1





# HAND SINKS

## WALL MOUNT HAND SINKS

MODEL: \_\_\_\_\_ PROJECT: \_\_\_\_\_ ITEM #: \_\_\_\_\_ QTY: \_\_\_\_\_

### PRODUCT IMAGES



### STANDARD FEATURES

- **Fabrication**  
20 gauge stainless steel. All seams tig welded and polished
- **Bowl**  
Deep drawn with stamped rim to prevent spillage
- **Wall Mounting Bracket**  
Offset design for added strength
- **Faucet**  
4" on center wall mount faucet included on most models
- **Drain**  
Stainless Steel
- **Drain with Overflow**  
Stainless Steel with plastic overflow tube and inlet
- **Plumbing**  
1/2" IPS hot and cold water. 1 1/2" IPS drain outlet. Install at 36" working height. 1/2" faucet supply 12" from floor. 1 1/2" drain line 23 1/4" from floor. (3 1/2" IPS drain on Model HS-20)
- **Low Lead Compliance**  
Low Lead Compliant faucet options are available to meet California AB-1953 and Vermont S152 standards

### OPTIONAL ACCESSORIES

- **H-100** Chrome Plated 1 1/2" IPS P-Trap
- **H-101** Deck Mount Soap Dispenser
- **H-102** Upgrade: Low Lead Wrist Handle Faucet
- **H-103** Wrist Handle Kit
- **H-104** Wall Mount Soap Dispenser
- **H-105** Wall Mount Towel Dispenser
- **H-106** One Side Splash (Specify Side)
- **H-107** Two Side Splashes
- **H-108** Stainless Steel Skirt
- **H-109** Upgrade: Low Lead Royal Series Faucet
- **H-110** Side Support Brackets
- **H-111** Soap & Towel Dispenser
- **H-200** Upgrade: Low Lead Commercial Series Faucet



### APPROVED BY:

Due to our commitment to continued product improvement, specifications are subject to change with

### Krowne Metal Corporation

100 Haul Rd. Wayne, NJ 07470 • Toll Free: (800) 631-0442 • Fax: (973) 872-1129

sales@krowne.com • www.krowne.com • www.facebook.com/KrowneMetal • www.twitter.com/KrowneMetal

KaTom Restaurant Supply, Inc.

Printed in the USA



18222786



# HAND SINKS

## WALL MOUNT HAND SINKS

MODEL: \_\_\_\_\_ PROJECT: \_\_\_\_\_ ITEM #: \_\_\_\_\_ QTY: \_\_\_\_\_

### WALL MOUNT HAND SINKS

|                 |                  |                  |
|-----------------|------------------|------------------|
| <b>HS-9</b><br> | <b>HS-2</b><br>  | <b>HS-4</b><br>  |
| <b>HS-5</b><br> | <b>HS-6</b><br>  | <b>HS-7</b><br>  |
| <b>HS-8</b><br> | <b>HS-10</b><br> | <b>HS-20</b><br> |



**CERTIFICATIONS:**

Printed in the USA

**APPROVED BY:**

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**Krowne Metal Corporation**

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KaTom Restaurant Supply, Inc.

Rev. 1/  
Nc



18222786



Model: **10-400L** Item #: \_\_\_\_\_ Date: \_\_\_\_\_  
 Project: \_\_\_\_\_ Qty: \_\_\_\_\_ Approved By: \_\_\_\_\_

## 4" Center Wall Mount Faucet w/ 3-1/2" Wide Gooseneck

**Silver Series**



### DESCRIPTION

- 4" Center Wall Mount Faucet with 3-1/2" Wide Gooseneck, Chrome plated finish, and 0.5 GPM aerator installed, 0.5 GPM

### STANDARD FEATURES

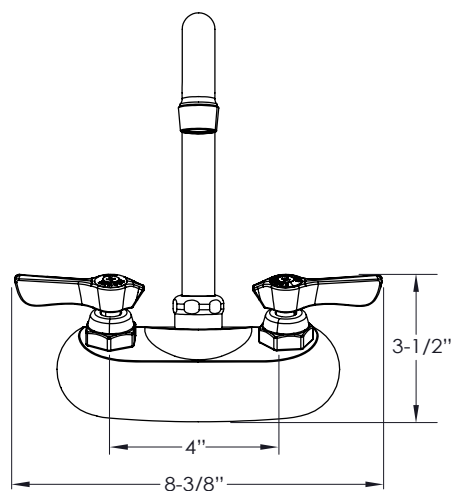
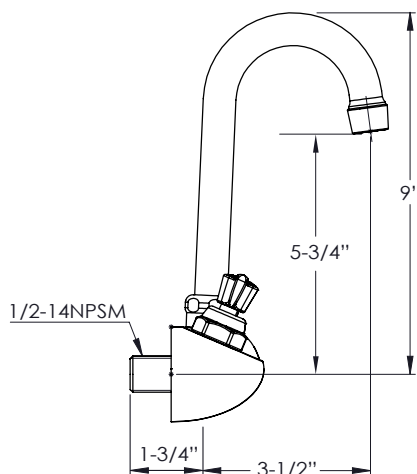
- 1/4 Turn ceramic valves
- 1/2" NPS male inlets on 4" centers
- Durable full range swing spout with double o-ring construction
- Chrome plated finish

### SPECIFICATIONS

|             |                               |
|-------------|-------------------------------|
| Water Inlet | 1/2" NPT                      |
| Temp. Range | 40°F - 180°F                  |
| Rough In    | 7/8" Round Holes on 4" Center |
| Flow Rate   | 0.5 GPM                       |
| Material    | Chrome Plated Brass           |
| Weight      | 2 lbs.                        |

### PRODUCT COMPLIANCE

- NSF-61 & NSF/ANSI 372
- ASME A112.18.1-2012/CSA BA125.1-12
- Massachusetts Approved
- CEC Approved



**WARNING:** Plumbing connections should be made by qualified personnel who will observe all applicable plumbing, sanitary and safety codes.

California Proposition 65

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Due to our commitment to continued product improvement, specifications are subject to change without notice.

Rev. 10/2020 • No. 10-400L



1225542

SK-2



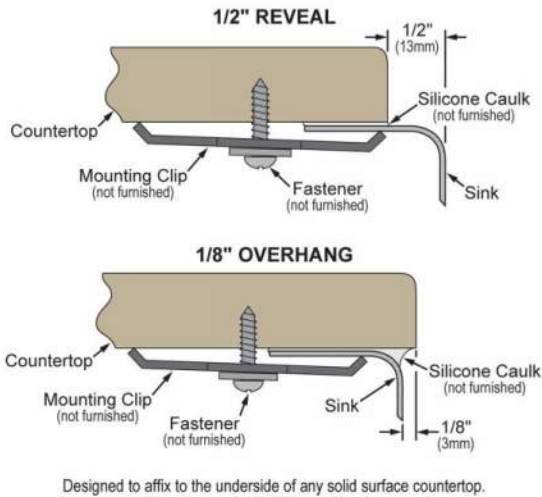
## PRODUCT SPECIFICATIONS

Lustertone® Classic Stainless Steel, 30-1/2" x 18-1/2" x 5-3/8" Single Bowl Undermount ADA Sink. Sink is manufactured from 18 gauge 304 Stainless Steel with a Lustrous Satin finish, Rear Center drain placement, and Bottom only pads.

|                              |                                     |
|------------------------------|-------------------------------------|
| <b>Installation Type:</b>    | Undermount                          |
| <b>Material:</b>             | 304 Stainless Steel                 |
| <b>Finish:</b>               | Lustrous Satin                      |
| <b>Gauge:</b>                | 18                                  |
| <b>Sound Deadening:</b>      | Bottom only pads                    |
| <b>Number of Bowls:</b>      | 1                                   |
| <b>Sink Dimensions:</b>      | 30-1/2" x 18-1/2" x 5-3/8"          |
| <b>Bowl 1 Dimensions:</b>    | 28" x 16" x 5-3/8"                  |
| <b>Drain Size:</b>           | 3-1/2" (89mm)                       |
| <b>Drain Location:</b>       | Rear Center                         |
| <b>Minimum Cabinet Size:</b> | 36"                                 |
| <b>Mounting Hardware:</b>    | Undermount brackets sold separately |
| <b>Template Included:</b>    | Yes                                 |
| <b>Cutout Template #:</b>    | <a href="#">1000001414</a>          |

Template is available for download at [elkay.com](#). CAD software will be required to open the template.

### Installation Profile:



A Century of Tradition and Quality. For more than 100 years, Elkay has been making innovative products and providing exceptional customer care. We take pride in offering plumbing products that make life easier, inspire change and leave the world a better place.



**Product Compliance:** ADA & ICC A117.1  
ASME A112.19.3/CSA B45.4



Sinks are listed by IAPMO® as meeting the applicable requirements of the Uniform Plumbing Code®, International Plumbing Code®, and National Plumbing Code of Canada.

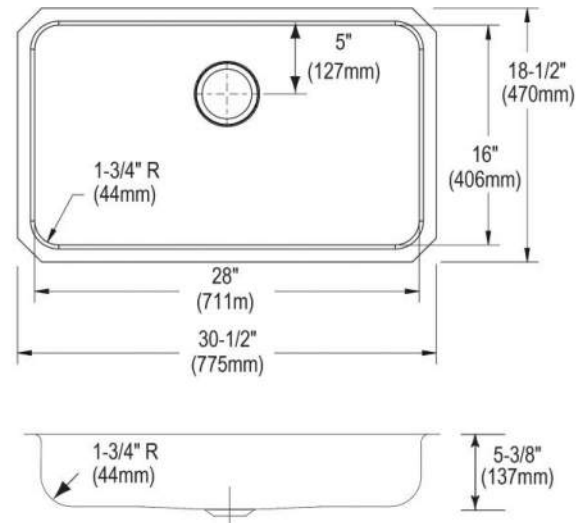


Complies with ADA & ICC A117.1 accessibility requirements when installed according to the requirements outlined in these standards.

[Clean and Care Manual \(PDF\)](#)

[Installation Instructions \(PDF\) - 1000005239](#)

[Warranty \(PDF\)](#)



PART: \_\_\_\_\_ QTY: \_\_\_\_\_  
PROJECT: \_\_\_\_\_  
CONTACT: \_\_\_\_\_  
DATE: \_\_\_\_\_  
NOTES: \_\_\_\_\_  
APPROVAL: \_\_\_\_\_

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970812

## PRODUCT SPECIFICATIONS

Elkay Avado Single Hole Bar Faucet with Pull-down Spray and Lever Handle. Faucet has a flow rate of 1.8 GPM, and is made of Brass material, with a Ceramic Disc valve. Faucet requires 1 faucet holes.

|                                |                                                                                |
|--------------------------------|--------------------------------------------------------------------------------|
| <b>Mounting Type:</b>          | Deck Mount                                                                     |
| <b>Special Features:</b>       | Solid Brass Construction                                                       |
| <b>Spray Type:</b>             | Pull Down                                                                      |
| <b>Finish:</b>                 | Black Stainless (BK), Chrome (CR),<br>Lustrous Steel (LS),<br>Matte Black (MB) |
| <b>Handle Type:</b>            | Lever Handle                                                                   |
| <b>Deck Clearance:</b>         | 7-1/2"                                                                         |
| <b>Spout Reach:</b>            | 7"                                                                             |
| <b>Spout Height:</b>           | 14-13/16"                                                                      |
| <b>Hole Drillings:</b>         | 1                                                                              |
| <b>Material:</b>               | Brass                                                                          |
| <b>Valve Type:</b>             | Ceramic Disc                                                                   |
| <b>Valve Connection:</b>       | 3/8" Female Compression Hose<br>Assembly                                       |
| <b>Flow Rate:</b>              | 1.8 GPM                                                                        |
| <b>Faucet Hole Size (min):</b> | 1-1/2"                                                                         |
| <b>Countertop Thickness:</b>   | 3                                                                              |
| <b>Spout Swing Rotation:</b>   | 360°                                                                           |
| <b>Spout Type:</b>             | Pull-down Spray                                                                |
| <b>Spray Functions:</b>        | Aerated, Nylon Hand Spray<br>Hose, Spray                                       |



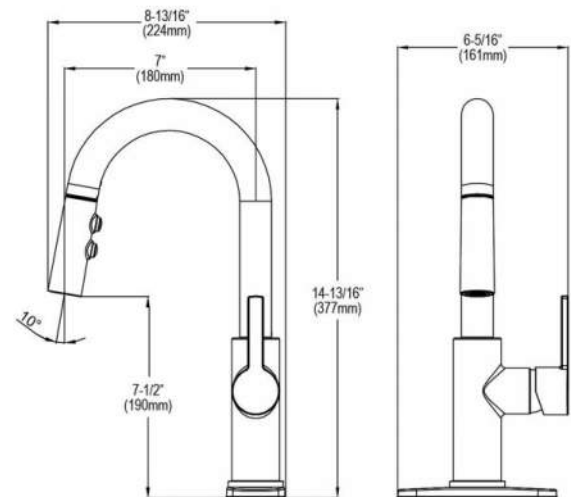
**Included with Product:**    **Optional Deck Plate/Escutcheon included**

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**Product Compliance:**    ADA & ICC A117.1  
ASME A112.18.1/CSA B125.1  
CEC  
NSF 372 (lead free)  
NSF 61



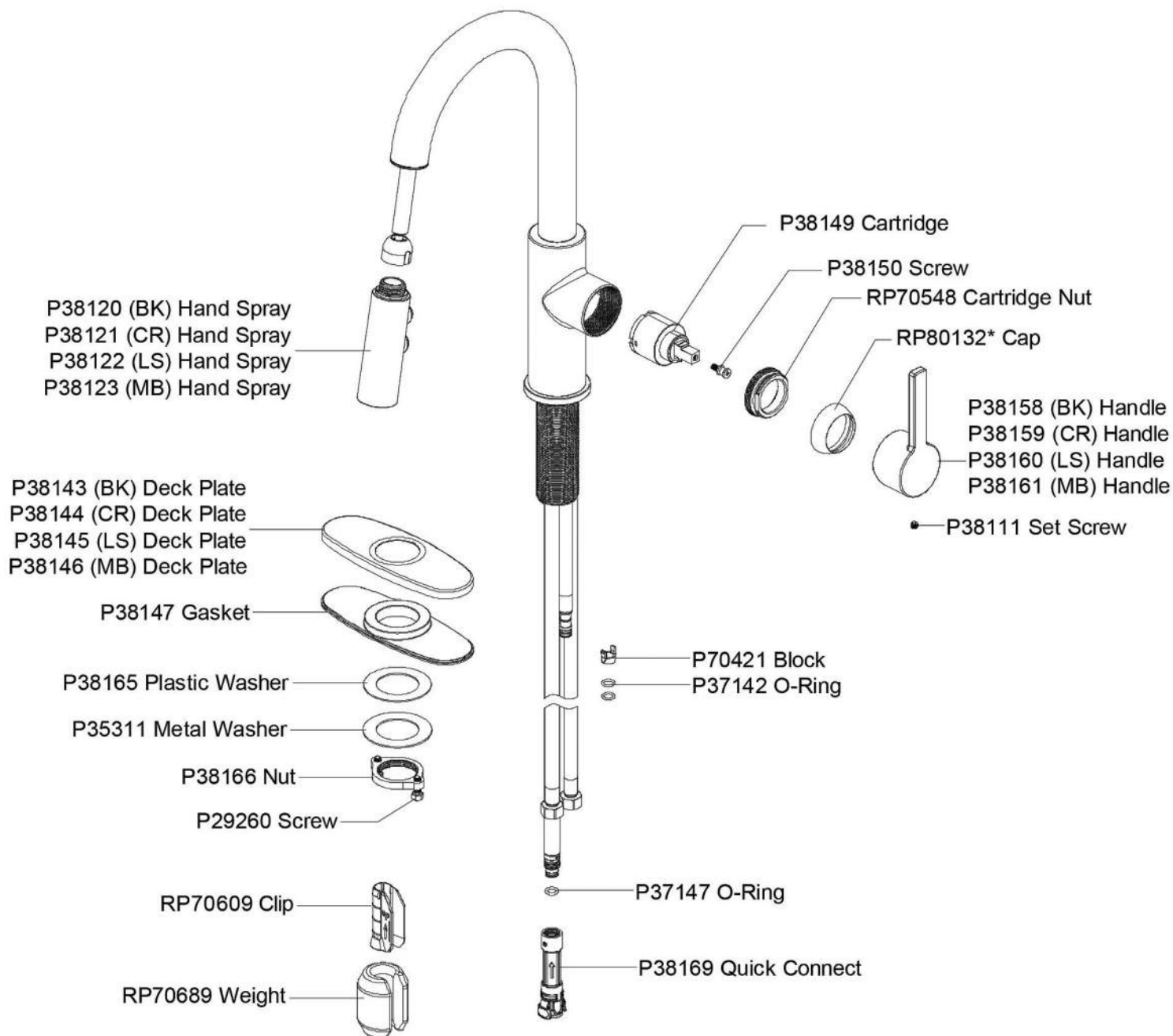
[Clean and Care Manual \(PDF\)](#)  
[Installation Instructions \(PDF\) - LKAV3032 - Install Sheet](#)  
[Warranty \(PDF\)](#)



PART: \_\_\_\_\_ QTY: \_\_\_\_\_  
PROJECT: \_\_\_\_\_  
CONTACT: \_\_\_\_\_  
DATE: \_\_\_\_\_  
NOTES: \_\_\_\_\_  
APPROVAL: \_\_\_\_\_

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2715468

# Food Waste Disposers



## Submittal Sheet

This popular model offers the following features and benefits:

- 1/2 Horsepower Heavy Duty Motor (Quiet Dura-Drive® Induction Motor)
- We Come To You® 2-Year In-Home Limited Warranty
- Rugged Galvanized Steel Construction (For Disposer Durability)
- Space-Saving Compact Design
- Available with or without a factory installed power cord.

## Sample Specification

Food Waste Disposer(s) shall be InSinkErator Badger® 5, continuous feed, with 1/2 H.P. motor, galvanized steel grinding elements with two stainless steel 360° swivel lugs. Self-service wrench.

Available with or without a factory installed power cord.

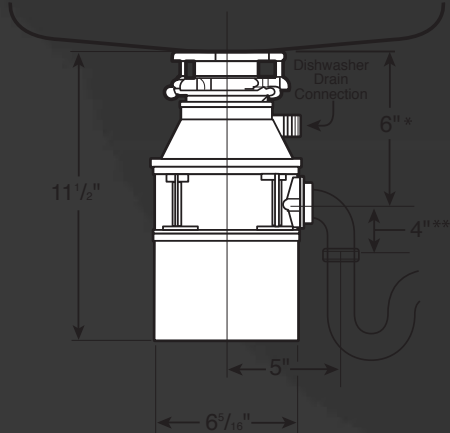
Warranty:  
2-Year parts and in-home service.

\* The complete InSinkErator warranty is included in the Care & Use Booklet packed with each unit.

## Specifications

|                              |                                                             |
|------------------------------|-------------------------------------------------------------|
| Type of Feed                 | Continuous                                                  |
| On/Off Control               | Wall Switch                                                 |
| Motor                        | Single Phase                                                |
| HP                           | 1/2                                                         |
| Volts                        | 120                                                         |
| HZ                           | 60                                                          |
| RPM                          | 1725                                                        |
| Amp. (Avg. Load)             | 6.3                                                         |
| Time Rating                  | Intermittent                                                |
| Lubrication                  | Permanently Lubricated Upper & Lower Bearings               |
| Shipping Weight (Approx.)    | 13 lbs. 10 oz. (without cord)<br>13 lbs. 14 oz. (with cord) |
| Unit Finish                  | Condor Grey                                                 |
| Overall Height               | 11-1/2"                                                     |
| Grind Chamber Capacity       | 26 oz.                                                      |
| Motor Protection             | Manual Reset Overload                                       |
| Average Water Usage          | Approx. 1 Gallon per Person Per Day                         |
| Average Electrical Usage     | 3-4 KWh per Year                                            |
| Drain Connection             | 1-1/2" Cushioned Slip Joint                                 |
| Dishwasher Drain Connection  | Yes                                                         |
| Factory Installed Power Cord | Available with or without                                   |

## Dimensions



\* Distance from bottom of sink to center line of disposer outlet. Add 1/2" when stainless steel sinks are used.  
\*\* Length of tailpipe from center line of disposer outlet to end of tailpipe.  
NOTE: Plumb waste line to prevent standing water in disposer motor housing.

## Job Specifications



Food waste disposers can provide an environmentally responsible alternative to transporting food waste to landfills. And they can help reduce greenhouse gas emissions. At capable wastewater treatment plants, food waste can be recycled to produce renewable energy. Additionally, capable wastewater treatment plants can process food waste into fertilizer. (Check the plant in your area.)



The Emerson logo is a trademark and service mark of Emerson Electric Co. The mounting collar configuration is a trademark of Emerson Electric Co.

1-800-558-5700  
www.insinkerator.com

InSinkErator may make improvements and/or change in the specifications at any time, in its sole discretion without notice or obligation and further reserves the right to change or discontinue models.



SK-3



### PRODUCT SPECIFICATIONS

Elkay Sturdibilt Stainless Steel 102" x 27-1/2" x 14" Floor Mount, Triple Compartment Scullery Sink with Drainboard. Sink is manufactured from 14 gauge 304 Stainless Steel with a Buffed Satin finish, Center drain placement. Sinks supported by (4) 16 gauge stainless steel, 1-5/8" O.D. tubular legs with bullet shaped feet adjustable up to 1".

|                             |                      |
|-----------------------------|----------------------|
| <b>Material:</b>            | 304 Stainless Steel  |
| <b>Special Features:</b>    | Drainboard           |
| <b>Finish:</b>              | Buffed Satin         |
| <b>Gauge:</b>               | 14                   |
| <b>Number of Bowls:</b>     | 3                    |
| <b>Sink Dimensions:</b>     | 102" x 27-1/2" x 44" |
| <b>Bowl 1 Dimensions:</b>   | 18" x 24" x 14"      |
| <b>Bowl 2 Dimensions:</b>   | 18" x 24" x 14"      |
| <b>Bowl 3 Dimensions:</b>   | 18" x 24" x 14"      |
| <b>Drainboard Location:</b> | Left and Right       |
| <b>Drainboard Size:</b>     | 24                   |
| <b>Backsplash Height:</b>   | 8                    |
| <b>Leg Type:</b>            | Stainless Steel      |
| <b>Drain Size:</b>          | 3-1/2" (89mm)        |
| <b>Drain Location:</b>      | Center               |

#### Custom Options

☐ Type 316 Stainless Steel

#### Drain Location

- ☐ Right Rear ☐ Left Rear  
☐ Center ☐ Center Rear

#### Overflow Location

- ☐ Front ☐ Left  
☐ Rear ☐ Right

#### Alternate Punching

- ☐ Faucet Model: ☐ Punch Required:

#### Sink Size

- ☐ Bowl Width: ☐ Drainboard Width:  
☐ Bowl Depth: ☐ Backsplash Height:

\*Options available if applicable

\*\*Drainboards over 25-1/2" wide furnished with one tubular leg.

\*\*\*Drainboards over 48" wide furnished with two tubular legs.



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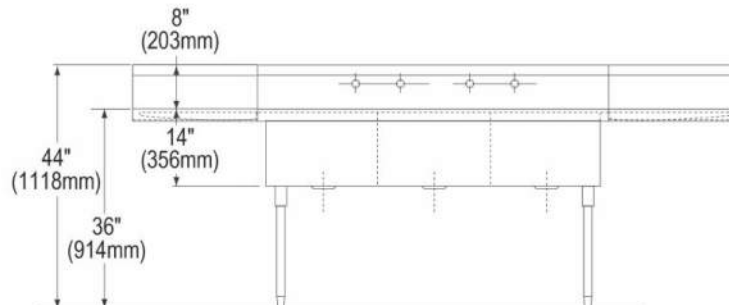
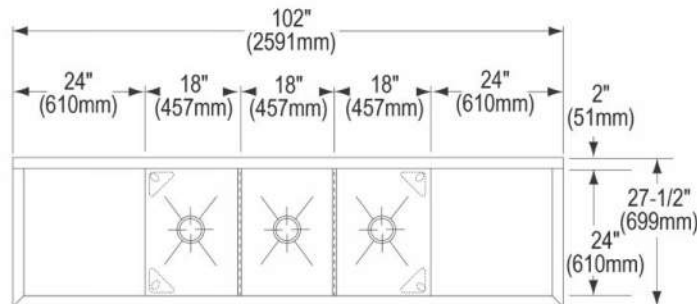


**Product Compliance:** ASME A112.19.3/CSA B45.4  
Buy American Act



Sinks are listed by IAPMO® as meeting the applicable requirements of the Uniform Plumbing Code®, International Plumbing Code®, and National Plumbing Code of Canada.

[Clean and Care Manual \(PDF\)](#)  
[Warranty \(PDF\)](#)



PART: \_\_\_\_\_ QTY: \_\_\_\_\_

PROJECT: \_\_\_\_\_

CONTACT: \_\_\_\_\_

DATE: \_\_\_\_\_

NOTES: \_\_\_\_\_

APPROVAL: \_\_\_\_\_

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Model: **17-109WL** Item #: \_\_\_\_\_ Date: \_\_\_\_\_  
Project: \_\_\_\_\_ Qty: \_\_\_\_\_ Approved By: \_\_\_\_\_

**8" Center Wall Mount Pre-Rinse with Add-On Faucet (12" Spout)    Royal Series Plumbing**



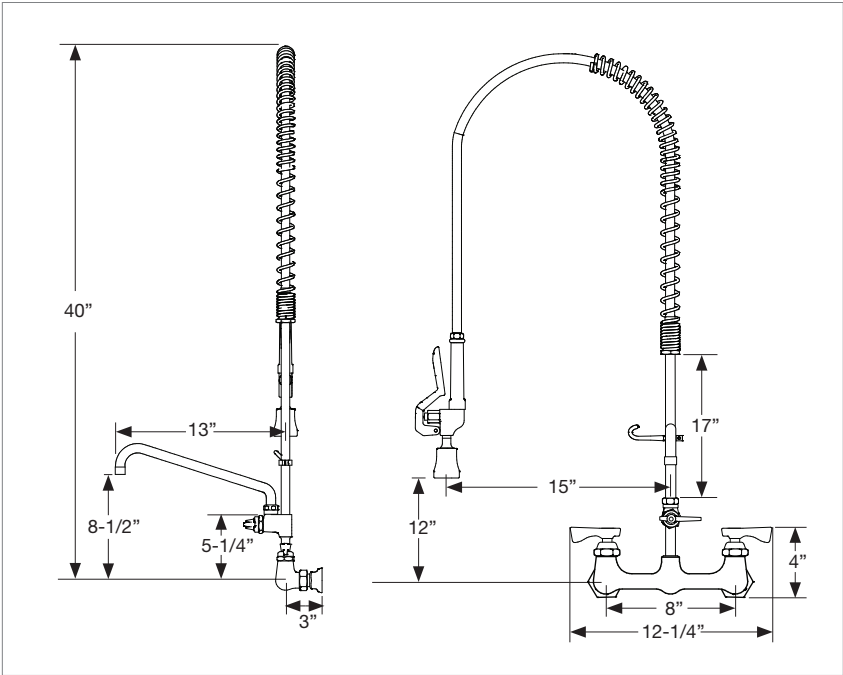
**Standard Features**

- Pre-assembled to cut installation time
- Includes add-on faucet with 12" spout
- Includes wall bracket
- 1/4 turn high performance ceramic cartridge valves with integral check
- 44" stainless steel hose with grip
- 38"H with 15" overhang
- 3/8" NPT x 18"L riser pipe with hook
- Mounting kit included
- Full replacement parts available

**Specifications**

|                      |                      |
|----------------------|----------------------|
| Water Inlet          | 1/2" NPT Female      |
| Temp. Range          | 40° - 180°F          |
| Centers              | 8" Center            |
| Mounting Style       | Wall Mount           |
| Rough In             | Two 7/8" Round Holes |
| Add-On Faucet        | 12" Spout, 1.8 GPM   |
| Spray Head Flow Rate | 1.15 GPM @ 60 PSI    |
| Material             | Solid Brass          |
| Finish               | Polished Chrome      |
| Weight               | 15 lbs.              |

| Model    | Description                                                               |
|----------|---------------------------------------------------------------------------|
| 17-109WL | Royal Series 8" Center Wall Mount Pre-Rinse with Add-On Faucet, 12" Spout |



**Product Compliance**

- NSF-61 & NSF/ANSI 372
- ASME A112.18.1-2011/CSA B125.1-11 & NSF/ANSI 372
- City of Los Angeles
- CEC Listed
- Commonwealth of Massachusetts



**WARNING:**  
Plumbing connections should be made by qualified personnel who will observe all applicable plumbing, sanitary and safety codes.

**California Prop 65 Warning:**  
This product can expose you to chemicals including lead and lead compounds, which are known to the State of California to cause cancer and birth defects or other reproductive harm. For more information go to [www.P65Warnings.ca.gov](http://www.P65Warnings.ca.gov).



1130255



Model: **17-109WL** Item #: \_\_\_\_\_ Date: \_\_\_\_\_  
Project: \_\_\_\_\_ Qty: \_\_\_\_\_ Approved By: \_\_\_\_\_

**8" Center Wall Mount Pre-Rinse with Add-On Faucet (12" Spout)    Royal Series Plumbing**



1130255

**Replacement Parts**

| No. | Model   | Description                     |
|-----|---------|---------------------------------|
| 1   | 21-162  | 24" Spring                      |
| 2   | 21-133L | 44" Hose with Grip              |
| 3   | 21-129L | 1.15 GPM Spray Head             |
| 4   | 21-165  | Hook Assembly                   |
| 5   | 21-161L | Spring Retainer                 |
| 6   | 21-137  | Wall Bracket                    |
| 7   | 21-160L | 18" Riser                       |
| 8   | 21-139L | Add-On Faucet with 12" Spout    |
| 9   | 21-108L | 8" Center Pre-Rinse Faucet Body |



**WARNING:**  
Plumbing connections should be made by qualified personnel who will observe all applicable plumbing, sanitary and safety codes.

**California Prop 65 Warning:**  
This product can expose you to chemicals including lead and lead compounds, which are known to the State of California to cause cancer and birth defects or other reproductive harm. For more information go to [www.P65Warnings.ca.gov](http://www.P65Warnings.ca.gov).



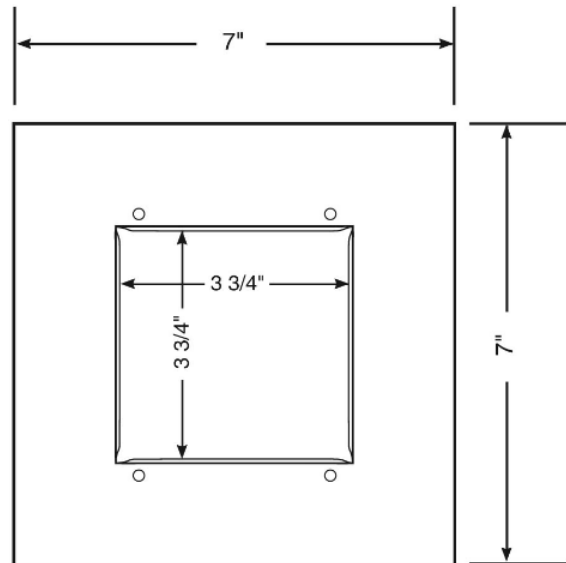
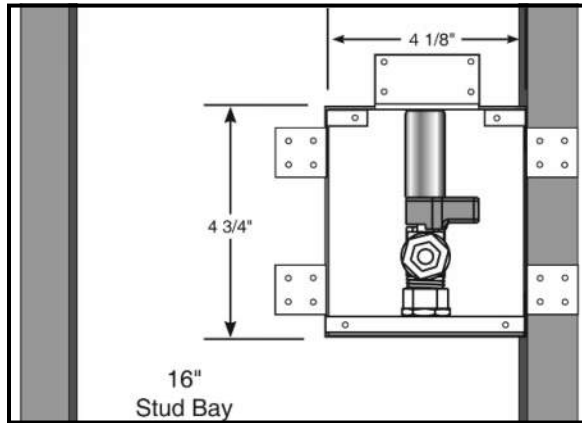
UB-1





## White Powder Coated Ice Maker Outlet Box

With Lead Free Hammer Arrestor Valves



1245652

### SPECIFICATION

Furnish and install recessed, white powder coated ice maker outlet box. Ice maker outlet box shall have a brass plated, quarter turn hammer arrestor valve meeting Lead-Free requirements. Valve shall feature either Mylar® tag and AB1953 stamping or white handle and check-mark logo for easy Lead-Free identification. Unit shall be Guy Gray™ product code checked below as manufactured by IPS Corporation.

| Product Code # | Model Number | Product Description                                               | Units/Case |
|----------------|--------------|-------------------------------------------------------------------|------------|
| 88531          | MIB1HAAB     | Brass Qtr Turn Hammer Arrestor Valve, Installed, 1/2" Sweat Conx. | 5          |
| 88533          | MIB2HAAB     | Brass Qtr Turn Hammer Arrestor Valve, Installed, 1/2" CPVC Conx.  | 5          |

**IPS**  
CORPORATION

www.ipscorp.com watertite@ipscorp.com

500 Distribution Parkway, Collierville, TN 38017, USA ■ TEL: 901-853-5001 ■ WATS: 800-888-8312 ■ FAX: 901-853-5001



WC-1





# Kingston™ Ultra

Wall-mount toilet bowl with top spud  
K-84325

## Features

- Elongated bowl offers added room and comfort
- Siphon jet
- 1-1/2" (38 mm) top spud
- 1.0 gpf (3.8 lpf), 1.1 gpf (4.2 lpf), 1.28 (4.8 lpf), or 1.6 gpf (6.0 lpf) depending on flushometer specified
- 11" x 8-1/4" (279 mm x 210 mm) water surface
- Wall-mounted vitreous china water closet bowls withstand a vertical force up to 1,000 lbs

## Material

- Vitreous china

## Technology

- Improved flush for toilet seat cover removal
- Re-engineered for strong flush performance in buildings with lower water pressure or flow
- Cast-on putty plate resists breakage
- New design eliminates gas leaks, splashing and paper shredding

## Installation

- Wall-mount
- This product replaces K-4325
- This product is not designed or intended for bariatric use



ADA

CSA B651

OBC

## Codes/Standards

ASME A112.19.2/CSA B45.1  
DOE - Energy Policy Act 1992  
EPA WaterSense®  
California Energy Commission (CEC)  
ADA  
ICC/ANSI A117.1  
CSA B651  
OBC

## KOHLER® One-Year Limited Warranty

See website for detailed warranty information.

## Available Colors/Finishes

*Color tiles intended for reference only.*

| Color                                                                               | Code | Description  |
|-------------------------------------------------------------------------------------|------|--------------|
|  | 0    | White        |
|  | 96   | Biscuit      |
|  | 7    | Black Black™ |

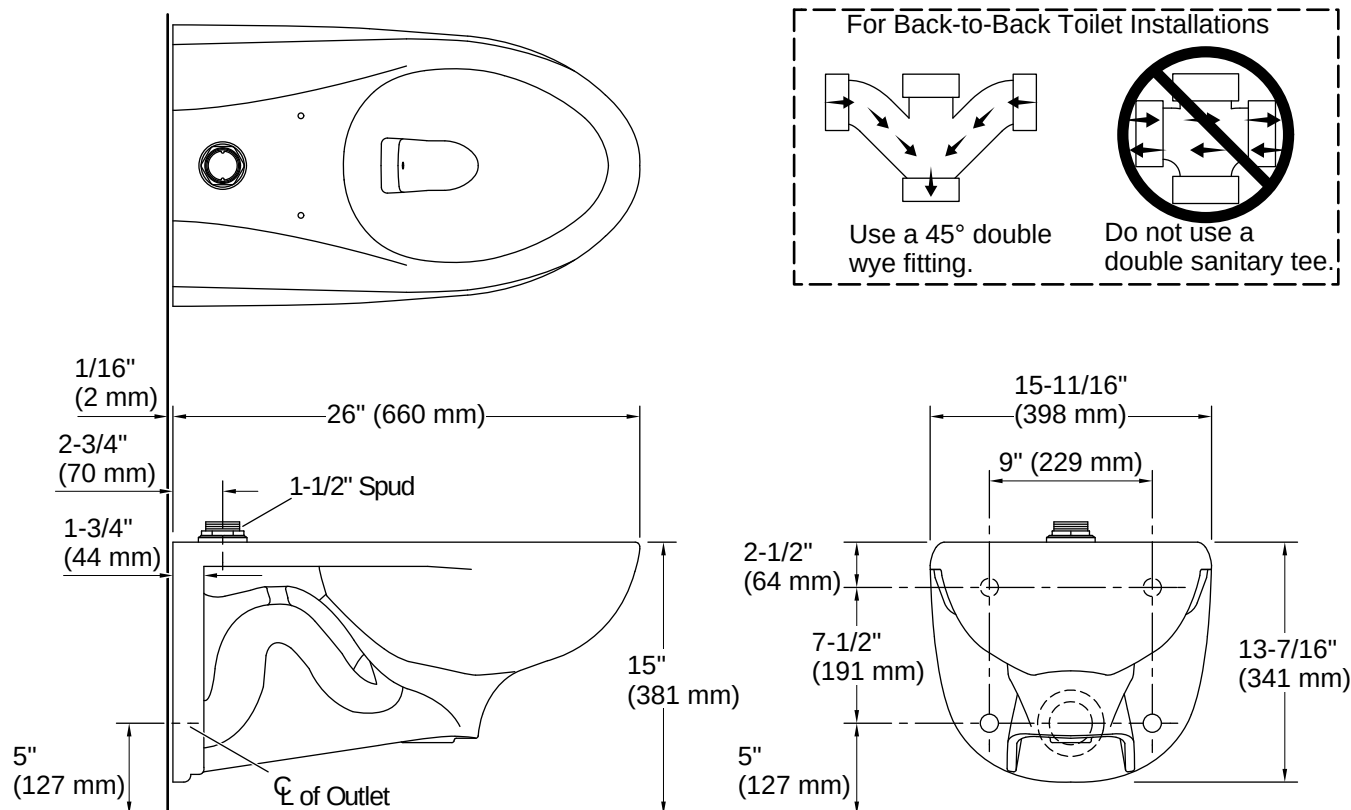
2002188



**KOHLER®**

# Kingston™ Ultra

Wall-mount toilet bowl with top spud  
K-84325



## Technical Information

All product dimensions are nominal.

|                       |                                |
|-----------------------|--------------------------------|
| Toilet type:          | Wall-hung                      |
| Waste Outlet:         | Wall                           |
| Bowl shape:           | Elongated                      |
| Flush type:           | Siphon                         |
| Spud size:            | 1 1/2", Inlet, Top             |
| Trap passageway:      | 2-1/8" (54 mm)                 |
| Water surface size:   | 11" x 8-1/4" (279 mm x 210 mm) |
| Rim to water surface: | 5-13/16" (148 mm)              |
| Seat-mounting holes:  | 5-1/2" (140 mm)                |

## Fixture Supply Requirements

|                       |                    |
|-----------------------|--------------------|
| Min static pressure:  | 35 psi (241.3 kPa) |
| Max static pressure:  | 80 psi (551.6 kPa) |
| Min flowing pressure: | 25 psi (172.4 kPa) |
| Min flow rate:        | 25 gpm (94.6 lpm)  |

## Notes

Install this product according to the installation instructions.

For back-to-back toilet installations: Use only a 45° double wye fitting.

Refer to manufacturer's instructions and local codes for flushometer requirements.

Ensure valve is braced properly.

Ensure pressure and flow are correct.

Product dimensions are nominal and may vary per ASME A112.19.2 tolerances.

This product should be used with a WaterSense labeled counterpart with the same rated flush volume to ensure the complete system meets the requirements of this specification for water efficiency and performance.

ADA, OBC, CSA B651 compliant when installed to the specific requirements of these regulations.

Plumbing codes may require elongated toilets

1-800-4KOHLER (1-800-456-4537)

Kohler Co. reserves the right to make revisions without notice to product specifications.

For the most current Specification Sheet, go to [www.kohler.com](http://www.kohler.com) USA or [www.kohler.ca](http://www.kohler.ca) Canada

2-11-2026 20:46 - US/CA

THE BOLD LOOK  
OF **KOHLER®**



2002188



ZTR6200

Battery Powered Exposed Sensor Flush Valve  
for Water Closets

TAG \_\_\_\_\_

Architectural/Engineering Specification:

Exposed, quiet piston-type, chrome plated flushometer valve with a polished exterior. Complete with chloramine resistant, filtered piston kit. The valve incorporates a battery powered solenoid actuator, automatic sensor with manual override push button, and robust vandal resistant metal cover with 10 degree angled sensor.

Product Features:

- Proprietary Dezincification resistant low lead brass alloy
- Control Stop has internal siphon-guard protection and vandal resistant stop cap, Sweat solder kit, wall flange with set screw.
- High back pressure Vacuum Breaker with one piece hex coupling nut
- Chloramine resistant Internal seals
- Adjustable tailpiece
- Spud coupling and flange for top spud connection

Compliance and Certification:

- ADA Compliant - Operational Controls



\*This product should be used with a WaterSense labeled counterpart with a compatible flush volume to ensure that the entire system meets the requirements for water efficiency and performance.

WHEN INSTALLING IN A HANDICAP STALL:

Per the ADA Guidelines (section 604.9.4) it is recommended that the grab bars be split or shifted to the wide side of the stall. If grab bars must be present over the valve, use the alternate ADA installation as shown on the next page.



Flow Options:

|      | Flush Volume<br>gpf [Lpf] | Dual Flush                   | WaterSense<br>Labeled* |
|------|---------------------------|------------------------------|------------------------|
| -ONE | 1.1 gpf [4.1 Lpf]         |                              | ✓                      |
| -EV  | 1.28 gpf [4.8 Lpf]        |                              | ✓                      |
| -WS1 | 1.6 gpf [6.1 Lpf]         | 1.6/1.1 Gpf<br>[6.1/4.1 Lpf] |                        |

Note: Dual Flush feature has 60 second threshold

Suffix Options:

- DF Dual Flush Option (See Chart Above)
- FM12 Floor Mount 12" Rough In
- LL 10 Year Long Life Battery
- YJ Split Ring Pipe Support
- YK Solid Ring Pipe Support

Valve Height Options:

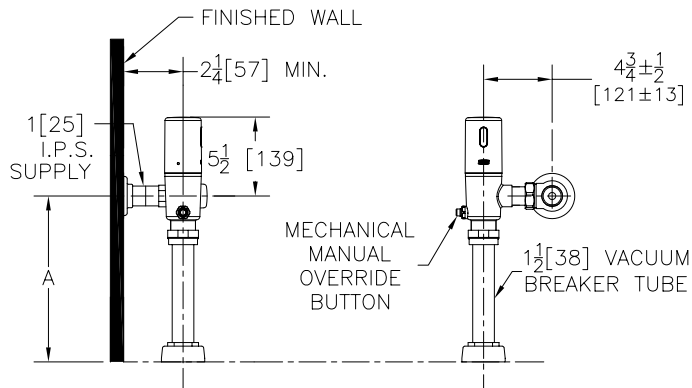
|          | A             |
|----------|---------------|
| Standard | 11 1/2" [292] |
| -1       | 16" [406]     |

1208094

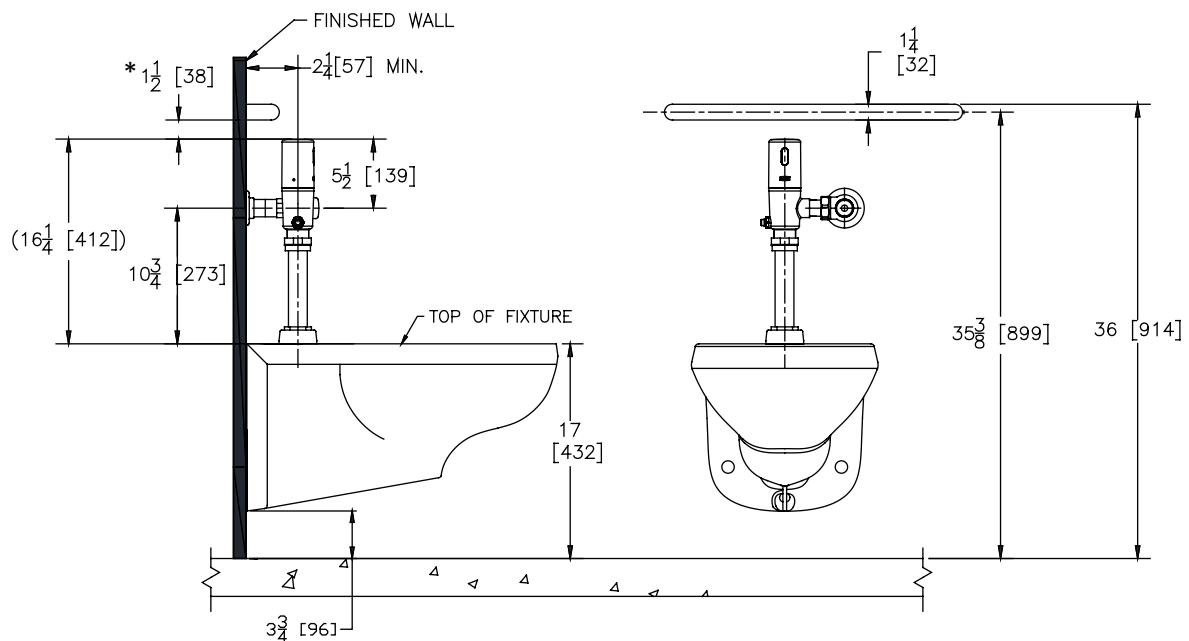
Architectural/Engineering Approval



### Rough-in/Overview dimensions



### Alternate ADA Installation

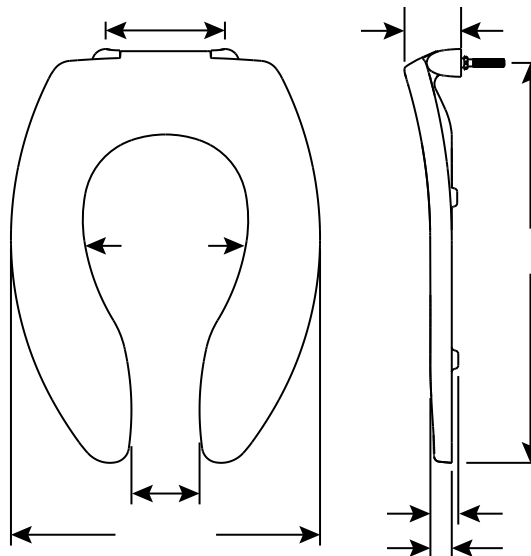


- Water supply rough-in to 10 3/4" (minimum) with grab bar at 36".
- ADA compliance met using 1 1/4" grab bar as shown.
- \*ADA compliant minimum gap between wall and grab bar 1 1/2".
- Does not meet California ADA compliance for grab bar height (33" grab bar centerline required)



# 1655CT / 1655SSCT

COMMERCIAL EXTRA HEAVY-DUTY  
PLASTIC TOILET SEAT



793988

## FEATURES:

- **STA-TITE®**  
**Commercial Fastening System™**  
Eliminates callbacks for loosened seats

## SPECIFICATIONS:

**Size:** Elongated

**Material:** Plastic

**Style:** Open Front less Cover

**Ring Bumpers:** Four

**Hinges:** Plastic Non Self-Sustaining Check Hinges (1655CT) or Self-Sustaining Check Hinges (1655SSCT) with Non-Corroding 300 Series Stainless Steel Posts and Pintles

**Hardware:** STA-TITE® Commercial Fastening System™

**Codes & Standards:**



Complies with  
IAPMO/ANSI Z124.5  
Performance Standards  
Heavy-Duty Commercial



Proudly Made in the USA

Eco-Friendly



## WC-1 CARRIER OPTIONS



G

DRAWING NUMBER

s0211Y, 0221Y (-M54)

SIZE

A

SCALE:

NONE

DATE:

4-3-12

APPROVED BY:

JM

CHECKED BY:

AM

DRAWN BY:

CR

FIGURE NUMBER

0211Y (-M54)  
0221Y (-M54)

SMITH® JAY R. SMITH MFG. CO.®

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MONTGOMERY, ALABAMA 36109-0237 (USA)  
TEL: 334-277-8520 FAX: 888-377-7818 www.jrsmith.com

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INTERNATIONAL

LOCATION

# ADJUSTABLE FIXTURE SUPPORTS FOR SIPHON JET WATER CLOSETS

**NOTES:**

- 1. Base Support must be securely anchored to the floor (with one 1/2"(13) bolt per base support provided by installer.)
- 2. All single supports shall have the rear support lug anchored by contractor to assure a rigid installation. The rear support completes the 3 point anchoring system for ultimate stability of the installation.
- 3. Dimensions shown in parentheses are in millimeters.
- 4. \*Dimension to top of vent.

| A<br>SIZE | D        |           |
|-----------|----------|-----------|
|           | MIN      | MAX       |
| 4(100)    | 3/4 (19) | 3 3/8(87) |

⊙ Dimension = 4"(100) (Regularly Furnished) consisting of nipple and Adjus-to-Wall coupling permitting adjustment from 3"(76) min. to 4"(100) max. Longer nipples are available, specify ⊙ dimension.

**NOTE:** Cast Iron Components are Duco Coated.

|                          |                          |                               |
|--------------------------|--------------------------|-------------------------------|
| <input type="checkbox"/> | Fig. 0211LY (-M54) ..... | NO-HUB Left Hand              |
| <input type="checkbox"/> | Fig. 0211RY (-M54) ..... | NO-HUB Right Hand             |
| <input type="checkbox"/> | Fig. 0211DY (-M54) ..... | NO-HUB Double                 |
| <input type="checkbox"/> | Fig. 0221LY (-M54) ..... | NO-HUB Left Hand Long Barrel  |
| <input type="checkbox"/> | Fig. 0221RY (-M54) ..... | NO-HUB Right Hand Long Barrel |
| <input type="checkbox"/> | Fig. 0221DY (-M54) ..... | NO-HUB Double Long Barrel     |

Conforms with the requirements of ASME A112.6.1M up to a 500 lbs (227kg) static load rating.

| REV. | DATE | DESCRIPTION | BY | CKD. BY | WEIGHT<br>POUNDS | VOLUME<br>CUBIC FEET | FIGURE NUMBER                              |
|------|------|-------------|----|---------|------------------|----------------------|--------------------------------------------|
|      |      |             |    |         |                  |                      | <b>0211Y (-M54)</b><br><b>0221Y (-M54)</b> |

**WARNING:** Cancer and Reproductive Harm - www.P65Warnings.ca.gov

1886448

C

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MONTGOMERY, ALABAMA 36109-0237 (USA)  
TEL: 334-277-8520 FAX: 888-377-7818 www.jrsmith.com



LOCATION

DRAWING NUMBER  
S0200BS

SIZE  
A

SCALE:  
NONE

DATE:  
2-28-86

APPROVED BY:  
JD

CHECKED BY:  
CR

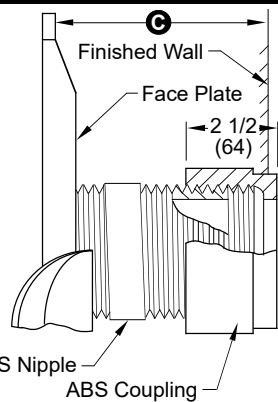
DRAWN BY:  
VGD

# MODIFICATIONS & VARIATIONS

## Ⓢ DIMENSION - ADJUS-TO-WALL TYPE OUTLET CONNECTION

The Smith Adjus-to-Wall outlet connection provides adjustment at the face of the wall at the time the fixture is set; thus, compensating for wall construction variations. The Ⓢ dimension is measured from the front of the face plate to the finished wall. To properly determine the length of nipple required, deduct 1" (25) from the Ⓢ dimension. When the Ⓢ dimension exceeds 7" (180), the -M40 Wide Pipe Chase Support is recommended. The Adjus-to-Wall coupling will provide 1/2" (13) horizontal adjustment in either direction with the coupling extended 5/16" (8) past face of the finished wall.

Ⓢ dimension = 4" (100) (regularly furnished) consisting of nipple and adjus-to-wall coupling permitting adjustment from 3" (76) min. to 4" (100) max.  
\*ABS (Regularly Furnished) other length PVC.



### NIPPLE LENGTHS AVAILABLE

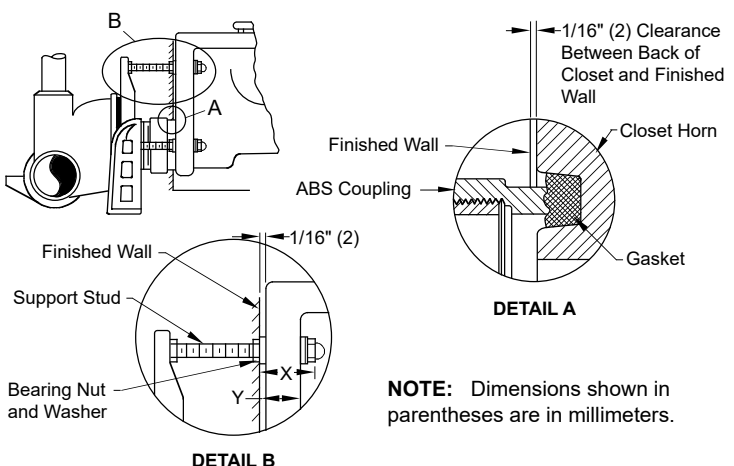
| LENGTH       | MIN          | Ⓢ MAX        |
|--------------|--------------|--------------|
| 2 1/2 (64)   | 2 1/4 (57)   | 3 1/4 (83)   |
| *3 1/4 (83)  | 3 (76)       | 4 (100)      |
| *8 1/4 (210) | 8 (205)      | 9 (230)      |
| 10 (255)     | 9 3/4 (250)  | 10 3/4 (275) |
| 12 (305)     | 11 3/4 (300) | 12 3/4 (325) |
| 15 (380)     | 14 3/4 (376) | 15 3/4 (400) |
| 18 (455)     | 17 3/4 (450) | 18 3/4 (475) |

NOTE: When nipple lengths between 3 1/4" (83) and 8 1/4" (210) are specified, the 8 1/4" (210) nipple will be supplied. Nipple must be cut to specified length by others.

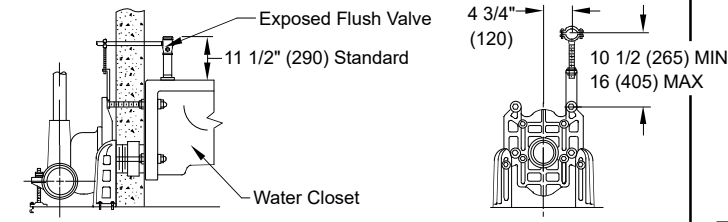
## RECOMMENDED SETTING OF CLOSET OUTLET CONNECTION AND FIXTURE STUDS

Manufacturers have slight differences in the depth of the closet horn and flange thickness of their closets. Usually the fixture support closet connection should extend 5/16" (8) beyond face of finished wall (Dimension "R") and the fixture studs 2 1/4" (57) (Dimension "X"). For exact dimensions please use the formulas below.

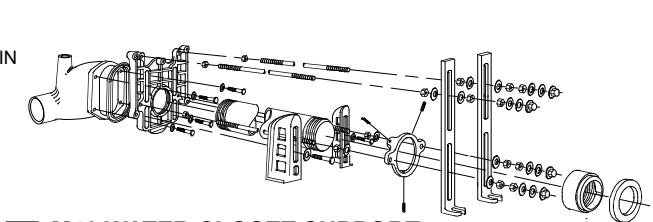
**DETAIL A** Formula for "R" is:  $S + 1/16" (2) - D = R$ .  
S = Depth of closet horn  
1/16" (2) = Distance closet is to set away from finished wall.  
D = 1/2" (13) for felt gasket and 3/8" (10) for neoprene gasket.  
R = Distance coupling should extend beyond finished wall.  
NOTE: When the fixture is installed, closet gasket must be compressed sufficiently to assure a gas and water-tight seal.



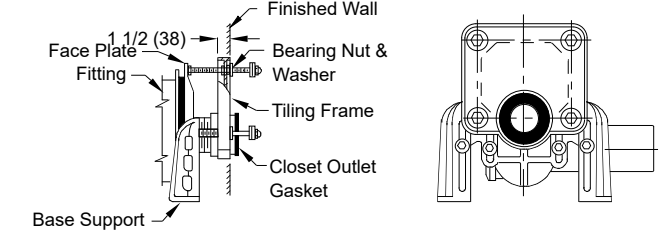
**DETAIL B** Formula for "X" is:  $Y + 1/16" (2) + 9/16" (14) = X$ .  
Y = Thickness of closet wall flange.  
1/16" (2) = Distance closet is to set away from finished wall.  
X = Distance supporting stud should extend beyond finished wall.  
NOTE: Bearing nuts and washers must be set to take full loading from the fixture allowing 1/16" (2) clearance between fixture and wall.



Provides support and alignment for the supply pipe to an exposed flush valve; must be specified for either 1" (25) copper tube or 1" (25) iron pipe.  
☐ SUFFIX -M12 EXPOSED FLUSH VALVE SUPPLY SUPPORT



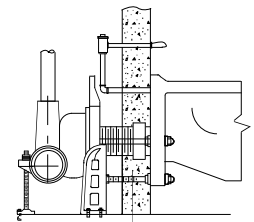
☐ -M40 WATER CLOSET SUPPORT FOR WIDE PIPE CHASE (USE WHEN Ⓢ DIMENSION EXCEEDS 7" (180) )



☐ SUFFIX -M13 TILING FRAME - POLYSTYRENE

## CONCEALED FLUSH VALVES

A concealed flush valve when used with a blowout water closet support will not clear the face plate. Therefore 4 1/2" (115) to 6 1/2" (165) is required from back of wall to face plate to permit the installation of the concealed flush valve piping.



0200/0300 Series Back Sheet

| FIGURE NUMBER | C<br>B<br>A | 12-10-03<br>6-1-95<br>7-1-93 | Revised Notes<br>Added Millimeters<br>Submittal Update | RN<br>EMB<br>EMB | CL<br>BS<br>BS | WEIGHT<br>POUNDS | VOLUME<br>CUBIC FEET | FIGURE NUMBER<br>0200/0300 Ser<br>Back Sheet |
|---------------|-------------|------------------------------|--------------------------------------------------------|------------------|----------------|------------------|----------------------|----------------------------------------------|
| REV.          | DATE        | DESCRIPTION                  | BY                                                     | CKD. BY          |                |                  |                      |                                              |



1466233

FIGURE NUMBER

0211Y (-M54)  
0221Y (-M54)

DRAWN BY:

CR

CHECKED BY:

AM

APPROVED BY:

JM

DATE:

4-3-12

SCALE:

NONE

SIZE

A

DRAWING NUMBER

s0211Y, 0221Y (-M54)

G

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LOCATION

# ADJUSTABLE FIXTURE SUPPORTS FOR SIPHON JET WATER CLOSETS

**NOTES:**

- 1. Base Support must be securely anchored to the floor (with one 1/2"(13) bolt per base support provided by installer.)
- 2. All single supports shall have the rear support lug anchored by contractor to assure a rigid installation. The rear support completes the 3 point anchoring system for ultimate stability of the installation.
- 3. Dimensions shown in parentheses are in millimeters.
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| A<br>SIZE | D        |           |
|-----------|----------|-----------|
|           | MIN      | MAX       |
| 4(100)    | 3/4 (19) | 3 3/8(87) |

⊕ Dimension = 4"(100) (Regularly Furnished) consisting of nipple and Adjust-to-Wall coupling permitting adjustment from 3"(76) min. to 4"(100) max. Longer nipples are available, specify ⊕ dimension.

**NOTE:** Cast Iron Components are Duco Coated.

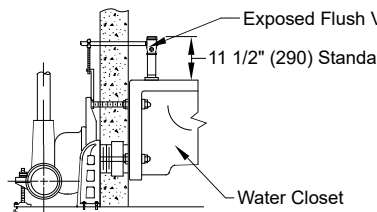
|                          |                          |                               |
|--------------------------|--------------------------|-------------------------------|
| <input type="checkbox"/> | Fig. 0211LY (-M54) ..... | NO-HUB Left Hand              |
| <input type="checkbox"/> | Fig. 0211RY (-M54) ..... | NO-HUB Right Hand             |
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| <input type="checkbox"/> | Fig. 0221LY (-M54) ..... | NO-HUB Left Hand Long Barrel  |
| <input type="checkbox"/> | Fig. 0221RY (-M54) ..... | NO-HUB Right Hand Long Barrel |
| <input type="checkbox"/> | Fig. 0221DY (-M54) ..... | NO-HUB Double Long Barrel     |

Conforms with the requirements of ASME A112.6.1M up to a 500 lbs (227kg) static load rating.

| REV. | DATE | DESCRIPTION | BY | CKD. BY | WEIGHT POUNDS | VOLUME CUBIC FEET | FIGURE NUMBER                |
|------|------|-------------|----|---------|---------------|-------------------|------------------------------|
|      |      |             |    |         |               |                   | 0211Y (-M54)<br>0221Y (-M54) |

**WARNING:** Cancer and Reproductive Harm - www.P65Warnings.ca.gov

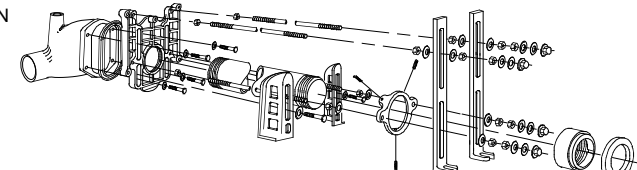
| <b>C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  | <b>LOCATION</b> |        |     |   |     |            |            |            |  |             |        |         |  |              |         |         |  |          |             |              |  |          |              |              |  |          |              |              |  |          |              |              |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------|--------|-----|---|-----|------------|------------|------------|--|-------------|--------|---------|--|--------------|---------|---------|--|----------|-------------|--------------|--|----------|--------------|--------------|--|----------|--------------|--------------|--|----------|--------------|--------------|--|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <h1 style="margin: 0;">MODIFICATIONS &amp; VARIATIONS</h1>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                   |                                                                                   |                                                                                   |                 |        |     |   |     |            |            |            |  |             |        |         |  |              |         |         |  |          |             |              |  |          |              |              |  |          |              |              |  |          |              |              |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <p><b>Ⓒ DIMENSION - ADJUS-TO-WALL TYPE OUTLET CONNECTION</b></p> <p>The Smith Adjus-to-Wall outlet connection provides adjustment at the face of the wall at the time the fixture is set; thus, compensating for wall construction variations. The Ⓒ dimension is measured from the front of the face plate to the finished wall. To properly determine the length of nipple required, deduct 1" (25) from the Ⓒ dimension. When the Ⓒ dimension exceeds 7" (180), the -M40 Wide Pipe Chase Support is recommended. The Adjus-to-Wall coupling will provide 1/2" (13) horizontal adjustment in either direction with the coupling extended 5/16" (8) past face of the finished wall.</p> <p>Ⓒ dimension = 4" (100) (regularly furnished) consisting of nipple and adjus-to-wall coupling permitting adjustment from 3" (76) min. to 4" (100) max.</p> <p>*ABS (Regularly Furnished) other length PVC.</p> </div> <div style="width: 50%;"> <p style="text-align: center;"><b>NIPPLE LENGTHS AVAILABLE</b></p> <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th>LENGTH</th> <th>MIN</th> <th>Ⓒ</th> <th>MAX</th> </tr> </thead> <tbody> <tr> <td>2 1/2 (64)</td> <td>2 1/4 (57)</td> <td>3 1/4 (83)</td> <td></td> </tr> <tr> <td>*3 1/4 (83)</td> <td>3 (76)</td> <td>4 (100)</td> <td></td> </tr> <tr> <td>*8 1/4 (210)</td> <td>8 (205)</td> <td>9 (230)</td> <td></td> </tr> <tr> <td>10 (255)</td> <td>9 3/4 (250)</td> <td>10 3/4 (275)</td> <td></td> </tr> <tr> <td>12 (305)</td> <td>11 3/4 (300)</td> <td>12 3/4 (325)</td> <td></td> </tr> <tr> <td>15 (380)</td> <td>14 3/4 (376)</td> <td>15 3/4 (400)</td> <td></td> </tr> <tr> <td>18 (455)</td> <td>17 3/4 (450)</td> <td>18 3/4 (475)</td> <td></td> </tr> </tbody> </table> <p><small>NOTE: When nipple lengths between 3 1/4" (83) and 8 1/4" (210) are specified, the 8 1/4" (210) nipple will be supplied. Nipple must be cut to specified length by others.</small></p> </div> </div> |                                                                                   |                                                                                   |                                                                                   |                 | LENGTH | MIN | Ⓒ | MAX | 2 1/2 (64) | 2 1/4 (57) | 3 1/4 (83) |  | *3 1/4 (83) | 3 (76) | 4 (100) |  | *8 1/4 (210) | 8 (205) | 9 (230) |  | 10 (255) | 9 3/4 (250) | 10 3/4 (275) |  | 12 (305) | 11 3/4 (300) | 12 3/4 (325) |  | 15 (380) | 14 3/4 (376) | 15 3/4 (400) |  | 18 (455) | 17 3/4 (450) | 18 3/4 (475) |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | LENGTH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MIN                                                                               | Ⓒ                                                                                 | MAX                                                                               |                 |        |     |   |     |            |            |            |  |             |        |         |  |              |         |         |  |          |             |              |  |          |              |              |  |          |              |              |  |          |              |              |  |
| 2 1/2 (64)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2 1/4 (57)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3 1/4 (83)                                                                        |                                                                                   |                                                                                   |                 |        |     |   |     |            |            |            |  |             |        |         |  |              |         |         |  |          |             |              |  |          |              |              |  |          |              |              |  |          |              |              |  |
| *3 1/4 (83)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3 (76)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4 (100)                                                                           |                                                                                   |                                                                                   |                 |        |     |   |     |            |            |            |  |             |        |         |  |              |         |         |  |          |             |              |  |          |              |              |  |          |              |              |  |          |              |              |  |
| *8 1/4 (210)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 8 (205)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 9 (230)                                                                           |                                                                                   |                                                                                   |                 |        |     |   |     |            |            |            |  |             |        |         |  |              |         |         |  |          |             |              |  |          |              |              |  |          |              |              |  |          |              |              |  |
| 10 (255)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 9 3/4 (250)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 10 3/4 (275)                                                                      |                                                                                   |                                                                                   |                 |        |     |   |     |            |            |            |  |             |        |         |  |              |         |         |  |          |             |              |  |          |              |              |  |          |              |              |  |          |              |              |  |
| 12 (305)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 11 3/4 (300)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 12 3/4 (325)                                                                      |                                                                                   |                                                                                   |                 |        |     |   |     |            |            |            |  |             |        |         |  |              |         |         |  |          |             |              |  |          |              |              |  |          |              |              |  |          |              |              |  |
| 15 (380)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 14 3/4 (376)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 15 3/4 (400)                                                                      |                                                                                   |                                                                                   |                 |        |     |   |     |            |            |            |  |             |        |         |  |              |         |         |  |          |             |              |  |          |              |              |  |          |              |              |  |          |              |              |  |
| 18 (455)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 17 3/4 (450)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 18 3/4 (475)                                                                      |                                                                                   |                                                                                   |                 |        |     |   |     |            |            |            |  |             |        |         |  |              |         |         |  |          |             |              |  |          |              |              |  |          |              |              |  |          |              |              |  |
| <p style="text-align: center;"><b>RECOMMENDED SETTING OF CLOSET OUTLET CONNECTION AND FIXTURE STUDS</b></p> <p>Manufacturers have slight differences in the depth of the closet horn and flange thickness of their closets. Usually the fixture support closet connection should extend 5/16" (8) beyond face of finished wall (Dimension "R") and the fixture studs 2 1/4" (57) (Dimension "X"). For exact dimensions please use the formulas below.</p> <p><b>DETAIL A</b> Formula for "R" is: <math>S + 1/16" (2) - D = R</math>.</p> <p>S = Depth of closet horn<br/> 1/16" (2) = Distance closet is to set away from finished wall.<br/> D = 1/2" (13) for felt gasket and 3/8" (10) for neoprene gasket.<br/> R = Distance coupling should extend beyond finished wall.<br/> NOTE: When the fixture is installed, closet gasket must be compressed sufficiently to assure a gas and water-tight seal.</p> <p><b>DETAIL B</b> Formula for "X" is: <math>Y + 1/16" (2) + 9/16" (14) = X</math>.</p> <p>Y = Thickness of closet wall flange.<br/> 1/16" (2) = Distance closet is to set away from finished wall.<br/> X = Distance supporting stud should extend beyond finished wall.<br/> NOTE: Bearing nuts and washers must be set to take full loading from the fixture allowing 1/16" (2) clearance between fixture and wall.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                   |                                                                                   |                                                                                   |                 |        |     |   |     |            |            |            |  |             |        |         |  |              |         |         |  |          |             |              |  |          |              |              |  |          |              |              |  |          |              |              |  |



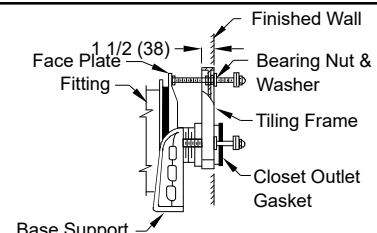
Exposed Flush Valve 4 3/4" (120)  
11 1/2" (290) Standard  
Water Closet

Provides support and alignment for the supply pipe to an exposed flush valve; must be specified for either 1" (25) copper tube or 1" (25) iron pipe.

☐ **SUFFIX -M12 EXPOSED FLUSH VALVE SUPPLY SUPPORT**



☐ **-M40 WATER CLOSET SUPPORT FOR WIDE PIPE CHASE (USE WHEN Ⓒ DIMENSION EXCEEDS 7" (180) )**

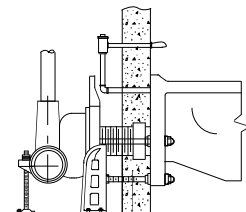


Finished Wall  
Face Plate 1 1/2 (38)  
Bearing Nut & Washer  
Fitting  
Tiling Frame  
Closet Outlet Gasket  
Base Support

☐ **SUFFIX -M13 TILING FRAME - POLYSTYRENE**

**CONCEALED FLUSH VALVES**

A concealed flush valve when used with a blowout water closet support will not clear the face plate. Therefore 4 1/2" (115) to 6 1/2" (165) is required from back of wall to face plate to permit the installation of the concealed flush valve piping.



DRAWING NUMBER  
**S0200BS**

SIZE  
**A**

SCALE:  
**NONE**

DATE:  
**2-28-86**

APPROVED BY:  
**JD**

CHECKED BY:  
**CR**

DRAWN BY:  
**VGD**

**0200/0300 Series Back Sheet**

FIGURE NUMBER

WE CAN ASSUME NO RESPONSIBILITY FOR USE OF SUPERSEDED OR VOID DATA

|      |          |                   |     |         |
|------|----------|-------------------|-----|---------|
| C    | 12-10-03 | Revised Notes     | RN  | CL      |
| B    | 6-1-95   | Added Millimeters | EMB | BS      |
| A    | 7-1-93   | Submittal Update  | EMB | BS      |
| REV. | DATE     | DESCRIPTION       | BY  | CKD. BY |

|                  |                      |                                                  |
|------------------|----------------------|--------------------------------------------------|
| WEIGHT<br>POUNDS | VOLUME<br>CUBIC FEET | FIGURE NUMBER<br><b>0200/0300 Ser Back Sheet</b> |
|------------------|----------------------|--------------------------------------------------|

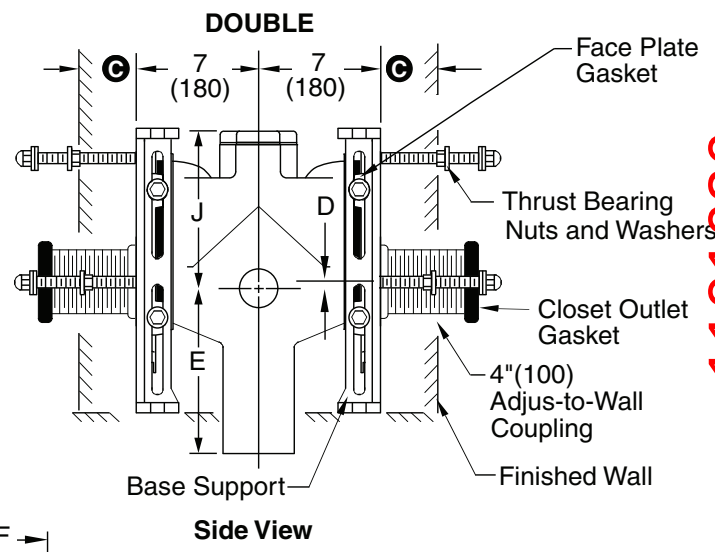
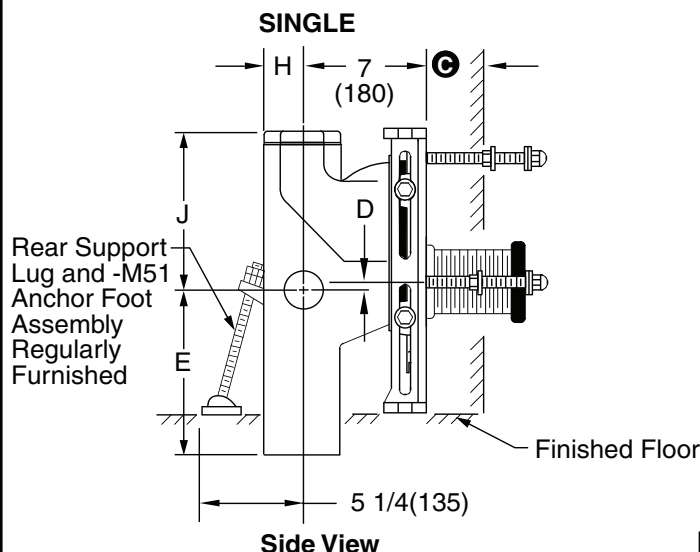


**WARNING:** Cancer and Reproductive Harm - www.P65Warnings.ca.gov



1466233

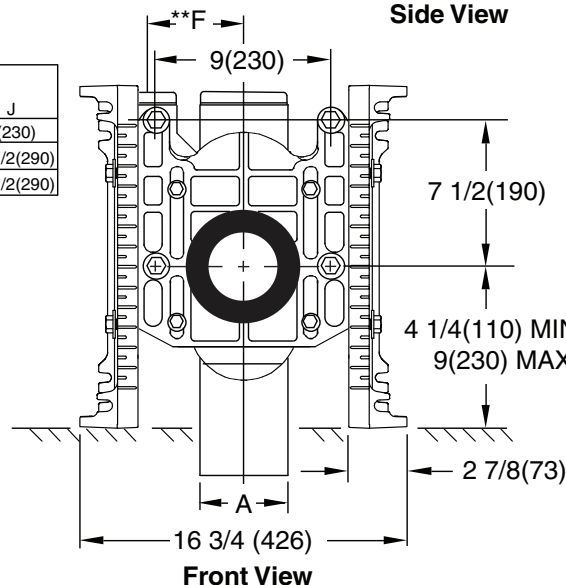
# ADJUSTABLE FIXTURE SUPPORTS FOR SIPHON JET WATER CLOSETS



| A<br>SIZE | D         |            | E           | **F        | H         | J           |
|-----------|-----------|------------|-------------|------------|-----------|-------------|
|           | MIN       | MAX        |             |            |           |             |
| 4(100)    | 1 1/2(38) | 4 1/4(110) | 14(355)     | 5 1/4(135) | 2 1/4(57) | 9(230)      |
| 5(125)    | 1 1/2(38) | 4 1/4(110) | 11 1/2(290) | 6 1/4(160) | 2 3/4(70) | 11 1/2(290) |
| 6(150)    | 1 1/2(38) | 4 1/4(110) | 11 1/2(290) | 6 1/4(160) | 3 1/4(83) | 11 1/2(290) |

NOTE: Base support must be firmly anchored to floor with one 1/2"(13) bolt per base support provided by contractor.

NOTE: All single supports shall have the rear support lug anchored by the contractor to assure a rigid installation. The rear support completes the 3-point anchoring system for ultimate stability of the installation.



NOTE: When used with concealed flush valve locate face plate behind supply pipe.

NOTE: Shallower minimum rough-in dimension available to accommodate lower rough bowls. Specify dimension required.

NOTE: 1 1/2(38) or 2(51) NPT inlet available. Add suffix -N

\*NOTE: 9"(230) core drilled hole is required for rough-in 4 1/4(110) thru 6 1/2(165)

\*\*NOTE: "F" dimension is from centerline of vent to centerline of stack.

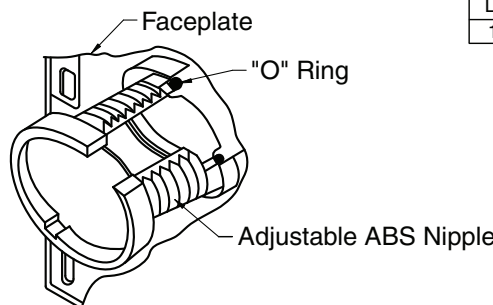
Fig. 0230Y (-M54-M58) . . . NO-HUB

Fig. 0230DY (-M54-M58) . . . NO-HUB DOUBLE

Conforms with the requirements of ASME A112.6.1M up to a 500 lbs (227kg) static load rating.

NOTE: Dimensions shown in parentheses are in millimeters.

NOTE: When nipple lengths between 3 1/4"(83) and 10 3/4"(275) are specified, the 10"(255) nipple will be supplied. The threaded end of nipple must be cut to specified length by others. Absolute minimum dimension is 1 1/2" (38).



| LENGTH  | MIN.        | MAX.         |
|---------|-------------|--------------|
| 10(255) | 9 3/4"(250) | 10 3/4"(275) |

\*ABS (Regularly Furnished).

|   |         |                                |     |    |        |            |               |
|---|---------|--------------------------------|-----|----|--------|------------|---------------|
| F | 6-16-26 | Revised NOTE                   | TB  | CL | WEIGHT | VOLUME     | FIGURE NUMBER |
| E | 12-7-22 | Added Note                     | KK  | CL | POUNDS | CUBIC FEET | 0230Y (-M54   |
| D | 8-19-20 | Rev. Draw./Dim: 16 7/16 16 3/4 | MW  | CL |        |            |               |
| C | 2-18-13 | Rev. Dim., Table               | TBW | CL |        |            |               |

| REV. | DATE | DESCRIPTION | BY | CKD. BY |
|------|------|-------------|----|---------|
|------|------|-------------|----|---------|



WARNING: Cancer and Reproductive Harm - www.P65Warnings.ca.gov



1101626

0230Y (-M54-M58)

FIGURE NUMBER

DRAWING NUMBER  
S0230Y (-M54-M58)

SIZE  
A

SCALE:  
NONE

DATE:  
12/09/04

APPROVED BY:  
AM

CHECKED BY:  
AM

DRAWN BY:  
JJ

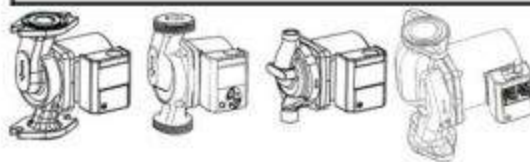
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DIMENSIONS ARE SUBJECT TO MANUFACTURERS TOLERANCE AND CHANGE WITHOUT NOTICE

PP-1



|             |  |                 |  |
|-------------|--|-----------------|--|
| JOB:        |  | REPRESENTATIVE: |  |
| UNIT TAG:   |  | ORDER NO.       |  |
| ENGINEER:   |  | SUBMITTED BY:   |  |
| CONTRACTOR: |  | APPROVED BY:    |  |
| DATE:       |  | DATE:           |  |
| DATE:       |  | DATE:           |  |



## Lead-Free<sup>†</sup> Bronze and Stainless Steel Bodied System Lubricated Circulators



### DESCRIPTION

A series of in-line wet rotor circulation pumps designed specifically for quiet operation in open (potable) water systems. These pumps have lead free<sup>†</sup> bronze or stainless steel bodies.

Automatic Timer and Aquastat accessories are available. (See Submittal A-128A for details).

### CONSTRUCTION MATERIALS

Pump Body:  
NBF: Lead-Free<sup>†</sup> Bronze  
SSF: Stainless Steel  
Bearings: Carbon

Impeller: Noryl  
Shaft: Ceramic

### OPERATING DATA

Maximum Working Pressure: 150 psi (10.3 Bar)  
Minimum Operating Temperature: 40°F (5°C)  
Maximum Operating Temperature:  
NBF-25, NBF-33, NBF-36, NBF-45: 225°F (107°C)  
All Others: 230°F (110°C)

<sup>†</sup>Contains less than 0.25% Lead content by weight on wetted surfaces.

| MODEL NUMBER        | PART NUMBER | CONNECTION                  | APPROX. SHPG. WT. Lbs. (Kg) | STANDARD 60 CYCLE 115 VOLT SINGLE PHASE |           |      | TAGGING INFORMATION | QUANTITY |
|---------------------|-------------|-----------------------------|-----------------------------|-----------------------------------------|-----------|------|---------------------|----------|
|                     |             |                             |                             | WATTS                                   | F.L. AMPS | RPM  |                     |          |
| NBF-8S/LW           | 103257LF    | 1/2" Sweat                  | 9.0 (4.1)                   | 39                                      | 0.38      | 2800 |                     |          |
| NBF-9U/LW           | 103258LF    | Union (See Following Page)  | 9.3 (4.2)                   | 41                                      | 0.40      | 2800 |                     |          |
| SSF-9U/LW           | 103360LF    | Union (See Following Page)  | 9.3 (4.2)                   | 41                                      | 0.40      | 2800 |                     |          |
| NBF-10S/LW          | 103259LF    | 1/2" Sweat                  | 9.0 (4.1)                   | 52                                      | 0.46      | 2800 |                     |          |
| NBF-12U/LW          | 103261LF    | Union (See Following Page)  | 9.3 (4.2)                   | 55                                      | 0.48      | 2800 |                     |          |
| SSF-12U/LW          | 103361LF    | Union (See Following Page)  | 9.3 (4.2)                   | 55                                      | 0.48      | 2800 |                     |          |
| NBF-12F/LW          | 103260LF    | Flange 3/4, 1, 1-1/4, 1-1/2 | 9.5 (4.3)                   | 55                                      | 0.48      | 2800 |                     |          |
| SSF-12F/LW          | 103358LF    | Flange 3/4, 1, 1-1/4, 1-1/2 | 9.5 (4.3)                   | 55                                      | 0.48      | 2800 |                     |          |
| NBF-18S             | 103316LF    | 1/2" Sweat                  | 9.0 (4.1)                   | 90                                      | 0.74      | 3000 |                     |          |
| NBF-22U             | 103255LF    | Union (See Following Page)  | 9.3 (4.2)                   | 92                                      | 0.80      | 2940 |                     |          |
| SSF-22U             | 103362LF    | Union (See Following Page)  | 9.3 (4.2)                   | 92                                      | 0.80      | 2940 |                     |          |
| NBF-22              | 103252LF    | Flange 3/4, 1, 1-1/4, 1-1/2 | 9.5 (4.3)                   | 92                                      | 0.80      | 2940 |                     |          |
| SSF-22              | 103357LF    | Flange 3/4, 1, 1-1/4, 1-1/2 | 9.5 (4.3)                   | 92                                      | 0.80      | 2940 |                     |          |
| NBF-25 <sup>†</sup> | 103418LF    | Flange 3/4, 1, 1-1/4, 1-1/2 | 10.4 (4.7)                  | 125                                     | 1.10      | 2950 |                     |          |
| NBF-33              | 103351LF    | Flange 3/4, 1, 1-1/4, 1-1/2 | 10.4 (4.7)                  | 125                                     | 1.10      | 2950 |                     |          |
| NBF-36 <sup>†</sup> | 103401LF    | Flange 3/4, 1, 1-1/4, 1-1/2 | 13.1 (6.0)                  | 270                                     | 2.30      | 3300 |                     |          |
| NBF-45 <sup>†</sup> | 103405LF    | Flange 1, 1-1/4, 1-1/2      | 14.5 (6.6)                  | 270                                     | 2.30      | 3300 |                     |          |

<sup>†</sup>3-speed circulators

### TYPICAL SPECIFICATIONS

The contractor shall furnish and install in-line circulating pumps as illustrated on the plans and in accordance with the following specifications:

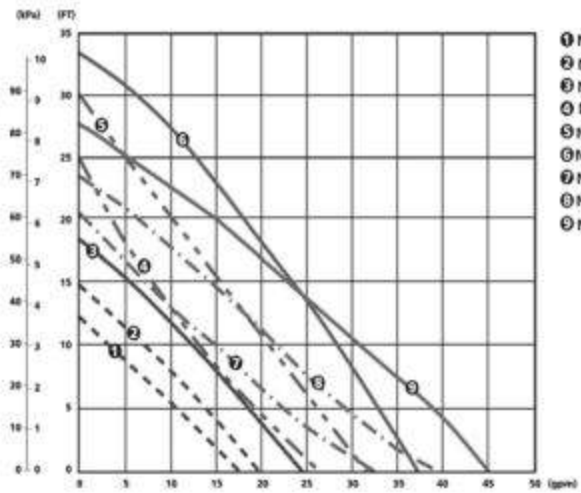
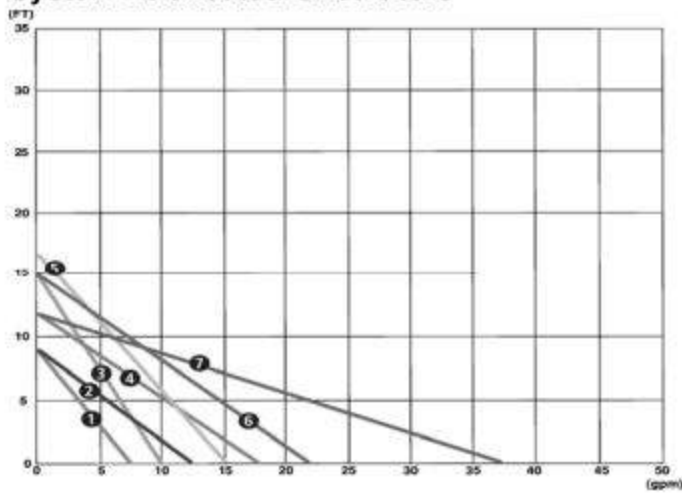
- The pumps shall be of the horizontal system lubricated type specifically designed and guaranteed for quiet operation.
- Pump to be suitable for \_\_\_\_\_°F (\_\_\_\_°C) [choose one: 225°F (107°C) for NBF-25, NBF-33, NBF-36, NBF-45 or 230°F (110°C) for all other circulators] operation at 150 psig (10.3 Bar) working pressure.
- The pumps shall have a ceramic shaft supported by carbon bearings. Bearings are to be lubricated by the circulating fluid.

- Pump body shall be lead-free bronze for NBF circulators or stainless steel for SSF circulators.
- Motor stator to be isolated from circulating fluid through use of stainless steel can. Rotor to be sheathed in stainless steel.
- Motors shall be non-overloading at any point on the pump curve. NBF-36 & NBF-45 to have built-in thermal protection. All other motors to have built-in impedance protection.
- NBF-25 has an optional check valve.

Pumps to have a capacity of \_\_\_\_\_ GPM at \_\_\_\_\_ foot head when powered by 115 volt, 60 cycle single phase electrical supply.

All pumps are to be Xylem - Bell & Gossett Model \_\_\_\_\_.

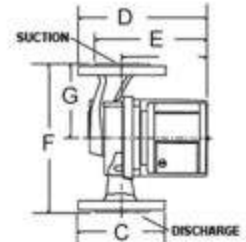
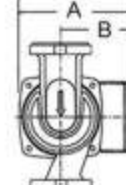
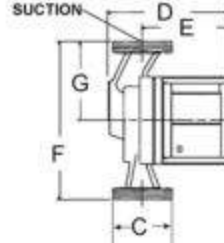
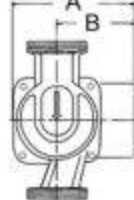
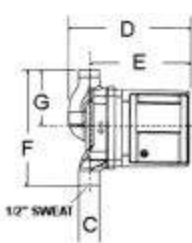
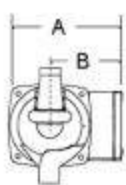
## System Lubricated Circulators



## HALF UNION CONNECTIONS

For NBF-9U/LW, NBF-12U/LW, NBF-22U, SSF-9U/LW, SSF-12U/LW, SSF-22U

| MODEL NUMBER | PART NUMBER | DESCRIPTION (SETS OF 2)      |
|--------------|-------------|------------------------------|
| UC-1/2S      | 113203LF    | 1/2" Bronze Union Sweat      |
| UC-3/4S      | 113201LF    | 3/4" Bronze Union Sweat      |
| UC-3/4NPT    | 113202LF    | 3/4" Bronze Union NPT Female |



| MODEL      | PART NUMBER | A<br>in (mm) | B<br>in (mm) | C<br>in (mm) | D<br>in (mm) | E<br>in (mm) | F<br>in (mm) | G<br>in (mm) |
|------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| NBF-8S/LW  | 103257LF    | 4-7/8 (124)  | 3-3/16 (81)  | 1/2 (13)     | 5-7/32 (132) | 4-9/32 (109) | 5 (127)      | 2-1/2 (63)   |
| NBF-9U/LW  | 103258LF    | 4-7/8 (124)  | 3-3/16 (81)  | 1-1/4 (32)   | 5-1/16 (129) | 3-11/16 (93) | 6-1/8 (156)  | 3-1/16 (78)  |
| SSF-9U/LW  | 103360LF    | 4-7/8 (124)  | 3-3/16 (81)  | 1-1/4 (32)   | 5-1/16 (129) | 3-11/16 (93) | 6-1/8 (156)  | 3-1/16 (78)  |
| NBF-10S/LW | 103259LF    | 4-7/8 (124)  | 3-3/16 (81)  | 1/2 (13)     | 5-7/32 (132) | 4-9/32 (109) | 5 (127)      | 2-1/2 (63)   |
| NBF-12U/LW | 103261LF    | 4-7/8 (124)  | 3-3/16 (81)  | 1-1/4 (32)   | 5-1/16 (129) | 3-11/16 (93) | 6-1/8 (156)  | 3-1/16 (81)  |
| SSF-12U/LW | 103361LF    | 4-7/8 (124)  | 3-3/16 (81)  | 1-1/4 (32)   | 5-1/16 (129) | 3-11/16 (93) | 6-1/8 (156)  | 3-1/16 (78)  |
| NBF-12F/LW | 103260LF    | 4-7/8 (124)  | 3-3/16 (81)  | 3-3/16 (81)  | 5-9/16 (141) | 3-11/16 (93) | 6-3/8 (162)  | 3-3/16 (82)  |
| SSF-12F/LW | 103358LF    | 4-7/8 (124)  | 3-3/16 (81)  | 3-3/16 (81)  | 5-9/16 (141) | 3-11/16 (93) | 6-3/8 (162)  | 3-3/16 (82)  |
| NBF-18S    | 103316LF    | 4-7/8 (124)  | 3-3/16 (81)  | 1/2 (13)     | 5-7/32 (132) | 4-9/32 (109) | 5 (127)      | 2-1/2 (63)   |
| NBF-22U    | 103255LF    | 4-7/8 (124)  | 3-3/16 (81)  | 1-1/4 (32)   | 5-1/16 (129) | 3-11/16 (93) | 6-1/8 (156)  | 3-1/16 (78)  |
| SSF-22U    | 103362LF    | 4-7/8 (124)  | 3-3/16 (81)  | 1-1/4 (32)   | 5-1/16 (129) | 3-11/16 (93) | 6-1/8 (156)  | 3-1/16 (78)  |
| NBF-22     | 103252LF    | 4-7/8 (124)  | 3-3/16 (81)  | 3-3/16 (81)  | 5-9/16 (141) | 3-11/16 (93) | 6-3/8 (162)  | 3-3/16 (82)  |
| NBF-25*    | 103418LF    | 5-1/8 (130)  | 3-3/16 (81)  | 3-3/16 (81)  | 6-3/16 (157) | 4-7/8 (124)  | 6-3/8 (162)  | 2-1/2 (63)   |
| NBF-33     | 103351LF    | 4-7/8 (124)  | 3-3/16 (81)  | 3-3/16 (81)  | 6-3/16 (157) | 3-11/16 (94) | 6-3/8 (162)  | 3-3/16 (82)  |
| NBF-36*    | 103401LF    | 5-3/4 (146)  | 3-9/16 (91)  | 3-3/16 (81)  | 6-3/16 (157) | 5-3/8 (137)  | 6-3/8 (162)  | 3-3/16 (82)  |
| NBF-45*    | 103405LF    | 5-3/4 (146)  | 3-9/16 (91)  | 3-7/16 (87)  | 7-3/8 (187)  | 5-1/2 (140)  | 8-1/2 (216)  | 4-1/4 (108)  |

Dimensions are subject to change. Not to be used for construction purposes unless certified.

Companion Flanges Available in Sizes: 3/4", 1", 1-1/4", and 1-1/2"

\* 3-speed circulators

Xylem Inc.  
8200 N. Austin Avenue  
Morton Grove, IL 60053  
Phone: (847)966-3700  
Fax: (847)965-8379  
www.xyleminc.com/brands/bellgossett

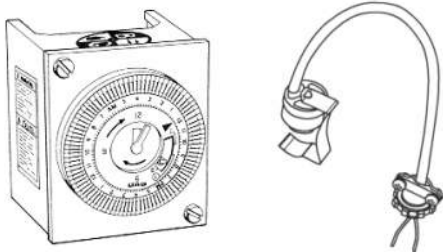
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997733

|                                                                                       |                                                                                                       |                                              |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>JOB:</b><br><br><br><br><b>UNIT TAG:</b><br><b>ENGINEER:</b><br><b>CONTRACTOR:</b> | <b>REPRESENTATIVE:</b><br><br><br><br><b>ORDER NO.</b><br><b>SUBMITTED BY:</b><br><b>APPROVED BY:</b> | <b>DATE:</b><br><b>DATE:</b><br><b>DATE:</b> |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------|



# Automatic Timer Kit

## Model TC-1

# Aquastat

### Models: AQS-1/2, AQS-3/4

532078

#### DESCRIPTION

**TC-1 AUTOMATIC TIMER KIT** - To increase the overall efficiency of a hot water recirculation system, the TC-1 timer control kit can be installed for use on any B&G NBF/NRF/SSF/PL and most 3-piece circulators. The TC-1 timer control is programmable to turn the circulator ON and OFF automatically at preset times. This permits the user to have the pump circulate hot water only during those times when high usage can be expected throughout the day.

**AQS-1/2 and AQS-3/4 AQUASTAT** - are designed to thermostatically turn any B&G NBF/NRF/SSF circulator ON and OFF. The AQS-1/2 or AQS-3/4 will switch the pump OFF at 120°F (48.9°C) and ON at 100°F (37.8°C). The aquastats are available in separate models that will sense the temperature for either 1/2" or 3/4" copper pipe.

**AUTOMATIC TIMER KIT and AQUASTAT COMBINATION** - The automatic timer and aquastat kits can be used in combination. When they are used together, the pump will only circulate water when the ON time conditions are met and when water temperature is low enough to cause the aquastat to switch ON.

#### OPERATIONAL LIMITS

##### TC-1 AUTOMATIC TIMER KIT:

Power Supply: 115/120 VAC, 60 HZ, 1Ø  
 Minimum Switch Interval: 15 minutes  
 Run Modes: ON (continuous run), OFF (at all times),  
 TIMER (run at programmed intervals)  
 Maximum Switch Current: 16 amps

##### AQS-1/2, AQS-3/4 AQUASTATS

Thermostatic Switch Modes: OFF (open) at 120°F (48.9°C) and ON (closed) at 100°F (37.8°C) operation of the aquastat is dependent on ambient settings. In setting where there is excessive heat loss or in installations where heat from the boiler or other components can affect the aquastat, the actual on and off temperatures may vary.

Thermostatic Element: Bimetal, senses surface temperature of outside diameter of pipe.

Pipe Size: AQS-1/2 clips onto 1/2" copper pipe or 3/8" steel pipe (OD of pipe 5/8"). AQS-3/4 clips onto 3/4" copper pipe or 1/2" steel pipe (OD of pipe 7/8").

Mounting: May be installed to sense temperature at the suction or discharge pipe of the pump.

#### CONSTRUCTION MATERIALS

##### TC-1 AUTOMATIC TIMER KIT:

Timer Body: Noryl Plastic

##### AQS-1/2, AQS-3/4 AQUASTATS:

Thermostat Covering: Epoxy (Environmentally Sealed)

Pipe Clip: Stainless Steel

Sensing Element: Bimetal

Wire Leads: Insulated 18" (457mm) #18 AWG

#### APPROX. SHIPPING WEIGHT

TC-1: 11oz. (.31Kg)

AQS-1/2: 5oz. (.14Kg)

AQS-3/4: 5oz. (.14Kg)

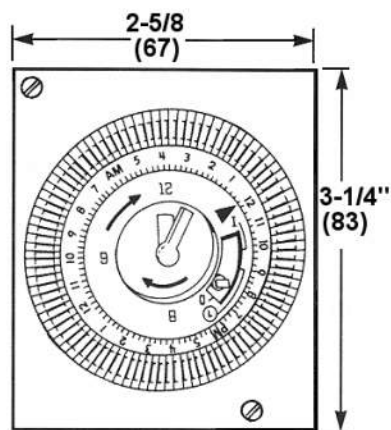
#### SCHEDULE

| MODEL NUMBER | PART NUMBER | SIZE INCHES | VOLTAGE RATING 60 HZ 1Ø | TAGGING INFORMATION | QUANTITY |
|--------------|-------------|-------------|-------------------------|---------------------|----------|
| TC-1         | 113210      | -           | 115-120                 |                     |          |
| AQS-1/2      | 113223      | 1/2         | 115-120                 |                     |          |
| AQS-3/4      | 113224      | 3/4         | 115-120                 |                     |          |

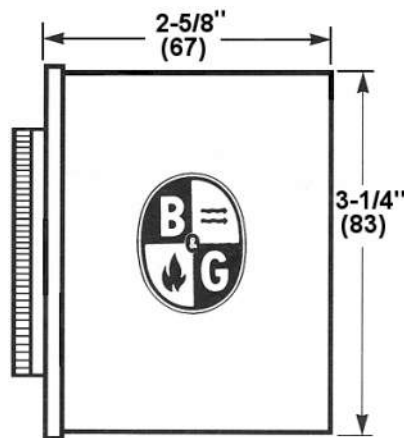
# AUTOMATIC TIMER KIT AQUASTATS

TIMER KIT DIMENSION IN INCHES (MM)

Spec Seq#: 42  
A-128D



FRONT VIEW



SIDE VIEW

## TYPICAL SPECIFICATIONS

Furnish and install as shown on the circulator plans and in accordance with the following specifications:

### AUTOMATIC TIMER KIT:

1. The timer kit shall be UL approved.
2. The timer kit shall be installed on the connection box of any Bell & Gossett NBF/NRF/SSF system lubricated circulator.
3. The timer kit will be suitable for 115/120V, 60 HZ operation.
4. The timer shall provide automatic ON-OFF control at minimum interval of every 15 minutes. It shall also have the option of providing manual ON-OFF control.

All units shall be Xylem - Bell & Gossett Model No.: TC-1

### AQUASTATS:

1. The aquastat shall be UL approved.
2. The aquastat shall be connected to the lead wires in the connection box of any Bell & Gossett NBF/NRF/SSF system lubricated circulator.
3. The aquastat will be suitable for 115/120V, 60 HZ operation.
4. The aquastat shall provide thermostat control to the circulator. It will turn OFF (open) at 120°F (48.9°C) and ON (closed) at 100°F (37.8°C) the use of insulation in installation with excessive heat or cooling may be necessary. Use of fiberglass insulation with R-values greater than 1.6 is recommended.

All units shall be Xylem - Bell & Gossett Model No.:  
(Choose: AQS-1/2 or AQS-3/4).

## AUTOMATIC TIMER KIT AND AQUASTAT COMBINATION:

1. The automatic timer kit and either aquastat model can be combined to provide automatic time and temperature control to any Bell & Gossett NBF/NRF/SSF circulator.
2. When the automatic timer and the aquastat are used together, the pump will only circulate water when the ON time conditions are met and when the water temperature is low enough to cause the aquastat to switch ON.

All units shall be Xylem - Bell & Gossett Model No.: TC-1 and Bell & Gossett Model No.: (Choose: AQS-1/2 or AQS-3/4).

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532078

# ISOLATION FLANGES

## Description

The isolation flange is a combination of an isolation ball valve and a companion flange for circulators. The isolation flange allows easy service or replacement of the circulator without the need to drain the system. The isolation flange fits the Bell & Gossett NRF/NBF/SSF wet rotors, Series PL, Series 100, HV, PR and LR circulators.

## Operating Data

Maximum Working Pressure: 150 PSI (10 bar)

Maximum Operating Temperature: 250°F (121°C)

## Materials of Construction

Valve Body: Lead-Free\* Brass

Flange: Chrome Plated Steel

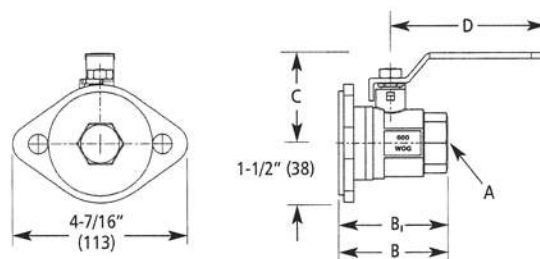
Ball: Chrome Plated Lead-Free\* Brass

Packing: PTFE

Seat Ring: PTFE

Stem: Lead-Free\* Brass

\*CSA certified to NSF/ANSI 372 that product contains less than 0.25% lead content by weight on wetted surface.



## Specifications

| Model Number | Size Inches   | Use with Following Circulators*                                                                                                                             | Dimensions - Inches (mm)<br>Following Circulators |                 |                |                | Approx. Shpg. Wt. lbs. (Kg) |
|--------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-----------------|----------------|----------------|-----------------------------|
|              |               |                                                                                                                                                             | A                                                 | B               | C              | D              |                             |
| 101221LF     | 3/4" NPT IF   | ecocirc 20-18 and ecocirc+ 20-18<br>ecocirc XL 20-35, 36-45, 55-45<br>NRF/NBF/SSF wet rotors<br>Series PL-30, PL-36, PL-55, PL-100<br>Series 100, PR and LR | 3/4" NPT                                          | 2-27/64" (61.5) | 2" (50.5)      | 4-47/64" (120) | 3.2 (1.5)                   |
| 101222LF     | 1" NPT IF     |                                                                                                                                                             | 1" NPT                                            | 2-57/64" (73.3) | 2-5/32" (54.7) | 4-47/64" (120) | 4.1 (1.9)                   |
| 101223LF     | 1-1/4" NPT IF |                                                                                                                                                             | 1-1/4" NPT                                        | 3-19/64" (84)   | 3" (76.2)      | 6-7/32" (158)  | 5.8 (26)                    |
| 101241LF     | 1-1/2" NPT IF |                                                                                                                                                             | 1-1/2" NPT                                        | 3-23/64" (85.5) | 3" (76.2)      | 6-7/32" (158)  | 6.1 (28)                    |
| 101226LF     | 3/4" SWT IF   |                                                                                                                                                             | 3/4" SWT                                          | 2-41/64" (67)   | 2" (50.5)      | 4-23/32" (120) | 3.2 (1.5)                   |
| 101227LF     | 1" SWT IF     |                                                                                                                                                             | 1" SWT                                            | 3" (75.3)       | 2-5/32" (54.7) | 4-23/32" (120) | 3.9 (1.8)                   |
| 101228LF     | 1-1/4" SWT IF |                                                                                                                                                             | 1-1/4" SWT                                        | 3-25/64" (86)   | 3" (76.2)      | 6-7/32" (158)  | 5.4 (25)                    |
| 101243LF     | 1-1/2" SWT IF |                                                                                                                                                             | 1-1/2" SWT                                        | 3-35/64" (90)   | 3" (76.2)      | 6-7/32" (158)  | 6 (27)                      |

"IF" = "Isolation Flange"

Note: Dimensions and weights are approximate and subject to change. Contact factory for certified dimensions. The part numbers and shipping weights are for two isolation flanges, capscrews and nuts.

\* Not for use with NRF/NBF-45.

# COMPANION FLANGES

## Flanges for Cast Iron Circulators

|                                                                                              | Size (NPT) | Master Carton of 12 Part No. | Set of 2 Part No. |
|----------------------------------------------------------------------------------------------|------------|------------------------------|-------------------|
| Series 100, PR<br>NRF-22, NRF-9F/LW,<br>NRF-33, NRF-36<br>PL-30, PL-36, PL-55,<br>ecocirc XL | 3/4"       | 101001                       | 101201            |
|                                                                                              | 1"         | 101002                       | 101202            |
|                                                                                              | 1-1/4"     | 101003                       | 101203            |
|                                                                                              | 1-1/2"     | 101004                       | 101204            |
| Series HV, PL-45<br>PL-50, PL-100, NRF-45<br>ecocirc XL                                      | 1"         | 101005                       | 101205            |
|                                                                                              | 1-1/4"     | 101006                       | 101206            |
|                                                                                              | 1-1/2"     | 101007                       | 101207            |

|                                      | Size (NPT) | Set of 2 Part No.* |
|--------------------------------------|------------|--------------------|
| PL-75, PL-130/2"<br>ecocirc XL 15-75 | 2"         | 101215             |
| PL-130/3"<br>ecocirc XL 40-275       | 2-1/2"     | 101219             |
|                                      | 3"         | 101217             |

\*Includes Fasteners

## Union Connection for NBF Circulators

|                                  | Union Connection | Set of Two |          |
|----------------------------------|------------------|------------|----------|
|                                  |                  | Model No.  | Part No. |
| NBF-22U, NBF-12U/LW<br>NBF-9U/LW | 1/2" sweat       | UC-1/2S    | 113203LF |
|                                  | 3/4" sweat       | UC-3/4S    | 113201LF |
|                                  | 3/4" NPT         | UC-3/4NPT  | 113202LF |

## Flanges for Bronze Circulators

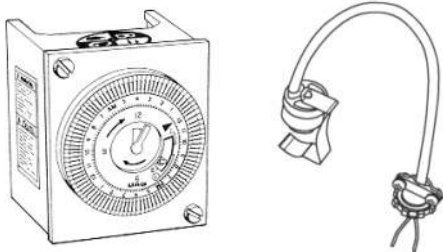
|                                                                                               | Size (NPT) | Master Carton of 12 Part No. | Set of 2 Part No. |
|-----------------------------------------------------------------------------------------------|------------|------------------------------|-------------------|
| Series 100B, PRAB,<br>NBF-22, NBF-12F/LW,<br>NBF-33, NBF-36<br>PL-30B, PL-36B,<br>ecocirc XLB | 3/4"       | 101511LF                     | 101501LF          |
|                                                                                               | 1"         | 101512LF                     | 101502LF          |
|                                                                                               | 1-1/4"     | 101513LF                     | 101503LF          |
|                                                                                               | 1-1/2"     | 1101514LF                    | 101504LF          |
| Series HV, PL-45B<br>PL-50B, PL-100B, NBF-45<br>ecocirc XLB                                   | 1"         | 101515LF                     | 101505LF          |
|                                                                                               | 1-1/4"     | 101516LF                     | 101506LF          |
|                                                                                               | 1-1/2"     | 101517LF                     | 101507LF          |

|                                         | Size (NPT) | Set of 2 Part No.* |
|-----------------------------------------|------------|--------------------|
| PL-75B, PL-130B/2"<br>ecocirc XLB 15-75 | 2"         | 101508LF           |
| PL-130B/3"<br>ecocirc XLB 40-275        | 2-1/2"     | 101510LF           |
|                                         | 3"         | 101509LF           |

\*Includes Fasteners



|                                                                                   |                                                                                                   |                                              |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>JOB:</b><br><br><br><b>UNIT TAG:</b><br><b>ENGINEER:</b><br><b>CONTRACTOR:</b> | <b>REPRESENTATIVE:</b><br><br><br><b>ORDER NO.</b><br><b>SUBMITTED BY:</b><br><b>APPROVED BY:</b> | <b>DATE:</b><br><b>DATE:</b><br><b>DATE:</b> |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------|



# Automatic Timer Kit

## Model TC-1

# Aquastat

### Models: AQS-1/2, AQS-3/4

684757

#### DESCRIPTION

**TC-1 AUTOMATIC TIMER KIT** - To increase the overall efficiency of a hot water recirculation system, the TC-1 timer control kit can be installed for use on any B&G NBF/NRF/SSF/PL and most 3-piece circulators. The TC-1 timer control is programmable to turn the circulator ON and OFF automatically at preset times. This permits the user to have the pump circulate hot water only during those times when high usage can be expected throughout the day.

**AQS-1/2 and AQS-3/4 AQUASTAT** - are designed to thermostatically turn any B&G NBF/NRF/SSF circulator ON and OFF. The AQS-1/2 or AQS-3/4 will switch the pump OFF at 120°F (48.9°C) and ON at 100°F (37.8°C). The aquastats are available in separate models that will sense the temperature for either 1/2" or 3/4" copper pipe.

**AUTOMATIC TIMER KIT and AQUASTAT COMBINATION** - The automatic timer and aquastat kits can be used in combination. When they are used together, the pump will only circulate water when the ON time conditions are met and when water temperature is low enough to cause the aquastat to switch ON.

#### OPERATIONAL LIMITS

##### TC-1 AUTOMATIC TIMER KIT:

Power Supply: 115/120 VAC, 60 HZ, 1Ø  
 Minimum Switch Interval: 15 minutes  
 Run Modes: ON (continuous run), OFF (at all times),  
 TIMER (run at programmed intervals)  
 Maximum Switch Current: 16 amps

##### AQS-1/2, AQS-3/4 AQUASTATS

Thermostatic Switch Modes: OFF (open) at 120°F (48.9°C) and ON (closed) at 100°F (37.8°C) operation of the aquastat is dependent on ambient settings. In setting where there is excessive heat loss or in installations where heat from the boiler or other components can affect the aquastat, the actual on and off temperatures may vary.

Thermostatic Element: Bimetal, senses surface temperature of outside diameter of pipe.

Pipe Size: AQS-1/2 clips onto 1/2" copper pipe or 3/8" steel pipe (OD of pipe 5/8"). AQS-3/4 clips onto 3/4" copper pipe or 1/2" steel pipe (OD of pipe 7/8").

Mounting: May be installed to sense temperature at the suction or discharge pipe of the pump.

#### CONSTRUCTION MATERIALS

##### TC-1 AUTOMATIC TIMER KIT:

Timer Body: Noryl Plastic

##### AQS-1/2, AQS-3/4 AQUASTATS:

Thermostat Covering: Epoxy (Environmentally Sealed)

Pipe Clip: Stainless Steel

Sensing Element: Bimetal

Wire Leads: Insulated 18" (457mm) #18 AWG

#### APPROX. SHIPPING WEIGHT

TC-1: 11oz. (.31Kg)

AQS-1/2: 5oz. (.14Kg)

AQS-3/4: 5oz. (.14Kg)

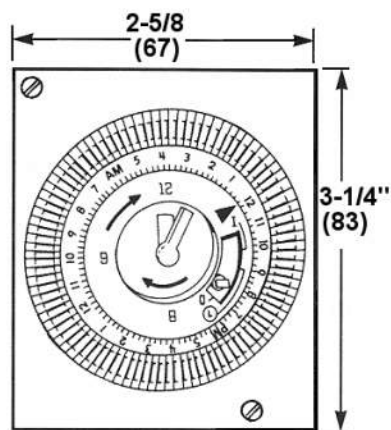
#### SCHEDULE

| MODEL NUMBER | PART NUMBER | SIZE INCHES | VOLTAGE RATING 60 HZ 1Ø | TAGGING INFORMATION | QUANTITY |
|--------------|-------------|-------------|-------------------------|---------------------|----------|
| TC-1         | 113210      | -           | 115-120                 |                     |          |
| AQS-1/2      | 113223      | 1/2         | 115-120                 |                     |          |
| AQS-3/4      | 113224      | 3/4         | 115-120                 |                     |          |

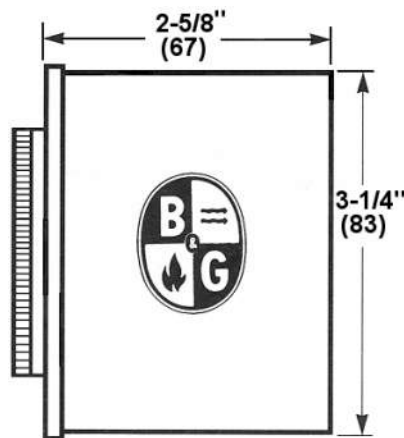
# AUTOMATIC TIMER KIT AQUASTATS

TIMER KIT DIMENSION IN INCHES (MM)

Spec Seq#: 44  
A-128D



FRONT VIEW



SIDE VIEW

## TYPICAL SPECIFICATIONS

Furnish and install as shown on the circulator plans and in accordance with the following specifications:

### AUTOMATIC TIMER KIT:

1. The timer kit shall be UL approved.
2. The timer kit shall be installed on the connection box of any Bell & Gossett NBF/NRF/SSF system lubricated circulator.
3. The timer kit will be suitable for 115/120V, 60 HZ operation.
4. The timer shall provide automatic ON-OFF control at minimum interval of every 15 minutes. It shall also have the option of providing manual ON-OFF control.

All units shall be Xylem - Bell & Gossett Model No.: TC-1

### AQUASTATS:

1. The aquastat shall be UL approved.
2. The aquastat shall be connected to the lead wires in the connection box of any Bell & Gossett NBF/NRF/SSF system lubricated circulator.
3. The aquastat will be suitable for 115/120V, 60 HZ operation.
4. The aquastat shall provide thermostat control to the circulator. It will turn OFF (open) at 120°F (48.9°C) and ON (closed) at 100°F (37.8°C) the use of insulation in installation with excessive heat or cooling may be necessary. Use of fiberglass insulation with R-values greater than 1.6 is recommended.

All units shall be Xylem - Bell & Gossett Model No.:  
(Choose: AQS-1/2 or AQS-3/4).

## AUTOMATIC TIMER KIT AND AQUASTAT COMBINATION:

1. The automatic timer kit and either aquastat model can be combined to provide automatic time and temperature control to any Bell & Gossett NBF/NRF/SSF circulator.
2. When the automatic timer and the aquastat are used together, the pump will only circulate water when the ON time conditions are met and when the water temperature is low enough to cause the aquastat to switch ON.

All units shall be Xylem - Bell & Gossett Model No.: TC-1 and Bell & Gossett Model No.: (Choose: AQS-1/2 or AQS-3/4).

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684757

FD-1



Location: \_\_\_\_\_

**TS****TOP SET HEELPROOF STRAINER**

**Specification:** MIFAB® Series TS(\*)-(\*\*) (\*specify strainer diameter, \*\* specify strainer material) MIFAB Top Set heelproof strainer that can be adjusted vertically and side to side after the pour with a 1/2" thick composite strainer protective cover and indicating whisker. Complete with securing and adjustable hardware and allen key vandal resistant stainless steel grate screws.

**Function:** Used in showers, kitchens, restrooms, and other non-membrane finished areas. Round adjustable heelproof strainer is used for all types of poured finished floors. The concrete is poured to the same level as the orange cap. After the pour, the cap is removed with channel locks and the three Phillips head screws underneath the grate can be engaged to raise the strainer vertically and to also adjust it side to side

VANDAL PROOF  
ALLEN KEY  
STAINLESS STEEL  
SCREWS

PROTECTIVE CAP  
WITH WHISKER

TOP SET  
HEELPROOF  
STRAINER

SCREWS THAT  
PROVIDE VERTICAL  
SIDE TO SIDE  
ADJUSTMENT  
AFTER THE  
CONCRETE POUR

|              |                                           |
|--------------|-------------------------------------------|
| <b>TS5-1</b> | 5" Round Nickel Bronze Top Set Strainer   |
| <b>TS5-3</b> | 5" Round Stainless Steel Top Set Strainer |
| <b>TS6-1</b> | 6" Round Nickel Bronze Top Set Strainer   |
| <b>TS6-3</b> | 6" Round Stainless Steel Top Set Strainer |

| STRAINER DIAMETER | FREE AREA                              | ANSI / ASME RATING<br>LOAD RATING Lbs. |                | WT. Lbs. | STRAINER ADJUSTMENT |             |
|-------------------|----------------------------------------|----------------------------------------|----------------|----------|---------------------|-------------|
|                   |                                        | N.B. (-1)                              | S.S. (-3)      |          | MIN                 | MAX         |
| <b>5" (127)</b>   | 8 in <sup>2</sup> (5200 <sup>2</sup> ) | Up to 2,000                            | 5,000 to 7,499 | 1.25     | 1/2" (13)           | 2 1/2" (64) |
| <b>6" (152)</b>   | 9 in <sup>2</sup> (5800 <sup>2</sup> ) | Up to 2,000                            | 5,000 to 7,499 | 1.75     | 11/16" (17)         | 2 1/2" (64) |

| SUFFIX     | OPTIONAL VARIATIONS                           |
|------------|-----------------------------------------------|
| <b>-1</b>  | Nickel Bronze                                 |
| <b>-3</b>  | Stainless Steel (Type 304)                    |
| <b>-6</b>  | Security Screws (TR25 Torx and pin - H-1039A) |
| <b>-49</b> | Polished Chrome                               |

CALIFORNIA PROPOSITION 65 WARNING: This product contains chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

Job Name: \_\_\_\_\_

Page No: \_\_\_\_\_

Section No: \_\_\_\_\_

Contractor: \_\_\_\_\_

Schedule No: \_\_\_\_\_

Purchase Order No: \_\_\_\_\_



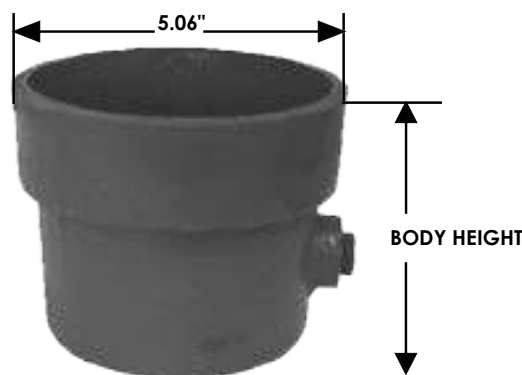
2421064

Location: \_\_\_\_\_

**A4 BODY****A4 BODY**

**Specification:** MIFAB's A4 body Series, 5" diameter body with ½" trap seal primer tapping and plug with secondary tapping to receive an optional secondary composite plug. Available in either cast iron, PVC and / or ABS material in various outlet connection sizes (see chart below).

**Function:** Universal body that is used with all MIFAB F1000, F1100, F1300, F1320, F1420, C1000, C1100, Series of floor and area drains and cleanouts that do not require a water-proofing membrane. The most common application for the A4 bodies is with the C1000 and C1100 cleanouts that require a secondary plug.



2726031

**Note:** No hub outlet connections are the standard outlet connection provided by MIFAB. No suffix is required to indicate the no hub outlet connection.

| PIPE SIZE | NO HUB (STANDARD) | PUSH ON (P) | PVC (-30)<br>ABS (-31) |
|-----------|-------------------|-------------|------------------------|
| 2"        | 4"                | 4 3/8"      | 3 5/8"                 |
| 3"        | 4"                | 4 3/8"      | 4 5/16"                |
| 4"        | 4"                | 4 3/8"      | 4 11/16"               |

| SUFFIX | OPTIONAL VARIATIONS                                                           |
|--------|-------------------------------------------------------------------------------|
| -11    | Acid Resistant Coated Epoxy Body                                              |
| -13    | Galvanized Body                                                               |
| -30    | PVC Connection Body (2", 3" or 4")                                            |
| -31    | ABS Connection Body (2", 3" or 4")                                            |
| -34    | Close plug (C1440 Series)                                                     |
| -90    | Threaded side outlet connection (2", 3" or 4").<br>Not available in ABS / PVC |

| SUFFIX | OPTIONAL VARIATIONS                                                         |
|--------|-----------------------------------------------------------------------------|
| -90NH  | No hub side outlet connection (2", 3" or 4").<br>Not available in ABS / PVC |
| P      | Push On Connection Body (2", 3" or 4")                                      |
| T      | Threaded Connection Body (2", 3" or 4")                                     |
| X      | Inside Caulk Connection Body (2", 3" or 4")                                 |

CALIFORNIA PROPOSITION 65 WARNING. This product contains chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

Job Name: \_\_\_\_\_

Page No: \_\_\_\_\_

Section No: \_\_\_\_\_

Contractor: \_\_\_\_\_

Schedule No: \_\_\_\_\_

Purchase Order No: \_\_\_\_\_



FD-2



Location: \_\_\_\_\_

**TSS****TOP SET SQUARE  
HEELPROOF STRAINER**

**Specification:** MIFAB® Series TSS(\*)-(\*\*) (\*specify strainer size, \*\* specify strainer material) MIFAB Top Set heelproof strainer that can be adjusted vertically and side to side after the pour with a ½" thick composite strainer protective cover and indicating whisker. Complete with securing and adjustable hardware and allen key vandal resistant stainless steel grate screws.

**Function:** Used in showers, kitchens, restrooms, and other non-membrane finished areas. Square adjustable heelproof strainer is used with floors that are tiled or have straight line patterns. The concrete is poured to the same level as the orange cap. After the pour, the cap is removed with channel locks and the three Phillips head screws underneath the grate can be engaged to raise the strainer vertically and to also adjust it side to side

U.S.  
PATENT  
#9,453,331

VANDAL PROOF ALLEN KEY  
STAINLESS STEEL SCREWS

PROTECTIVE CAP  
WITH WHISKER

TOP SET HEELPROOF  
STRAINER

SCREWS THAT  
PROVIDE VERTICAL  
SIDE TO SIDE  
ADJUSTMENT AFTER  
THE CONCRETE POUR

|               |                                                 |
|---------------|-------------------------------------------------|
| <b>TSS5-1</b> | 5" x 5" Square Nickel Bronze Top Set Strainer   |
| <b>TSS5-3</b> | 5" x 5" Square Stainless Steel Top Set Strainer |
| <b>TSS6-1</b> | 6" x 6" Square Nickel Bronze Top Set Strainer   |
| <b>TSS6-3</b> | 6" x 6" Square Stainless Steel Top Set Strainer |

| STRAINER<br>SIZE     | FREE AREA                               | ANSI / ASME RATING<br>LOAD RATING Lbs. |                | WT. Lbs. | STRAINER ADJUSTMENT |              |
|----------------------|-----------------------------------------|----------------------------------------|----------------|----------|---------------------|--------------|
|                      |                                         | N.B. (-1)                              | S.S. (-3)      |          | MIN                 | MAX          |
| <b>5" x 5" (127)</b> | 9 in <sup>2</sup> (5200 <sup>2</sup> )  | Up to 2,000                            | 5,000 to 7,499 | 2.00     | 1/2" (13)           | 2 9/16" (65) |
| <b>6" x 6" (152)</b> | 21 in <sup>2</sup> (7700 <sup>2</sup> ) | Up to 2,000                            | 5,000 to 7,499 | 3.00     | 11/16" (17)         | 2 9/16" (65) |

| SUFFIX     | OPTIONAL VARIATIONS                           |
|------------|-----------------------------------------------|
| <b>-1</b>  | Nickel Bronze                                 |
| <b>-3</b>  | Stainless Steel (Type 304)                    |
| <b>-6</b>  | Security Screws (TR25 Torx and pin - H-1039A) |
| <b>-49</b> | Polished Chrome                               |

CALIFORNIA PROPOSITION 65 WARNING: This product contains chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

Job Name: \_\_\_\_\_

Page No: \_\_\_\_\_

Section No: \_\_\_\_\_

Contractor: \_\_\_\_\_

Schedule No: \_\_\_\_\_

Purchase Order No: \_\_\_\_\_



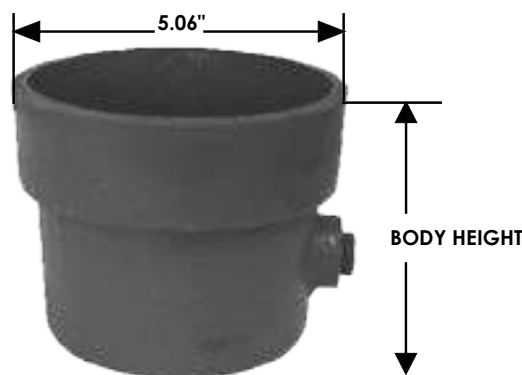
2514481

Location: \_\_\_\_\_

**A4 BODY****A4 BODY**

**Specification:** MIFAB's A4 body Series, 5" diameter body with ½" trap seal primer tapping and plug with secondary tapping to receive an optional secondary composite plug. Available in either cast iron, PVC and / or ABS material in various outlet connection sizes (see chart below).

**Function:** Universal body that is used with all MIFAB F1000, F1100, F1300, F1320, F1420, C1000, C1100, Series of floor and area drains and cleanouts that do not require a water-proofing membrane. The most common application for the A4 bodies is with the C1000 and C1100 cleanouts that require a secondary plug.



2726031

**Note:** No hub outlet connections are the standard outlet connection provided by MIFAB. No suffix is required to indicate the no hub outlet connection.

| PIPE SIZE | NO HUB (STANDARD) | PUSH ON (P) | PVC (-30)<br>ABS (-31) |
|-----------|-------------------|-------------|------------------------|
| 2"        | 4"                | 4 3/8"      | 3 5/8"                 |
| 3"        | 4"                | 4 3/8"      | 4 5/16"                |
| 4"        | 4"                | 4 3/8"      | 4 11/16"               |

| SUFFIX | OPTIONAL VARIATIONS                                                           |
|--------|-------------------------------------------------------------------------------|
| -11    | Acid Resistant Coated Epoxy Body                                              |
| -13    | Galvanized Body                                                               |
| -30    | PVC Connection Body (2", 3" or 4")                                            |
| -31    | ABS Connection Body (2", 3" or 4")                                            |
| -34    | Close plug (C1440 Series)                                                     |
| -90    | Threaded side outlet connection (2", 3" or 4").<br>Not available in ABS / PVC |

| SUFFIX | OPTIONAL VARIATIONS                                                         |
|--------|-----------------------------------------------------------------------------|
| -90NH  | No hub side outlet connection (2", 3" or 4").<br>Not available in ABS / PVC |
| P      | Push On Connection Body (2", 3" or 4")                                      |
| T      | Threaded Connection Body (2", 3" or 4")                                     |
| X      | Inside Caulk Connection Body (2", 3" or 4")                                 |

CALIFORNIA PROPOSITION 65 WARNING. This product contains chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

Job Name: \_\_\_\_\_

Page No: \_\_\_\_\_

Section No: \_\_\_\_\_

Contractor: \_\_\_\_\_

Schedule No: \_\_\_\_\_

Purchase Order No: \_\_\_\_\_



FD-3



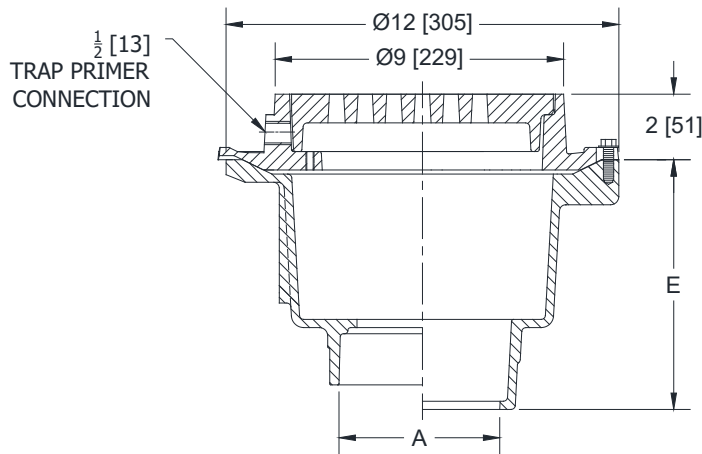
**Z511****9 [229] DIAMETER HEAVY-DUTY DRAIN W/  
INTEGRAL TRAP PRIMER CONNECTION**

Spec Seq#: 52

SPECIFICATION SHEET

TAG \_\_\_\_\_

Design and Dimensional Data (inches and [ mm ]) are Subject to Manufacturing Tolerances and Change Without Notice



2102082

| 'A'<br>Pipe Size In. [mm] | Approx. Wt.<br>Lbs. [kg] | Grate Open Area<br>Sq. In. [cm <sup>2</sup> ] |
|---------------------------|--------------------------|-----------------------------------------------|
| 2, 3, 4, [51, 76, 102]    | 40 [18]                  | 18 [116]                                      |

**ENGINEERING SPECIFICATION: ZURN Z511**

9 [229mm] Diameter top drain, Dura-Coated cast iron deep sump body with bottom outlet, seepage pan, and combination membrane flashing clamp and frame with integral trap primer and plug for heavy-duty ductile iron deep flange slotted grate.

**OPTIONS** (Check/specify appropriate options)**PIPE SIZE**

3, 4 [76, 102]  
3, 4 [76, 102]  
3, 4 [76, 102]  
2, 3, 4 [51, 76, 102]

(specify size/type) **OUTLET**

\_\_\_ IC Inside Caulk  
\_\_\_ IG Inside Gasket  
\_\_\_ IP Threaded  
\_\_\_ NL Neo-Loc

**'E' BODY HT. DIM.**

7-5/8 [194]  
7-5/8 [194]  
7-5/8 [194]  
6-7/8 [175]

**PREFIXES**

\_\_\_ Z D.C.C.I. Body and Top\*  
\_\_\_ ZN D.C.C.I. Body with Polished Nickel Bronze Top (Add 3/16 [5]) to 2 [51] Dim. and 3/4 [19] to 9 [229] Dim.)

**SUFFIXES**

|                                                                                           |                                                                                            |
|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| ___ -AR Acid Resistant Epoxy Coated Cast Iron                                             | ___ -TC Neo-Loc Test Cap Gasket<br>(2, 3, 4 [51, 76, 102] NL Bottom Outlet Only)           |
| ___ -BT Bell Trap                                                                         | ___ -TS Top Secured with Slotted Screws                                                    |
| ___ -F Extension Frame (Specify Total Height Req'd.)                                      | ___ -V Backwater Valve (For 3 & 4 [76 & 102] IC &<br>IP or 3 [76] IG Connections only)     |
| ___ -G Galvanized Cast Iron                                                               | ___ -VP Vandal-Proof Secured Top                                                           |
| ___ -HP Heel-Proof Grate                                                                  | ___ -W Winter Closure Plug (For 3 & 4 [76 & 102]<br>IC & IP or 3 [76] IG Connections only) |
| ___ -J Auxiliary Inlet Connection<br>(Specify 1-1/2 [38] or 2 [51] IP, Number & Location) | ___ -Y Sediment Bucket                                                                     |
| ___ -S Secondary Strainer                                                                 |                                                                                            |
| ___ -SC Solid Cover w/ Lift Handle                                                        |                                                                                            |
| ___ -SG Solid Gasketed Cover                                                              |                                                                                            |

\* Regularly furnished unless otherwise specified

⚠ **WARNING:** Cancer and Reproductive Harm - [www.P65Warnings.ca.gov](http://www.P65Warnings.ca.gov)⚠ **ADVERTENCIA:** Cáncer y daño reproductivo - [www.P65Warnings.ca.gov](http://www.P65Warnings.ca.gov)⚠ **AVERTISSEMENT:** Cancer et effets néfastes sur la reproduction - [www.P65Warnings.ca.gov](http://www.P65Warnings.ca.gov)

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[www.zurn.com](http://www.zurn.com)

Rev. L  
Date: 05/15/2024  
C.N. No. 146062  
Patent [zurn.com/patents](http://zurn.com/patents)  
Prod. | Dwg. No. Z511



FS-1



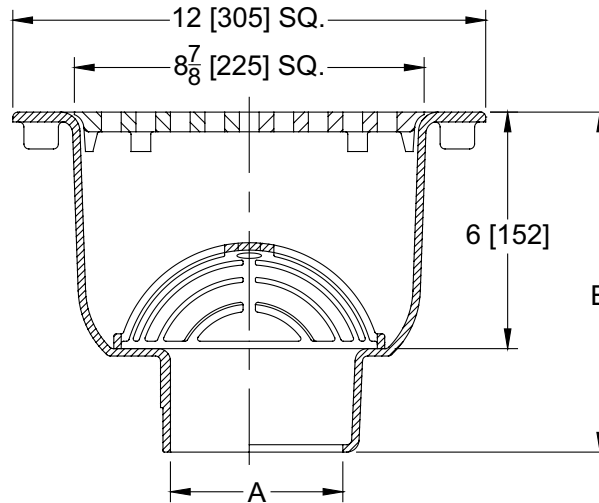
**Z1900****12 x 12 [305 x 305] A.R.E. SANI-FLOR RECEPTOR**  
**6 [152] SUMP DEPTH**

Spec Seq#: 54

SPECIFICATION SHEET

TAG \_\_\_\_\_

Design and Dimensional Data (inches and [ mm ]) are Subject to Manufacturing Tolerances and Change Without Notice



| A<br>Pipe size<br>in [mm] | Approx.<br>Weight<br>lbs [kg] | Grate<br>Open Area<br>in <sup>2</sup> [cm <sup>2</sup> ] |
|---------------------------|-------------------------------|----------------------------------------------------------|
| 2, 3, 4 [51, 76, 102]     | 32 [15]                       | 31 [200]                                                 |

**ENGINEERING SPECIFICATION: ZURN Z1900**

Sani-Flor Receptor 12" x 12" x 6" [305mm x 305mm x 152mm] deep cast iron body and square, light-duty grate with 1/2" [13mm] slotted openings, white acid resisting porcelain enamel interior and top, complete with white ABS anti-splash interior bottom dome strainer.

**Options** (Check/specify appropriate options)**PIPE SIZE**

3, 4 [76, 102]  
2, 3, 4 [51, 76, 102]  
2, 3, 4 [51, 76, 102]

(Specify Size/Type) **OUTLET**

\_\_\_ IC Inside Caulk  
\_\_\_ NH No-Hub  
\_\_\_ NL Neo-Loc

**'E' BODY HEIGHT DIMENSION****Z****ZN/ZS**

8-5/8 [219] 9-1/8 [232]  
8-5/8 [219] 9-1/8 [232]  
8 [203] 8-1/2 [216]

**PREFIXES**

\_\_\_ Z Cast Iron Body with White A.R.E. Interior\*  
\_\_\_ ZN Cast Iron Body with White A.R.E. Interior, 12-1/2 [317] Square N.B. Frame and Full Grate with 1/2 [13] Openings (Add 1/2 [13] to 12 [305] Dim.)  
\_\_\_ ZS Cast Iron Body with White A.R.E. Interior, 12-1/2 [317] Square Stainless Steel Frame and Full Grate with 1/2 [13] Openings (Add 1/2 [13] to 12 [305] Dim.)

**SUFFIXES**

|          |                                                                       |         |                                                                                       |
|----------|-----------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------|
| ___ -DX  | Dex-O-Tex Flange (ZN Only)                                            | ___ -4  | Full Grate w/ Center Opening (2 [51] Dia. for Z) and (3-1/32 [77] Square for ZN & ZS) |
| ___ -HD  | Stainless Steel Frame w/ X-Heavy Duty Stainless Steel Grate (ZS Only) | ___ -6  | Grate w/ 6 [152] Dia. x 6 [152] High Funnel                                           |
| ___ -HP  | Heel-Proof Grate (ZN Only)                                            | ___ -8  | Grate w/ 8-7/8 x 3-5/8 x 3-3/4 [225 x 92 x 95] High Oval Funnel                       |
| ___ -K   | Anchor Flange with Seepage Holes                                      | ___ -11 | Vandal-Proof Secured Top (ZN & ZS Only)                                               |
| ___ -KC  | Anchor Flange with Seepage Holes and Clamp Collar                     | ___ -15 | Solid Loose Set Cover (Z & ZN Only)                                                   |
| ___ -LD  | (Less) Bottom Dome Strainer                                           | ___ -19 | Full Hinged Grate (ZN Only)                                                           |
| ___ -P   | 1/2 [13] Trap Primer Connection (See Z1023)                           | ___ -23 | Aluminum Bucket                                                                       |
| ___ -SA  | Stabilizer Assembly (See Z1903)                                       | ___ -25 | White A.R.C. Bucket                                                                   |
| ___ -TC  | Neo-Loc Test Cap Gasket (2, 3, 4 [51, 76, 102] NL Bottom Outlet Only) | ___ -31 | Stainless Steel Mesh Liner for Bucket                                                 |
| ___ -VPS | Vandal-Proof Strainer                                                 | ___ -32 | Aluminum Dome Strainer                                                                |
| ___ -1   | (Less) Grate                                                          | ___ -33 | White A.R.C. Anti-Splash Bottom Dome Strainer                                         |
| ___ -2   | 1/2 Grate                                                             |         |                                                                                       |
| ___ -3   | 3/4 Grate                                                             |         |                                                                                       |

**⚠ WARNING: Cancer and Reproductive Harm - [www.P65Warnings.ca.gov](http://www.P65Warnings.ca.gov)****⚠ ADVERTENCIA: Cáncer y daño reproductivo - [www.P65Warnings.ca.gov](http://www.P65Warnings.ca.gov)****⚠ AVERTISSEMENT: Cancer et effets néfastes sur la reproduction - [www.P65Warnings.ca.gov](http://www.P65Warnings.ca.gov)**

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Rev. N  
Date: 01/11/23  
C.N. No. 144899  
Patent [zurn.com/patents](http://zurn.com/patents)  
Prod. | Dwg. No. Z1900



1401756

FS-2



## 63M Mop Service Basin – 24" x 24" x 10"



**The 63M is engineered and built to provide years of service.** A complete range of accessories are available to outfit your Model 63M Mop Basin.

### Features

- High impact resistant, one piece molded structural fiberglass.
- Generous 10" high side walls to contain splashes.
- Removable strainer.
- Integral molded-in-drain with seal for installing to 3" ABS, PVC (Sch. 80) or cast iron.
- Available drain seals for connecting 2" pipe or 3" no-hub soil pipe.
- Convenient anchor brackets for securing to wall.

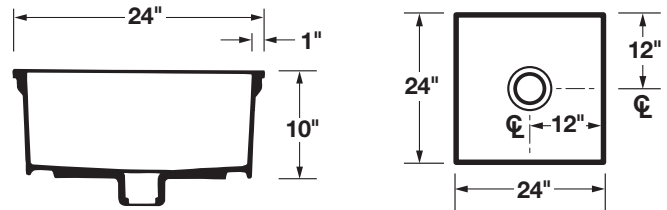
### Code Accepted

- International Association of Plumbing and Mechanical Officials (IAPMO)® Listed
- Warnock Hersey® (Canada) Listed—CSA Standards: Mop Basins – #B45.0
- American National Standards Institute (ANSI) – Specification ANSI Z 124-2022
- American Society of Testing and Materials (ASTM) – E-84
- CAN/ULC S102-07 Burn and Smoke Test

### Specifications

**General:** Mop Service Basin Model 63M, as manufactured by E.L. Mustee & Sons, Inc. One-piece, molded fiberglass made with matched metal molds. Height shall be 10" with not less than 1" wide shoulder. Size 24" x 24". Drain shall be integrally molded with drain seal for installation of 3" ABS, PVC (Sch. #80) and iron pipe. Removable stainless steel strainer. Shall meet ANSI Specifications Z 124-2022. Color: White.

| Model | Color | Length | Width |
|-------|-------|--------|-------|
| 63M   | White | 24"    | 24"   |



**Accessories** – for complete details and specifications visit [www.mustee.com](http://www.mustee.com)



- 1 63.600A SERVICE FAUCET** – Chrome plated brass service sink faucet, 1/2" pipe eccentric inlets on 8" centers. Vacuum breaker integral stops, chrome plated (H&C) handles, 3/4" hose end spout with pail hook, top reinforcing bar and mounting bracket. Shall meet ANSI Specifications A112.1.1, Section 2.9 and A112.18.1M, CSA Listed.

- 2 65.700 HOSE and HOSE HOLDER** – Heavy-duty 5/8" diameter, reinforced 31" rubber hose with brass couplings on one end. Spring loaded, molded rubber hose holder mounts on stainless steel wall plate.

- 3 65.600 MOP HANGER** – Three spring-loaded rubber holders mounted to 3" x 22" stainless steel wall plate. Wall anchors and mounting screws included.

- 4 BUMPER GUARDS** – Protect rims, snap-on

#### High impact resistant vinyl

63.401 – 20 3/4" long white, fits Models 63M and 65M  
65.403 – 32 3/4" long white, fits Model 65M

#### Stainless steel

63.403 – 20 3/4" long, fits Models 63M and 65M  
65.404 – 32 3/4" long, fits Model 65M

- 5 DURAGUARD™ WALL GUARDS** – Protect wall surfaces, 12" height, 20 gauge, #304 stainless steel.  
67.2424 – two panels and bracket for 24" x 24" corner  
67.2436 – two panels and bracket for 24" x 36" corner

- 6 OPTIONAL SEALS** – Order separately.

65.308 – for 2" PVC, ABS and iron  
65.309 – for 3" PVC, ABS and iron, include mop basin  
65.311 – for 3" soil pipe (no hub)

# 65M Mop Service Basin – 24" x 36" x 10"



The **65M** is engineered and built to provide years of service. A complete range of accessories are available to outfit your Model 65M Mop Basin.

## Features

- High impact resistant, one piece molded structural fiberglass.
- Elevated, self-draining shelf and removable strainer.
- Generous 10" high side walls to contain splashes.
- Integral molded-in-drain with seal for installing to 3" ABS, PVC (Sch. 80) or cast iron.
- Available drain seals for connecting 2" pipe or 3" no-hub soil pipe.
- Convenient anchor brackets for securing to wall.

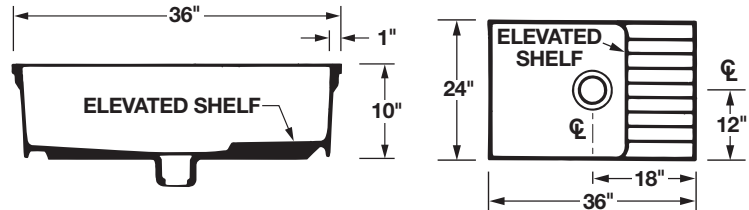
## Code Accepted

- International Association of Plumbing and Mechanical Officials (IAPMO)<sup>®</sup> Listed 
- Warnock Hersey<sup>®</sup> (Canada) Listed—CSA Standards: Mop Basins – #B45.0 
- American National Standards Institute (ANSI) – Specification ANSI Z 124-2022 
- American Society of Testing and Materials (ASTM) – E-84 
- CAN/ULC S102-07 Burn and Smoke Test

## Specifications

**General:** Mop Service Basin Model 65M, as manufactured by E.L. Mustee & Sons, Inc. One-piece, molded fiberglass made with matched metal molds. Height shall be 10" with not less than 1" wide shoulder. Size 24" x 36" with integrally molded, self-draining shelf and drain with seal for installation of 3" ABS, PVC (Sch. #80) and iron pipe. Removable stainless steel strainer. Shall meet ANSI Specifications Z 124-2022. Color: White.

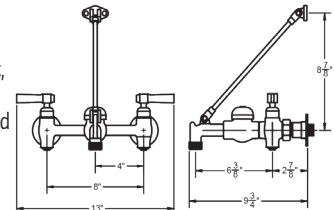
| Model | Color | Length | Width |
|-------|-------|--------|-------|
| 65M   | White | 24"    | 36"   |



**Accessories** – for complete details and specifications visit [www.mustee.com](http://www.mustee.com)



- 1 63.600A SERVICE FAUCET** – Chrome plated brass service sink faucet, 1/2" pipe eccentric inlets on 8" centers. Vacuum breaker integral stops, chrome plated (H&C) handles, 3/4" hose end spout with pail hook, top reinforcing bar and mounting bracket. Shall meet ANSI Specifications A112.1.1, Section 2.9 and A112.18.1M, CSA Listed. 



- 2 65.700 HOSE and HOSE HOLDER** – Heavy-duty 5/8" diameter, reinforced 31" rubber hose with brass couplings on one end. Spring loaded, molded rubber hose holder mounts on stainless steel wall plate.



- 3 65.600 MOP HANGER** – Three spring-loaded rubber holders mounted to 3" x 22" stainless steel wall plate. Wall anchors and mounting screws included.



- 4 BUMPER GUARDS** – Protect rims, snap-on

### High impact resistant vinyl

63.401 – 20 3/4" long white, fits Models 63M and 65M  
65.403 – 32 3/4" long white, fits Model 65M

### Stainless steel

63.403 – 20 3/4" long, fits Models 63M and 65M  
65.404 – 32 3/4" long, fits Model 65M

- 5 DURAGUARD<sup>™</sup> WALL GUARDS** – Protect wall surfaces, 12" height, 20 gauge, #304 stainless steel.

67.2424 – two panels and bracket for 24" x 24" corner  
67.2436 – two panels and bracket for 24" x 36" corner

- 6 OPTIONAL SEALS** – Order separately.

65.308 – for 2" PVC, ABS and iron  
65.309 – for 3" PVC, ABS and iron, included with mop basin  
65.311 – for 3" soil pipe (no hub)

PRV



Job Name \_\_\_\_\_

Contractor \_\_\_\_\_

Job Location \_\_\_\_\_

Approval \_\_\_\_\_

Engineer \_\_\_\_\_

Contractor's P.O. No. \_\_\_\_\_

Approval \_\_\_\_\_

Representative \_\_\_\_\_

**LEAD FREE\***

## Series LFN223B, LFN223BS Super Capacity Water Pressure Reducing Valves\*\*

**Sizes: 2½" – 3"**

Series LFN223B and LFN223BS Super Capacity Water Pressure Reducing Valves are designed to reduce incoming water pressure to a sensible level to protect plumbing system components and reduce water consumption. The LFN223B/LFN223BS features Lead Free\* construction to comply with Lead Free\* installation requirements. This series is suitable for water supply pressures up to 300psi (20.7 bar) and may be adjusted from 25 – 75psi (172 – 517 kPa). The standard setting is 50psi (345 kPa). Series LFN223B features an enlarged diaphragm, spring cage and seat orifice for super capacity performance. Series LFN223BS has the same options as the LFN223B, except it is furnished with a Lead Free\* strainer. All parts are quickly and easily serviceable without removing the valve from the line. The standard bypass feature permits the flow of water back through the valve into the main when pressures, due to thermal expansion on the outlet side of the valve, exceed the pressure in the main supply.

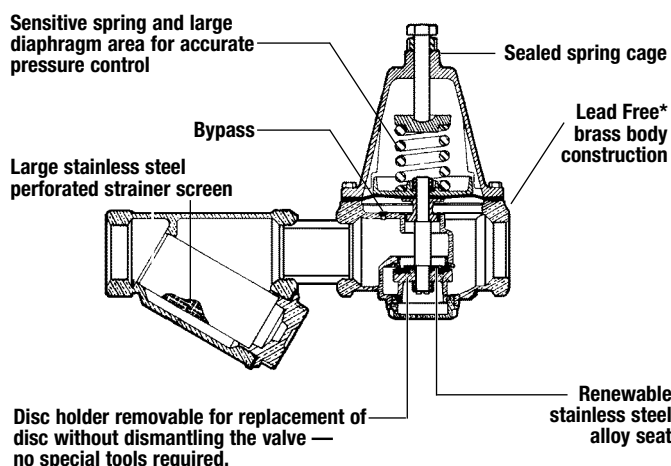
### Features

- Enlarged diaphragm, spring cage and seat orifice for super capacity performance
- Lead Free\* cast copper silicon alloy body construction
- Serviceable in line
- Series LFN223BS furnished with separate Lead Free\* strainer
- Standard bypass feature controls thermal expansion pressure\*\*\*
- Sealed spring cage on all models for accessible outdoor or pit installations

### Specifications

A Lead Free\* Water Pressure Reducing Valve shall be installed on the water service pipe near its entrance to the building where supply main pressure exceeds 60psi (413 kPa) to reduce it to 50psi (345 kPa) or lower. The water pressure reducing valve shall be constructed using Lead Free\* materials. Lead Free\* regulators shall comply with state codes and standards, where applicable, requiring reduced lead content. Provision shall be made to permit the bypass flow of water back through the valve into the main when pressures, due to thermal expansion on the outlet side of the valve, exceed the pressure in the main supply. Pressure reducing valves with built-in bypass check valves and strainer will be acceptable. Approved valves shall comply with ASSE 1003. Valve shall be a Watts Series LFN223B or LFN223BS (with strainer).

Series LFN223B

Available with or without  
separate strainer

\*The wetted surface of this product contacted by consumable water contains less than 0.25% of lead by weight.

\*\*A water saving test program concluded that reducing the supply pressure from 80-50psi (551-345 kPa) resulted in a water savings of 30%.

\*\*\*The bypass feature will not prevent the pressure relief valve from opening on the hot water supply system with pressure above 150psi (10.3 bar).

### NOTICE

The information contained herein is not intended to replace the full product installation and safety information available or the experience of a trained product installer. You are required to thoroughly read all installation instructions and product safety information before beginning the installation of this product.

### NOTICE

Inquire with governing authorities for local installation requirements

Watts product specifications in U.S. customary units and metric are approximate and are provided for reference only. For precise measurements, please contact Watts Technical Service. Watts reserves the right to change or modify product design, construction, specifications, or materials without prior notice and without incurring any obligation to make such changes and modifications on Watts products previously or subsequently sold.



1240786

## Models

|                        |                                                                      |
|------------------------|----------------------------------------------------------------------|
| <b>2½" LFN223M2-B</b>  | NPT threaded female inlet x NPT threaded female outlet               |
| <b>3" LFN223M1-B</b>   | NPT threaded female inlet x NPT threaded female outlet               |
| <b>2½" LFN223M2-BS</b> | NPT threaded female inlet with strainer x NPT threaded female outlet |
| <b>3" LFN223M1-BS</b>  | NPT threaded female inlet with strainer x NPT threaded female outlet |

For ½" – 2½" threaded connections, refer to literature ES-LF223.

For 3" flanged connections, refer to literature ES-LF223F.

## Materials

|                  |                                      |
|------------------|--------------------------------------|
| Body:            | Lead Free* cast copper silicon alloy |
| Seat:            | Replaceable stainless steel alloy    |
| Strainer Screen: | Stainless steel (model LFN223S)      |
| Diaphragm:       | Reinforced Buna-N                    |
| Valve Disc:      | EPDM                                 |

## Pressure – Temperature

Temperature Range: 33°F – 160°F (0.5°C – 71°C)

Maximum Working Pressure: 300psi (20.7 bar)

Adjustable Reduced Pressure Range: 25 – 75psi (172 – 517 kPa)

Standard Reduced Pressure Setting: 50psi (345 kPa)

## Standards



Both 2½" and 3" are listed to ASSE 1003 through IAPMO (UPC).

The 2½" is also listed to ASSE 1003 through ASSE.

Both 2½" and 3" are listed to NSF 61-G

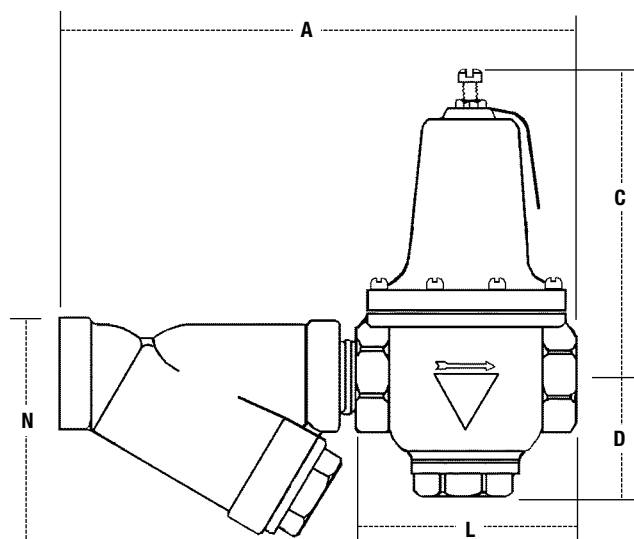
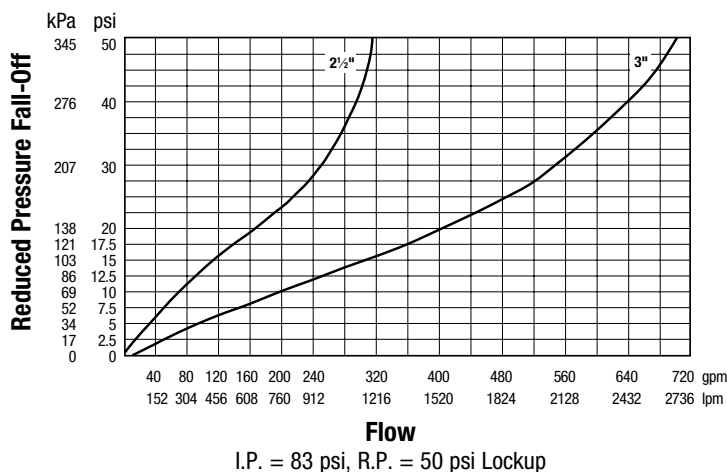
## Options

HP High pressure range 75 – 125psi (172 – 862 kPa)

2½" HP, LP, G and GC options

3" HP option

## Capacity



## Dimensions – Weights

| SIZE | DIMENSIONS   |     |     |     |     |     | WEIGHT |     |              |     |
|------|--------------|-----|-----|-----|-----|-----|--------|-----|--------------|-----|
|      | A (LFN223BS) |     | C   |     | D   |     | L      |     | N (LFN223BS) |     |
|      | in.          | mm  | in. | mm  | in. | mm  | in.    | mm  | in.          | mm  |
| 2½   | 17           | 432 | 10¾ | 273 | 2⅞  | 73  | 7⅞     | 200 | 5            | 127 |
| 3    | 20¾          | 527 | 12¾ | 324 | 4⅞  | 105 | 10½    | 267 | 6¾           | 172 |



EWH-1





The new degree of comfort.®



Spec Seq#: 62  
Commercial Electric  
Light-Duty Water Heaters

## Light-Duty Commercial Electric Water Heaters

*With easy field-level conversion for multiple light duty applications*

### Flexible design for a variety of light duty applications

- 30 - 119.9-Gallon models
- 208/240/277/480 Voltages available on all models
- 30 to 50-Gallon models available in 3kW-12 kW, tall and short
- 66, 80 and 119.9-Gallon models only available in 12.1 kW
- Factory shipped in simultaneous configuration

| FEATURES & BENEFITS                      |                                                                                                                                  |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Terminal Block                           | Easy to convert standard models from simultaneous to non-simultaneous, single phase to three phase                               |
|                                          | Field convertible to meet any light-duty application need                                                                        |
|                                          | Available on 30 - 119.9-gallon models                                                                                            |
| Wiring Options                           | Custom field level configuration based on application requirements                                                               |
|                                          | Simultaneous and non-simultaneous wiring                                                                                         |
|                                          | Single & three phase available                                                                                                   |
| Energy Efficient Design                  | Reduces energy consumption & standby heat loss with 2-1/2" of rigid polyurethane foam insulation                                 |
| Designed for Long Life & Top Performance | Proprietary protective steel formulation with a unique coat of high temperature porcelain enamel                                 |
|                                          | Design maximizes corrosion resistance & life of anode rod                                                                        |
|                                          | Patented corrosion-resistant elements include double layer of magnesium oxide and copper to resist corrosion                     |
|                                          | Durable factory-installed full flow brass drain valve                                                                            |
| Automatic Temperature Control            | Automatic Temperature Control - Surface mounted thermostat automatically cycles on and off to maintain desired water temperature |
|                                          | 170°F Maximum temperature setting                                                                                                |
| Warranty                                 | 3-year limited tank, upgradable to 5-years and a 1-year limited parts warranty*                                                  |

\*See Commercial Warranty Certificate for complete information.

**All models come with a terminal block. Models with power requirements over 48 Amps come with a fused terminal block. See Electrical Characteristics Chart on Back.**



**Rheem Light-Duty**  
30 to 119.9-Gallon Capacities  
208/240/277/480 Voltages  
3kW - 12.1 kW  
Electric



**Consult factory for certification listing.**

**Safety and Construction** | These products are design certified by Underwriters Laboratories (UL) to meet UL safety standards as electric storage tank water heaters. All models are North Carolina and Massachusetts Code compliant.  
**Certified for 150 PSI maximum working pressure.**

See specifications chart on back.



INTEGRATED AIR & WATER

12/21 FORM NO. RR102

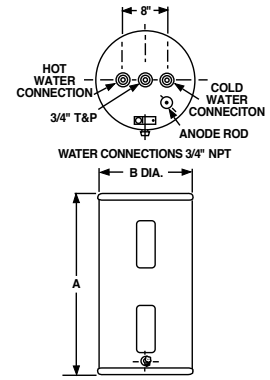




The new degree of comfort.®



| DIMENSIONAL INFORMATION (All dimensions shown in English and Metric) |            |            |                        |     |                          |     |        |      |        |     |                      |      |     | FIRST HOUR DEL. G.P.H. | UNIFORM ENERGY FACTOR (UEF) | STL |
|----------------------------------------------------------------------|------------|------------|------------------------|-----|--------------------------|-----|--------|------|--------|-----|----------------------|------|-----|------------------------|-----------------------------|-----|
| MODEL NUMBER                                                         | MIN. WATTS | MAX. WATTS | TANK CAPACITY (LISTED) |     | TANK CAPACITY (MEASURED) |     | A      |      | B      |     | APPROX. SHIPPING WT. |      |     |                        |                             |     |
|                                                                      |            |            | GAL.                   | LTR | GAL.                     | LTR | INCHES | MM   | INCHES | MM  | lbs.                 | kgs. |     |                        |                             |     |
| ELD30-TB                                                             | 3,000      | 12,000     | 30                     | 114 | 27                       | 102 | 47-1/2 | 1207 | 19     | 483 | 125                  | 57   | 46  | 0.92                   | N/A                         |     |
| ELD40-TB                                                             | 3,000      | 12,000     | 40                     | 151 | 36                       | 136 | 48-1/4 | 1226 | 20-1/4 | 514 | 140                  | 64   | 55  | 0.93                   | N/A                         |     |
| ELD52-TB                                                             | 3,000      | 12,000     | 50                     | 189 | 45                       | 170 | 58-5/8 | 1489 | 20-1/4 | 514 | 155                  | 70   | 63  | 0.93                   | N/A                         |     |
| ELD66-TB                                                             | 12,100     | 12,100     | 65                     | 246 | 58.5                     | 221 | 59-3/8 | 1508 | 22-1/4 | 565 | 185                  | 84   | N/A | N/A                    | 0.76                        |     |
| ELD80-TB                                                             | 12,100     | 12,100     | 80                     | 303 | 72                       | 273 | 59-5/8 | 1514 | 23-3/4 | 603 | 220                  | 100  | N/A | N/A                    | 0.68                        |     |
| ELD120-TB                                                            | 12,100     | 12,100     | 119.9                  | 454 | 108                      | 409 | 62-7/8 | 1597 | 28-1/4 | 718 | 335                  | 152  | N/A | N/A                    | 0.55                        |     |
| ELDS30-TB                                                            | 3,000      | 12,000     | 28                     | 106 | 25                       | 95  | 30     | 762  | 23     | 584 | 130                  | 59   | 43  | 0.92                   | N/A                         |     |
| ELDS40-TB                                                            | 3,000      | 12,000     | 36                     | 136 | 33                       | 125 | 31-1/2 | 800  | 24-1/4 | 616 | 150                  | 68   | 46  | 0.92                   | N/A                         |     |
| ELDS52-TB                                                            | 3,000      | 12,000     | 47                     | 178 | 43                       | 163 | 32     | 813  | 26-1/4 | 667 | 180                  | 82   | 57  | 0.93                   | N/A                         |     |



| PRODUCT AVAILABILITY |      | Electric Light Duty models are available with terminal block in both fused and non-fused configurations.<br>Fused configuration is Standard in all water heaters with power requirements over 48 Amps. |     |         |     |     |      |     |         |     |     |      |     |         |     |     |      |     |         |     |     |
|----------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|-----|-----|------|-----|---------|-----|-----|------|-----|---------|-----|-----|------|-----|---------|-----|-----|
| SIM OR<br>NON-SIM    | SIZE | 208V                                                                                                                                                                                                   |     |         |     |     | 240V |     |         |     |     | 277V |     |         |     |     | 480V |     |         |     |     |
|                      |      | 3/3                                                                                                                                                                                                    | 4/4 | 4.5/4.5 | 5/5 | 6/6 | 3/3  | 4/4 | 4.5/4.5 | 5/5 | 6/6 | 3/3  | 4/4 | 4.5/4.5 | 5/5 | 6/6 | 3/3  | 4/4 | 4.5/4.5 | 5/5 | 6/6 |
|                      | 30   | ✓                                                                                                                                                                                                      | ✓   | ✓       | ✓   | ✓   | ✓    | ✓   | ✓       | ✓   | ✓   | ✓    | ✓   | ✓       | ✓   | ✓   | ✓    | ✓   | ✓       | ✓   | ✓   |
|                      | 40   | ✓                                                                                                                                                                                                      | ✓   | ✓       | ✓   | ✓   | ✓    | ✓   | ✓       | ✓   | ✓   | ✓    | ✓   | ✓       | ✓   | ✓   | ✓    | ✓   | ✓       | ✓   | ✓   |
|                      | 50   | ✓                                                                                                                                                                                                      | ✓   | ✓       | ✓   | ✓   | ✓    | ✓   | ✓       | ✓   | ✓   | ✓    | ✓   | ✓       | ✓   | ✓   | ✓    | ✓   | ✓       | ✓   | ✓   |
|                      | 66   |                                                                                                                                                                                                        |     |         |     | ✓   |      |     |         |     | ✓   |      |     |         |     | ✓   |      |     |         |     | ✓   |
|                      | 80   |                                                                                                                                                                                                        |     |         |     | ✓   |      |     |         |     | ✓   |      |     |         |     | ✓   |      |     |         |     | ✓   |
|                      | 120  |                                                                                                                                                                                                        |     |         |     | ✓   |      |     |         |     | ✓   |      |     |         |     | ✓   |      |     |         |     | ✓   |

- Fused or Non-Fused, Fused if > 48 Amps • Terminal Block allows for conversions from Simultaneous element usage to non-Simultaneous for maximum energy savings.
- ELD-C Models 66/80/120 still available without terminal block. • All models are factory shipped with sim configuration.

| ELECTRICAL CHARACTERISTICS |                    |                                                                        |     |     |     |                                                               |     |     |     |                                                                              |     |     |                                                                                                    |     |     |     |
|----------------------------|--------------------|------------------------------------------------------------------------|-----|-----|-----|---------------------------------------------------------------|-----|-----|-----|------------------------------------------------------------------------------|-----|-----|----------------------------------------------------------------------------------------------------|-----|-----|-----|
| ELEMENT<br>WATTAGE         | ELEMENT<br>WATTAGE | NON-SIMULTANEOUS WIRING                                                |     |     |     | SIMULTANEOUS WIRING                                           |     |     |     |                                                                              |     |     |                                                                                                    |     |     |     |
|                            |                    | SINGLE/THREE PHASE OPERATION<br>FULL LOAD CURRENT IN AMPERES           |     |     |     | 2-WIRE SINGLE PHASE OPERATION<br>FULL LOAD CURRENT IN AMPERES |     |     |     | 4-WIRE THREE PHASE<br>OPERATION FULL LOAD<br>CURRENT IN AMPERES              |     |     | 4-WIRE SINGLE PHASE OPERATION<br>FULL LOAD CURRENT IN AMPERES<br>(DOUBLE BRANCH CIRCUIT)           |     |     |     |
| UPPER                      | LOWER              | TERMINALS L1 AND L2 FOR<br>SINGLE PHASE,<br>L1, L2, AND T2 FOR 3 PHASE |     |     |     | TERMINALS L1 AND L2                                           |     |     |     | THERE'S NO 4 WIRE 3 PHASE<br>THIS IS 3 PHASE SIM<br>TERMINALS L1, L2, AND T2 |     |     | TERMINALS L1 AND L2 FIRST CIRCUIT<br>TERMINALS T1 AND T2<br>SECOND CIRCUIT                         |     |     |     |
|                            |                    | 208                                                                    | 240 | 277 | 480 | 208                                                           | 240 | 277 | 480 | 208                                                                          | 240 | 480 | 208                                                                                                | 240 | 277 | 480 |
| 3000                       | 3000               | 14                                                                     | 13  | 11  | 6   | 29                                                            | 25  | 22  | 13  | 25                                                                           | 22  | 11  | SAME AS NON SIM,<br>BUT THERE WILL BE TWO<br>SETS OF INCOMING POWER.<br>THIS IS THE FACTORY WIRING |     |     |     |
| 4000                       | 4000               | 19                                                                     | 17  | 14  | 8   | 38                                                            | 33  | 29  | 17  | 33                                                                           | 29  | 14  |                                                                                                    |     |     |     |
| 4500                       | 4500               | 22                                                                     | 19  | 16  | 9   | 43                                                            | 38  | 32  | 19  | 37                                                                           | 32  | 16  |                                                                                                    |     |     |     |
| 5000                       | 5000               | 24                                                                     | 21  | 18  | 10  | 48                                                            | 42  | 36  | 21  | 42                                                                           | 36  | 18  |                                                                                                    |     |     |     |
| 6000                       | 6000               | 29                                                                     | 25  | 22  | 13  | 58                                                            | 50  | 43  | 25  | 50                                                                           | 43  | 22  |                                                                                                    |     |     |     |
| 6050                       | 6050               |                                                                        |     |     |     | 58                                                            | 50  | 44  | 25  | 50                                                                           | 44  | 22  |                                                                                                    |     |     |     |

| RECOVERY CAPACITIES (Recovery in U.S. Gallons/Hr. (GPH) and Liters/Hr. (LPH) at various temperature rises) |                         |     |               |     |               |     |                |     |                |     |                     |     |               |     |               |     |                |     |                |     |
|------------------------------------------------------------------------------------------------------------|-------------------------|-----|---------------|-----|---------------|-----|----------------|-----|----------------|-----|---------------------|-----|---------------|-----|---------------|-----|----------------|-----|----------------|-----|
| ELEMENT<br>WATTAGE<br>UPPER/LOWER                                                                          | NON-SIMULTANEOUS WIRING |     |               |     |               |     |                |     |                |     | SIMULTANEOUS WIRING |     |               |     |               |     |                |     |                |     |
|                                                                                                            | 40°F / (22°C)           |     | 60°F / (33°C) |     | 80°F / (45°C) |     | 100°F / (56°C) |     | 120°F / (67°C) |     | 40°F / (22°C)       |     | 60°F / (33°C) |     | 80°F / (45°C) |     | 100°F / (56°C) |     | 120°F / (67°C) |     |
|                                                                                                            | GPH                     | LPH | GPH           | LPH | GPH           | LPH | GPH            | LPH | GPH            | LPH | GPH                 | LPH | GPH           | LPH | GPH           | LPH | GPH            | LPH | GPH            | LPH |
| 3000/3000                                                                                                  | 30                      | 115 | 20            | 77  | 15            | 58  | 12             | 46  | 10             | 38  | 61                  | 230 | 41            | 153 | 30            | 115 | 24             | 92  | 20             | 77  |
| 4000/4000                                                                                                  | 41                      | 153 | 27            | 102 | 20            | 77  | 16             | 61  | 14             | 51  | 81                  | 307 | 54            | 205 | 41            | 153 | 32             | 123 | 27             | 102 |
| 4500/4500                                                                                                  | 46                      | 173 | 30            | 115 | 23            | 86  | 18             | 69  | 15             | 58  | 91                  | 345 | 61            | 230 | 46            | 173 | 36             | 138 | 30             | 115 |
| 5000/5000                                                                                                  | 51                      | 192 | 34            | 128 | 25            | 96  | 20             | 77  | 17             | 64  | 101                 | 384 | 68            | 256 | 51            | 192 | 41             | 153 | 34             | 128 |
| 6000/6000                                                                                                  | 61                      | 230 | 41            | 153 | 30            | 115 | 24             | 92  | 20             | 77  | 122                 | 460 | 81            | 307 | 61            | 230 | 49             | 184 | 41             | 153 |
| 6050/6050                                                                                                  |                         |     |               |     |               |     |                |     |                |     | 122                 | 460 | 81            | 307 | 61            | 230 | 49             | 184 | 41             | 153 |

Consult factory for certification requirements.

### Recommended Specifications (for trade reference only)

Water heater(s) shall be model \_\_\_\_\_, manufactured by Rheem, having electrical input of \_\_\_\_\_ kW and a recovery rate of \_\_\_\_\_ GPH at a 100°F temperature rise. Water heater(s) shall have a storage capacity of \_\_\_\_\_ gallons. Water heater(s) shall have the UL seal of certification and be factory equipped with an CSA/ ASME rated temperature and pressure relief valve. Tank(s) interior shall be coated with a high temperature porcelain enamel and furnished with

a magnesium anode rod rigidly supported. Water heater(s) shall meet or exceed the energy factor requirements of ASHRAE. Tanks shall have a working pressure rating of 150 psi, and shall be completely assembled. Water heater(s) shall be equipped with copper, resistored, "screw-in" type elements. Tank shall be insulated with 2-1/2" of rigid polyurethane foam insulation. Water heater(s) shall be equipped with surface mounted thermostats each with an integral, manual reset, high limit control. Water heater(s) shall be covered by a three year limited warranty against tank leaks.

In keeping with its policy of continuous progress and product improvement, Rheem reserves the right to make changes without notice.

Rheem Water Heating • 1115 Northmeadow Parkway, Suite 100  
Roswell, Georgia 30076 • www.rheem.com



INTEGRATED AIR & WATER



1918115

# GAS METER



**ONICON**

Flow and Energy Measurement

# F-5500

## THERMAL MASS FLOW METER

ONICON's  
F-5500 Series  
Thermal Mass  
Flow Meters  
provide  
accurate and  
reliable flow  
measurement  
for natural gas,  
compressed  
air, and other  
industrial gas  
applications.



2496658

• Natural Gas • Compressed Air • Industrial Gases •

## F-5500

### THERMAL MASS FLOW METER



2496658



### DESCRIPTION

ONICON's F-5000 Series Thermal Mass Flow Meters provide accurate mass flow measurement of natural gas, compressed air and other industrial gases. The proprietary sensor design measures mass flow directly and does not require additional pressure or temperature compensation to deliver accurate flow rate and total data.

The F-5500 is available as an inline or an insertion style meter and includes an easy to operate user interface/display. The standard version of the meter is provided with a 4-20 mA analog output and an RS485 interface that is field configurable for BACnet® MS/TP or Modbus® RTU. A second output signal configuration is available that includes a 4-20 mA analog output and a programmable pulse output. The pulse output model is also available with HART.

### APPLICATIONS

Accurate sub-metering of natural gas & propane for:

- Tenant space usage
- Boiler efficiency
- Campus monitoring

Also ideal for monitoring:

- Compressed Air
- Medical gases
- Other industrial gases

### CALIBRATION

Every ONICON flow meter is wet calibrated in a flow laboratory against standards that are directly traceable to NIST\*. A certificate of calibration accompanies every meter.

\* National Institute of Standards and Technology

\*\* Installations must comply with federal, state and municipal building codes. Review all proposed combustible gas installations with your local code enforcement officials before attempting to install.

### FEATURES

**BACnet MS/TP or MODBUS RTU** - The standard F-5500 includes an RS485 output that provides BACnet MS/TP or Modbus RTU. Data reported to the network includes flow rate and total, temperature and elapsed time since reset.

**User Friendly Interface / Display** - The bright, easy-to-read, backlit display and intuitive menu structure simplify page navigation and allow for field programming. Free utility software is also available for programming and data logging.

**Provides for Field Validation of Calibration** - F-5500 internal diagnostic functions include a zero flow calibration check. This fast, easy to perform test allows for field validation of the factory zero flow calibration. The utility software provided with the meter allows you to print a certificate validating the test results.

**Insertion Meters Can Be Installed Without Interrupting Gas Service\*\*** - ONICON's hot tap design allows for installation without interruption to the gas service. The meter can also be removed for service without disrupting flow.

**Highly Accurate Over a Wide Operating Range** - Our proprietary direct digital control sensing circuitry is very stable yet highly responsive to changes in flow. This design allows for accurate flow measurement over a very wide operating range (over 1000:1 for the inline version). It also makes the meter ideal for measuring low flow rates.

**Excellent Value** - ONICON insertion style meters are accurate, easy-to-use and reliable. They are also priced independently of pipe size. This makes them an excellent value, particularly in larger diameter pipes.



# F-5500

## THERMAL MASS FLOW METER



### SPECIFICATIONS\*

| F-5500 THERMAL MASS FLOW METER                |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FLOW ACCURACY                                 | NATURAL GAS/PROPANE GAS<br>±1.0% of reading from 500 – 7,000 SFPM (14:1 turndown)<br>±2.0% of reading from 100 – 500 SFPM (5:1 turndown)<br>COMPRESSED AIR & OTHER HIGH VELOCITY CALIBRATIONS<br>±1.0% of reading ± 0.5% of full scale over a 100:1 turndown                                                                                                                                                                                       |
| TEMPERATURE ACCURACY                          | ±1.0°F over the range of -40°F to 250°F                                                                                                                                                                                                                                                                                                                                                                                                            |
| OVERALL FLOW RANGE                            | 15 - 35,000 SFPM                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| SENSING METHOD                                | Thermal mass flow utilizing direct digital control sensing circuitry                                                                                                                                                                                                                                                                                                                                                                               |
| PIPE SIZE RANGE                               | INSERTION STYLE<br>1½ - 24" nominal diameter<br>INLINE STYLE<br>¾ - 6" nominal diameter<br>Note: Inline meters larger than 2.5" are shipped disassembled in two separate boxes.                                                                                                                                                                                                                                                                    |
| INPUT POWER                                   | 12 - 28 VDC, 6W minimum power                                                                                                                                                                                                                                                                                                                                                                                                                      |
| TEMPERATURE RANGE                             | FLUID<br>-40°F to 250°F<br>AMBIENT<br>-40°F to 158°F                                                                                                                                                                                                                                                                                                                                                                                               |
| MAXIMUM OPERATING PRESSURE                    | INSERTION STYLE<br>Process adapter fitting - 60 psig (4.1 barg) max<br>High pressure adapter fitting - 150 psig (10.3 barg) max<br>INLINE STYLE<br>ANSI Class 150 flanges - 230 psig at 100°F (16 barg)<br>NPT - 300 psig (20.7 barg)<br>All stainless steel ferrules                                                                                                                                                                              |
| PRESSURE DROP (at 2500 SFPM, 70°F and 2 PSIG) | INSERTION STYLE<br>Less than ½" W.C. (H2O) in 1½" diameter pipes, decreasing in larger pipes<br>INLINE STYLE (with built-in flow conditioner)<br>Less than ½" W.C (H2O) in 2" and larger diameter meters<br>Less than 0.9" W.C (H2O) in 1" and 1½" diameter meters                                                                                                                                                                                 |
| PROGRAMMING/MEMORY                            | Factory programmed for specific application. Field programming available through mini-USB interface and utility program.<br>Non-volatile memory retains all program parameters and totalized values in the event of power loss.                                                                                                                                                                                                                    |
| OUTPUT SIGNALS PROVIDED                       | Analog output: 4-20 mA<br>Select from one of the following options as the second output: <ul style="list-style-type: none"> <li>• RS485 interface: BACnet MS/TP or Modbus RTU (field selectable)</li> <li>• Programmable pulse output: Field selectable as scaled pulse or alarm (Isolated open collector output)</li> <li>• 4-20 mA with HART FSK (Only available with programmable pulse output)</li> </ul>                                      |
| ELECTRONICS ENCLOSURE                         | NEMA 4X Weathertight aluminum enclosure                                                                                                                                                                                                                                                                                                                                                                                                            |
| ELECTRICAL CONNECTIONS                        | Enclosed terminal blocks, cable access through two ¾" NPT conduit fittings                                                                                                                                                                                                                                                                                                                                                                         |
| MATERIAL                                      | Wetted metal components: 316 stainless steel                                                                                                                                                                                                                                                                                                                                                                                                       |
| APPROVALS                                     | FM (USA) FMc (CAN): Approved<br>Class 1, Div 1, Groups B, C, D;<br>Class 2, Div 1, Groups E, F, G;<br>Class 3, Div 1; T4, Ta = -40°C to 70°C;<br>Class 1, Zone 1, AEx/Ex db IIB = H2 T4; Gb Ta = -40°C to 70°C;<br>Type 4X, IP66/67<br>EMC Directive; 2014/30/EU<br>Emissions and Immunity Testing: EN61326-1:2013<br>Massachusetts Board of State Examiners of Plumbers and Gasfitters<br>Canadian Registration Number (CRN) according to CSA BS1 |

\* SPECIFICATIONS subject to change without notice.

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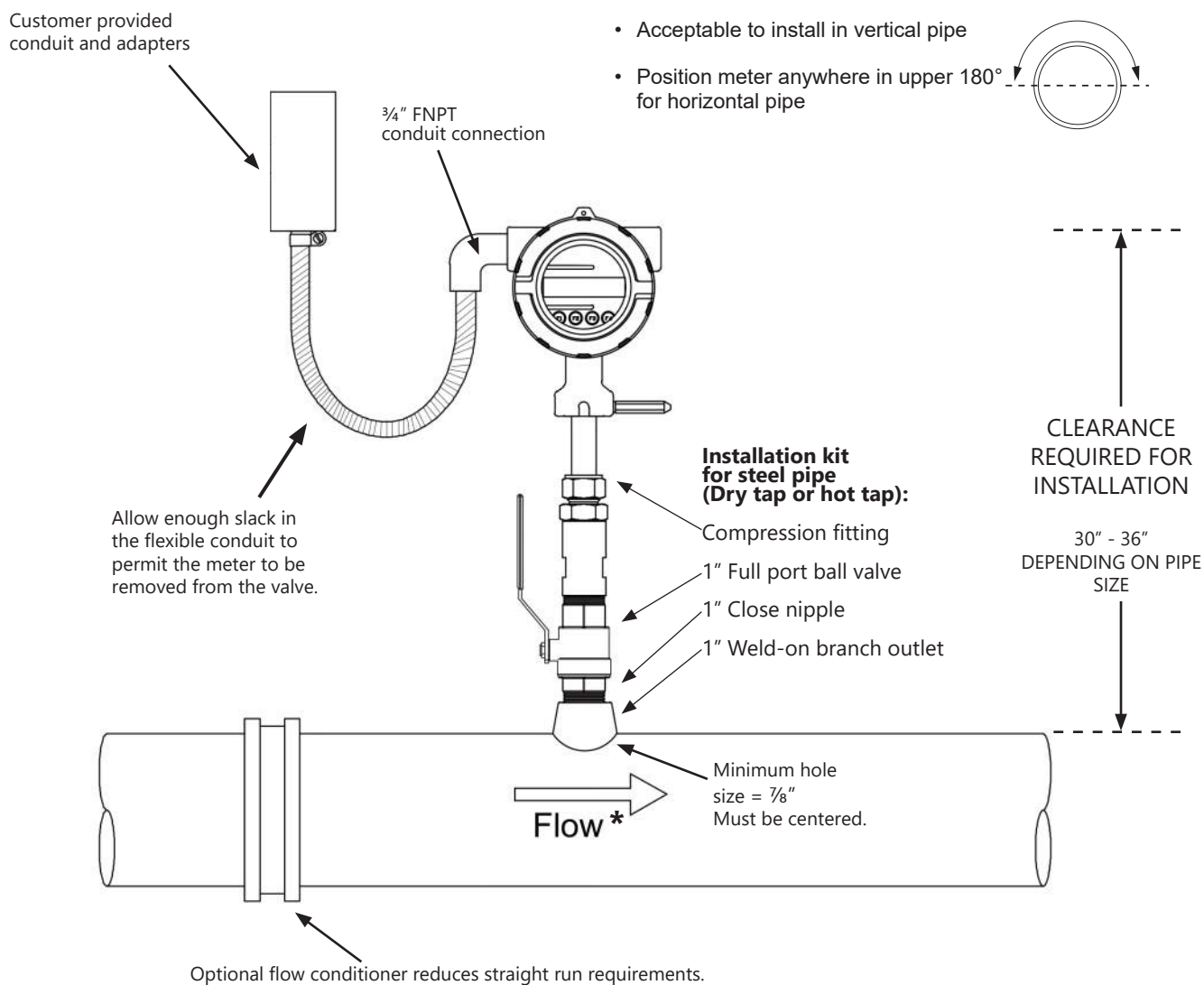


# F-5500

## THERMAL MASS FLOW METER



### TYPICAL INSERTION METER INSTALLATION



2496658

\*Standard orientation. Contact ONICON for other options.

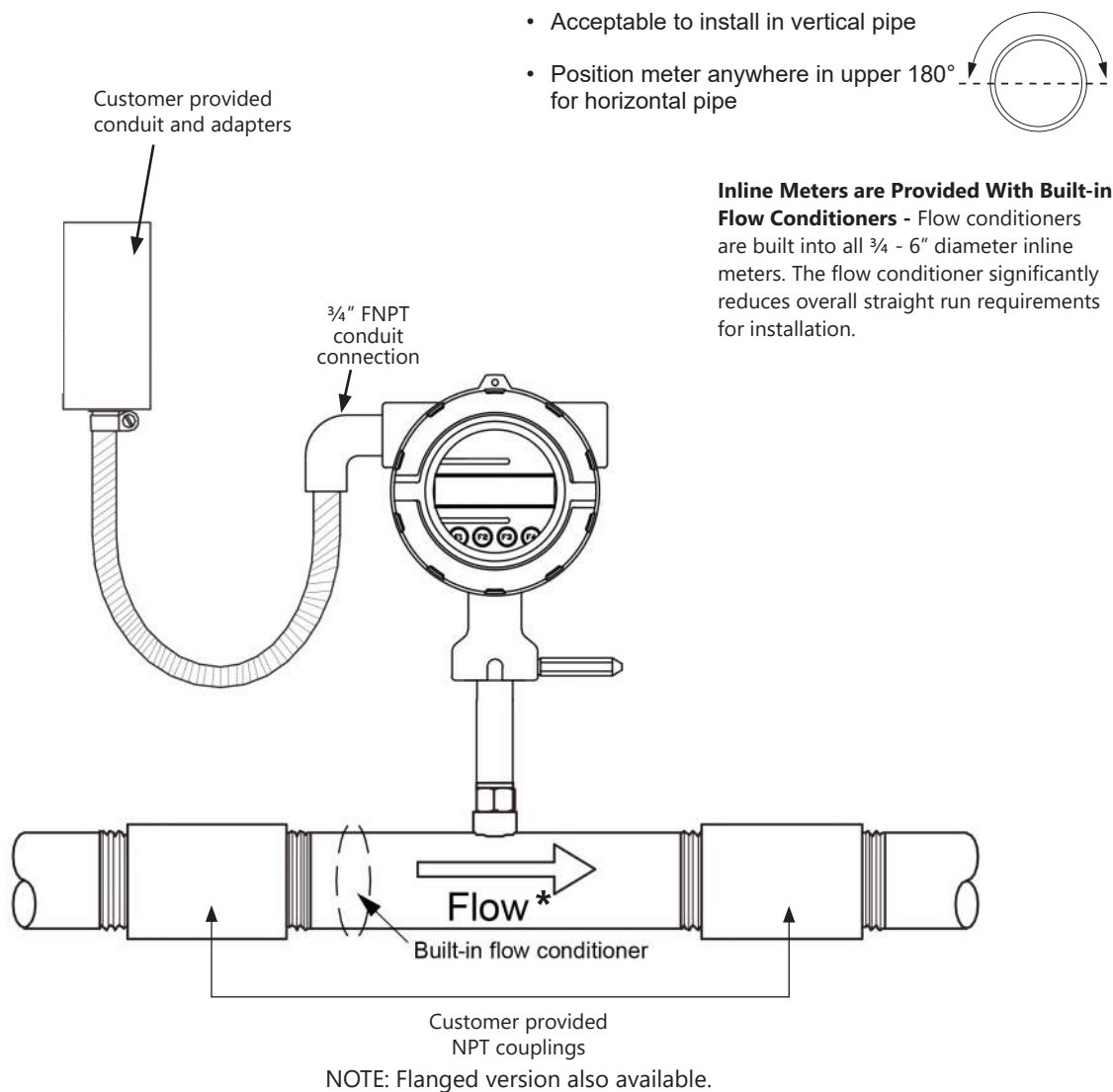


# F-5500

## THERMAL MASS FLOW METER



### TYPICAL INLINE METER INSTALLATION



2496658

\*Standard orientation. Contact ONICON for other options.



# F-5500

## THERMAL MASS FLOW METER



### METER ORDERING INFORMATION

#### Meter Model Number Coding = F-55AA-BCDE-FGGH

#### F-55 = Thermal Mass Flow Meter with Display

##### AA = Pipe Diameter

|                |             |
|----------------|-------------|
| 00 = Insertion | 02 = 2"     |
| 34 = 3/4"      | 25 = 2 1/2" |
| 01 = 1"        | 03 = 3"     |
| 13 = 1 1/4"    | 04 = 4"     |
| 15 = 1 1/2"    | 06 = 6"     |

##### B = Output Signals

- 2 = Loop powered 4-20 mA & scaled pulse
- 3 = Loop powered 4-20 mA & RS485
- 4 = Loop powered 4-20 mA with HART & scaled pulse

##### C = Input Power

- 1 = 12 - 28 VDC

##### D = Electronics Enclosure Mounting Configuration

- 1 = Integral enclosure with LCD

##### E = Process Connection

- 4 = Insertion
- 5 = Threaded MNPT (3/4 - 3" pipe sizes only)
- 6 = ANSI Class 150 flanges

##### F = Flow Conditioner

- 1 = Insertion without flow conditioner
- 2 = Insertion with flow conditioner
- 3 = Inline with flow conditioner

##### GG = Pipe Size Range

- 00 = Inline Meter
- 15 = 1 1/2 - 6" nominal diameter
- 18 = >6" nominal diameter

##### H = Process Adapter Fitting

- 0 = Standard (0-60 psig)
- 1 = High pressure (0-150 psig)
- 9 = Inline meter

##### GAS TYPE

|                  |                     |
|------------------|---------------------|
| NG = Natural Gas | HE = Helium Gas     |
| ME = Methane Gas | NI = Nitrogen Gas   |
| PG = Propane Gas | AR = Argon Gas      |
| AI = Air         | CD = Carbon Dioxide |
| O2 = Oxygen Gas  | BU = Butane         |
| HY = Hydrogen    |                     |

### ACCESSORY ORDERING INFORMATION

#### Install Kits for Carbon Steel Piping Systems

| Model Number  | Description                                                                                 |
|---------------|---------------------------------------------------------------------------------------------|
| INSTL0094-FMH | Insertion flow meter hot tap installation kit, wetted materials are bronze, brass and steel |

#### OPERATING RANGE FOR COMMON NATURAL GAS PIPE SIZES 15 to 7,000 SFPM in schedule 40 pipe

| Pipe Size<br>(Inches) | Flow Rate (SCFH) |         |
|-----------------------|------------------|---------|
|                       | Min              | Max     |
| 3/4                   | 3.3              | 1,560   |
| 1                     | 5.4              | 2,521   |
| 1 1/4                 | 9.3              | 4,362   |
| 1 1/2                 | 13               | 5,938   |
| 2                     | 21               | 9,740   |
| 2 1/2                 | 30               | 13,964  |
| 3                     | 46               | 21,562  |
| 4                     | 80               | 37,130  |
| 5                     | 125              | 58,350  |
| 6                     | 181              | 84,263  |
| 8                     | 313              | 145,912 |

#### OPERATING RANGE FOR COMMON COMPRESSED AIR PIPE SIZES 15 to 35,000 SFPM in schedule 40 pipe

| Pipe Size<br>(Inches) | Flow Rate (SCFM) |        |
|-----------------------|------------------|--------|
|                       | Min              | Max    |
| 3/4                   | 0.06             | 130    |
| 1                     | 0.09             | 210    |
| 1 1/4                 | 0.16             | 364    |
| 1 1/2                 | 0.21             | 395    |
| 2                     | 0.35             | 816    |
| 2 1/2                 | 0.5              | 1,160  |
| 3                     | 0.77             | 1,800  |
| 4                     | 1.33             | 3,090  |
| 5                     | 2.08             | 4,860  |
| 6                     | 3                | 7,020  |
| 8                     | 5.2              | 12,200 |
| 10                    | 8.2              | 19,200 |



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# ONICON INSTALLATION HARDWARE SELECTION GUIDE

ONICON installation kits are specifically designed for use with ONICON insertion flow and BTU (energy) meters. Their use is recommended with ONICON products.



*F-1000 Series Turbine Flow Meter with INST0005-FMD Dry Ta*

2052116



## INSTALLATION HARDWARE SELECTION GUIDE

### How to use this selection guide?

1. Select the type of installation hardware required, either Dry Tap or Hot Tap.
2. Choose the pipe material.
3. Choose the Installation Kit based on material preference and/or pipe size.

### What does Dry Tap or Hot Tap mean?

The terms "Dry Tap" and "Hot Tap" refer to the installation method of the isolation valve kit only.

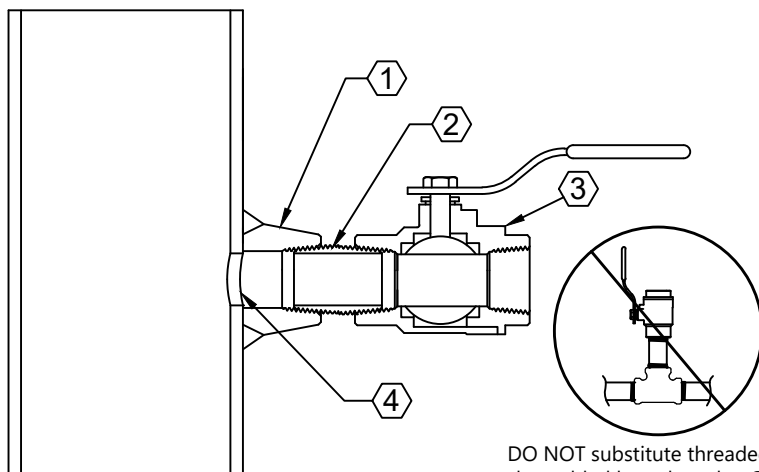
### Dry Tap Installation Hardware

For new construction or scheduled shut down, the installation kit is installed into a dry, or unfilled system.

### Hot Tap Installation Hardware

For installations which require the access hole in the pipe to be drilled through the installation valve, via a hot tap drilling machine, while the hydronic system is full and pressurized.

Below is an example of Dry Tap Installation Kit for Welded Steel Pipe:



1. 1" NPT welded branch outlet, typical for steel piping system installation.
2. 1" threaded close nipple
3. 1" full port ball valve
4. 1" minimum dia. access hole

DO NOT substitute threaded tees for the welded branch outlet. Contact ONICON if you need installation hardware for threaded pipe.

### Important Note:

**Installation hardware materials vary based on pipe material, pipe size and dry tap vs hot tap versions.**

See next pages for the complete list of installation kits offered by ONICON. If you have any questions or need assistance, please contact ONICON at (727) 447-6 or sales@onicon.com.



## INSTALLATION HARDWARE SELECTION GUIDE

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# INSTALLATION HARDWARE SELECTION GUIDE

## DRY TAP INSTALLATION HARDWARE



For F-1000 Series, FSM-3 and F-3500 Flow Meters

| Pipe Material                                           | Model Number  | Description                                                                                                                                                              | Items Included with Kit                                                                                                                                                      | Notes                                                 |
|---------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Carbon Steel/<br/>Black Iron<br/>(welded pipe)</b>   | INSTL0001-FMD | Insertion flow meter installation kit for pipe size range from 1.25" to 42". Wetted materials are bronze, brass and steel, for use in carbon steel piping systems.       | <ul style="list-style-type: none"> <li>1" Full port bronze ball valve</li> <li>1" Brass close nipple</li> <li>1" Weld-on carbon steel branch outlet</li> </ul>               | Designed for use in hydronic systems.                 |
|                                                         | INSTL01DW-FMD | Insertion flow meter installation kit for pipe size range from 1.25" to 42". Wetted materials are bronze, CU and low lead brass, for use in carbon steel piping systems. | <ul style="list-style-type: none"> <li>1" Full port bronze ball valve</li> <li>1" Brass close nipple</li> <li>1" Weld-on carbon steel branch outlet</li> </ul>               | NSF certified for use in domestic water applications. |
|                                                         | INSTL0005-FMD | Insertion flow meter installation kit for pipe size range from 1.25" to 42". Wetted materials are SS, for use in carbon steel piping systems.                            | <ul style="list-style-type: none"> <li>1" Full port SS ball valve</li> <li>1" SS close nipple</li> <li>1" Weld-on carbon steel branch outlet</li> </ul>                      | Designed for use in hydronic systems.                 |
| <b>Carbon Steel/<br/>Black Iron<br/>(threaded pipe)</b> | INSTL0007-FMD | Insertion flow meter installation kit for pipe size range from 1.25" to 2.5". Wetted materials are bronze, brass and steel, for use in carbon steel piping systems.      | <ul style="list-style-type: none"> <li>1" Full port bronze ball valve</li> <li>1" Brass close nipple</li> <li>(Pipe dia x 12" long) threaded nipple 1" NPT outlet</li> </ul> | Designed for use in hydronic systems.                 |
|                                                         | INSTL0008-FMD | Insertion flow meter installation kit for pipe size 1.25" to 2.5". Wetted materials are bronze, brass and steel, for use in carbon steel piping systems.                 | <ul style="list-style-type: none"> <li>1" Full port SS ball valve</li> <li>1" SS close nipple</li> <li>(Pipe dia x 12" long) threaded nipple 1" NPT outlet</li> </ul>        | Designed for use in hydronic systems.                 |
| <b>Copper<br/>Tube</b>                                  | INSTL0003-FMD | Insertion flow meter installation kit for pipe size range from 1.25" to 2". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems.        | <ul style="list-style-type: none"> <li>1" Full port bronze ball valve</li> <li>1" Copper street adapter with MNPT threads</li> <li>Copper tee with 1" outlet</li> </ul>      | NSF certified for use in domestic water applications. |
|                                                         | INSTL0004-FMD | Insertion flow meter installation kit for pipe size range from 2.5" to 3". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems.         | <ul style="list-style-type: none"> <li>1" Full port bronze ball valve</li> <li>1" Copper street adapter with MNPT threads</li> <li>Copper tee with 1" outlet</li> </ul>      | NSF certified for use in domestic water applications. |
|                                                         | INSTL0009-FMD | Insertion flow meter installation kit for pipe size 4". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems.                            | <ul style="list-style-type: none"> <li>1" Full port bronze ball valve</li> <li>1" Copper street adapter with MNPT threads</li> <li>Copper tee with 1" outlet</li> </ul>      | NSF certified for use in domestic water applications. |
| <b>Stainless<br/>Steel</b>                              | INSTL0018-FMD | Insertion flow meter installation kit for pipe size range from 1.25" to 42". Wetted materials are SS, for use in SS piping systems.                                      | <ul style="list-style-type: none"> <li>1" Full port SS ball valve</li> <li>1" SS close nipple</li> <li>1" SS weld-on branch outlet</li> </ul>                                | Designed for use in process water applications.       |

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## INSTALLATION HARDWARE SELECTION GUIDE

### DRY TAP INSTALLATION HARDWARE



For System-10, System-20 and F-4300 BTU Meters (For use with current (mA) based temperature sensors)

| Pipe Material                       | Model Number  | Description                                                                                                                                                              | Items Included with Kit                                                                                                                                                                                                                     | Notes                                                                 |
|-------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Steel Pipe Welded</b>            | INSTL0032-TSD | Temperature sensor (pair) installation kit for pipe size range from 0.75" to 5". Wetted materials are bronze, brass and steel, for use in carbon steel piping systems.   | <ul style="list-style-type: none"> <li>• (2) Brass thermowells</li> <li>• (2) Weld-on carbon steel branch outlets, ½" NPT</li> <li>• (2) Transmitter enclosures</li> </ul>                                                                  | Designed for use in hydronic systems.                                 |
|                                     | INSTL0034-TSD | Temperature sensor (pair) installation kit for pipe size range from 0.75" to 42". Wetted materials are SS, for use in carbon steel piping systems.                       | <ul style="list-style-type: none"> <li>• (2) SS thermowells</li> <li>• (2) Weld-on carbon steel branch outlets, ½"</li> <li>• (2) Transmitter enclosures</li> </ul>                                                                         | Designed for use in hydronic systems.                                 |
|                                     | INSTL0066-TSD | Temperature sensor (pair) installation kit for pipe size range from 1.5" to 42". Wetted materials are SS, for use in carbon steel piping systems.                        | <ul style="list-style-type: none"> <li>• (2) SS thermowells</li> <li>• (2) Weld-on carbon steel branch outlets, 1"</li> <li>• (2) High temperature transmitter enclosure assemblies</li> </ul>                                              | Designed for use in hydronic systems, and in HTHW systems (T ≥ 300°F) |
| <b>Steel Pipe Threaded</b>          | INSTL0052-TSD | Temperature sensor (pair) installation kit for pipe size range from 0.75" to 2.5". Wetted materials are bronze, brass and steel, for use in carbon steel piping systems. | <ul style="list-style-type: none"> <li>• (2) Brass thermowells</li> <li>• (2) Brass reducer bushings, ½"</li> <li>• (2) Transmitter enclosures</li> </ul>                                                                                   | Designed for use in hydronic systems.                                 |
| <b>Steel /PVC Pipe with Saddles</b> | INSTL0045-TSD | Temperature sensor (pair) installation kit for pipe size range from 1.5" to 8". Wetted materials are SS, for use in PVC piping systems.                                  | <ul style="list-style-type: none"> <li>• (2) SS thermowells</li> <li>• (2) Brass reducer bushings, 1" x ½"</li> <li>• (2) Ductile iron saddles for Sch std./40/80 pipe with 1" NPT outlets</li> <li>• (2) Transmitter enclosures</li> </ul> | Designed for use in hydronic systems.                                 |

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## INSTALLATION HARDWARE SELECTION GUIDE

### DRY TAP INSTALLATION HARDWARE



For System-10, System-20 and F-4300 BTU Meters (For use with current (mA) based temperature sensors)

| Pipe Material                          | Model Number  | Description                                                                                                                                                      | Items Included with Kit                                                                                                                                                                                | Notes                                           |
|----------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>Copper Tees<br/>(brass wells)</b>   | INSTL0036-TSD | Temperature sensor (pair) installation kit for pipe size range from 0.75" to 2". Wetted materials are bronze, brass and steel, for use in copper piping systems. | <ul style="list-style-type: none"> <li>• (2) Copper tees with 1/2" outlet</li> <li>• (2) Brass thermowells</li> <li>• (2) Copper street adapter, 1/2"</li> <li>• (2) Transmitter enclosures</li> </ul> | Designed for use in hydronic systems.           |
|                                        | INSTL0037-TSD | Temperature sensor (pair) installation kit for pipe size range from 2.5" to 3". Wetted materials are bronze, brass and steel, for use in copper piping systems.  | <ul style="list-style-type: none"> <li>• (2) Copper tees with 1/2" outlet</li> <li>• (2) Brass thermowells</li> <li>• (2) Copper street adapter, 1/2"</li> <li>• (2) Transmitter enclosures</li> </ul> | Designed for use in hydronic systems.           |
|                                        | INSTL0038-TSD | Temperature sensor (pair) installation kit for pipe size 4". Wetted materials are bronze, brass and steel, for use in copper piping systems.                     | <ul style="list-style-type: none"> <li>• (2) Copper tees with 1/2" outlet</li> <li>• (2) Brass thermowells</li> <li>• (2) Copper street adapter, 1/2"</li> <li>• (2) Transmitter enclosures</li> </ul> | Designed for use in hydronic systems.           |
| <b>Copper Tees<br/>(SS wells)</b>      | INSTL0083-TSD | Temperature sensor (pair) installation kit for pipe size range from 0.75" to 2". Wetted materials are SS, for use in copper piping systems.                      | <ul style="list-style-type: none"> <li>• (2) Copper tees with 1/2" outlet</li> <li>• (2) SS thermowells</li> <li>• (2) Copper street adapter, 1/2"</li> <li>• (2) Transmitter enclosures</li> </ul>    | Designed for use in hydronic systems.           |
|                                        | INSTL0086-TSD | Temperature sensor (pair) installation kit for pipe size range from 2.5" to 3". Wetted materials are SS, for use in copper piping systems.                       | <ul style="list-style-type: none"> <li>• (2) Copper tees with 1/2" outlet</li> <li>• (2) SS thermowells</li> <li>• (2) Copper street adapter, 1/2"</li> <li>• (2) Transmitter enclosures</li> </ul>    | Designed for use in hydronic systems.           |
| <b>Stainless Steel<br/>Pipe Welded</b> | INSTL0061-TSD | Temperature sensor (pair) installation kit for pipe size range from 0.75" to 42". Wetted materials are SS, for use in SS piping systems.                         | <ul style="list-style-type: none"> <li>• (2) SS thermowells</li> <li>• (2) Weld-on stainless steel branch outlets, 1/2"</li> <li>• (2) Transmitter enclosures</li> </ul>                               | Designed for use in process water applications. |

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# INSTALLATION HARDWARE SELECTION GUIDE

## DRY TAP INSTALLATION HARDWARE



For System-20, System-40 and F-4300 BTU Meters (For use with RTD temperature sensor pair)

| Pipe Material               | Model Number  | Description                                                                                                                                                                                                      | Items Included with Kit                                                                                                           | Notes                                                                                                                                                                                                                                                |
|-----------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Steel Pipe Threaded</b>  | INSTL4001-TSD | Temperature sensor (pair) installation kit for pipe size range from 0.5" to 1". Wetted material is low lead brass, for use in carbon steel piping systems. NSF certified for use in domestic water applications. | <ul style="list-style-type: none"> <li>(2) Lead free brass NPT thermowells</li> </ul>                                             | NPT threads sizes will match meter size. For use with customer supplied threaded steel tees. Suitable for use with SYS-40 where G=1 only. Refer to System-40 Selection and Ordering Guide for the meter ordering information.                        |
|                             | INSTL4002-TSI | Temperature sensor installation kit for pipe size range from 0.5" to 1". Wetted material is low lead brass, for use in carbon steel piping systems. Designed for use in hydronic systems.                        | <ul style="list-style-type: none"> <li>(1) Lead free brass M10 x NPT reducer bushing</li> </ul>                                   | NPT thread size will match meter size. For use with customer supplied threaded steel tee. Suitable for use with SYS-40 where temperature sensor option G=0 only. Refer to System-40 Selection and Ordering Guide for the meter ordering information. |
| <b>Copper Tube</b>          | INSTL4003-TSD | Temperature sensor (pair) installation kit for pipe size range from 0.5" to 1". Wetted material is low lead brass, for use in copper piping systems. NSF certified for use in domestic water applications.       | <ul style="list-style-type: none"> <li>(2) Lead free brass flush thermowells</li> </ul>                                           | Bushing sizes will match meter size. For use with customer supplied copper tees. Suitable for use with SYS-40 where G=1 only. Refer to System-40 Selection and Ordering Guide for the meter ordering information.                                    |
|                             | INSTL4004-TSI | Temperature sensor installation kit for pipe size range from 0.5" to 1". Wetted material is low lead brass, for use in copper piping systems. Designed for use in hydronic systems.                              | <ul style="list-style-type: none"> <li>(1) Lead free brass M10 x flush reducer bushing</li> </ul>                                 | Bushing size will match meter size. For use with customer supplied copper tee. Suitable for use with SYS-40 where temperature sensor option G=0 only. Refer to System-40 Selection and Ordering Guide for the meter ordering information.            |
| <b>Steel / PVC / Copper</b> | INSTL4005-TSD | Temperature sensor (pair) installation kit for pipe size range from 0.5" to 2.5". Wetted material is low lead brass. NSF certified for use in domestic water applications.                                       | <ul style="list-style-type: none"> <li>(2) 5.0 mm dia., lead free, brass, 1/2" NPT thermowells</li> <li>(2) Set screws</li> </ul> | Suitable for use with with F-4300, SYS-20, and SYS-40 (where G =2) BTU meters. Refer to SYS-40 or F-4300 Ordering Guides for the meter ordering information.                                                                                         |

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## INSTALLATION HARDWARE SELECTION GUIDE

### DRY TAP INSTALLATION HARDWARE



For System-20, System-40 and F-4300 BTU Meters (For use with RTD temperature sensor pair)

| Pipe Material                | Model Number  | Description                                                                                                                                                                                                                    | Items Included with Kit                                                                                                                                                                                                  | Notes                                                                                                            |
|------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Copper Tube</b>           | INSTL4007-TSD | Temperature sensor (pair) installation kit for pipe size range from 1.25" to 2.5". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems. NSF certified for use in domestic water applications. | <ul style="list-style-type: none"> <li>• (2) 5.0 mm dia., lead free, brass, 1/2" NPT thermowells</li> <li>• (2) 1"x 1/2" street adapters</li> <li>• (2) Set screws</li> </ul>                                            | For use with customer supplied tees with 1" outlets. Suitable for use with F-4300, SYS-20 and SYS-40 BTU meters. |
| <b>Kits with Copper Tees</b> | INSTL4008-TSD | Temperature sensor (pair) installation kit for pipe size range from 1.25" to 2". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems. NSF certified for use in domestic water applications.   | <ul style="list-style-type: none"> <li>• (2) 5.0 mm dia., lead free, brass, 1/2" NPT thermowells</li> <li>• (2) 1"x 1/2" street adapters</li> <li>• (2) Copper tees with 1" outlets</li> <li>• (2) Set screws</li> </ul> | Suitable for use with F-4300, SYS-20 and SYS-40 BTU meters.                                                      |
|                              | INSTL4009-TSD | Temperature sensor (pair) installation kit for pipe size 2.5". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems. NSF certified for use in domestic water applications.                     | <ul style="list-style-type: none"> <li>• (2) 5.0 mm dia., lead free, brass, 1/2" NPT thermowells</li> <li>• (2) 1"x 1/2" street adapters</li> <li>• (2) Copper tees with 1" outlets</li> <li>• (2) Set screws</li> </ul> | Suitable for use with F-4300, SYS-20 and SYS-40 BTU meters.                                                      |
| <b>Steel</b>                 | INSTL4021-TSD | Temperature sensor (pair) installation kit for pipe size range from 3" to 48". Wetted materials are SS, for use in carbon steel piping systems. Designed for use in hydronic systems.                                          | <ul style="list-style-type: none"> <li>• (2) 6.0 mm dia., SS, 1/2" NPT thermowells</li> <li>• (2) Carbon steel 1/2" NPT branch outlets</li> </ul>                                                                        | Suitable for use with F-4300 and SYS-20 BTU meters.                                                              |

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# INSTALLATION HARDWARE SELECTION GUIDE

## HOT TAP INSTALLATION HARDWARE



For F-1000 Series, FSM-3 and F-3500 Flow Meters

| Pipe Material                                         | Model Number  | Description                                                                                                                                                                      | Items Included with Kit                                                                                                                                                                                                           | Notes                                                 |
|-------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Carbon Steel/<br/>Black Iron<br/>(welded pipe)</b> | INSTL0002-FMH | Insertion flow meter hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are bronze, brass and steel, for use in carbon steel piping systems.       | <ul style="list-style-type: none"> <li>• 1 1/4" Full port bronze ball valve</li> <li>• 1 1/4" Brass close nipple</li> <li>• 1 1/4" x 1" Brass reducing bushing</li> <li>• 1 1/4" Weld-on carbon steel branch outlet</li> </ul>    | Designed for use in hydronic systems.                 |
|                                                       | INSTL02DW-FMH | Insertion flow meter hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are bronze, CU and low lead brass, for use in carbon steel piping systems. | <ul style="list-style-type: none"> <li>• 1 1/4" Full port bronze ball valve</li> <li>• 1 1/4" SS close nipple</li> <li>• 1 1/4" x 1" SS reducing bushing</li> <li>• 1 1/4" Weld-on carbon steel branch outlet</li> </ul>          | NSF certified for use in domestic water applications. |
|                                                       | INSTL0006-FMH | Insertion flow meter hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are SS, for use in carbon steel piping systems.                            | <ul style="list-style-type: none"> <li>• 1 1/4" Full port SS ball valve</li> <li>• 1 1/4" SS close nipple</li> <li>• 1 1/4" x 1" SS reducing bushing</li> <li>• 1 1/4" Weld-on carbon steel branch outlet</li> </ul>              | Designed for use in hydronic systems.                 |
| <b>Carbon Steel or<br/>PVC Pipe<br/>w/Saddles</b>     | INSTL019A-FMH | Insertion flow meter hot tap installation kit for pipe size range from 1.5" to 6". Wetted materials are bronze, CU and low lead brass, for use in PVC piping systems.            | <ul style="list-style-type: none"> <li>• 1 1/4" Full port bronze ball valve</li> <li>• 1 1/4" Brass close nipple</li> <li>• 1 1/4" x 1" Brass reducing bushing</li> <li>• Ductile iron saddle with 1 1/4" outlet</li> </ul>       |                                                       |
|                                                       | INSTL019B-FMH | Insertion flow meter hot tap installation kit for pipe size range from 8" to 14". Wetted materials are bronze, CU and low lead brass, for use in PVC piping systems.             | <ul style="list-style-type: none"> <li>• 1 1/4" Full port bronze ball valve</li> <li>• 1 1/4" Brass close nipple</li> <li>• 1 1/4" x 1" Brass reducing bushing</li> <li>• Ductile iron saddle with 1 1/4" outlet</li> </ul>       |                                                       |
| <b>Copper Tube<br/>w/Saddles</b>                      | INSTL0021-FMH | Insertion flow meter hot tap installation kit for pipe size 2". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems.                            | <ul style="list-style-type: none"> <li>• 1 1/4" Full port bronze ball valve</li> <li>• 1 1/4" Brass close nipple</li> <li>• 1 1/4" x 1" Brass reducing bushing</li> <li>• Ductile iron or SS saddle with 1 1/4" outlet</li> </ul> | NSF certified for use in domestic water applications. |
|                                                       | INSTL0022-FMH | Insertion flow meter hot tap installation kit for pipe size range from 2.5" to 6". Wetted materials are bronze, CU and low lead brass, for use in copper piping systems.         | <ul style="list-style-type: none"> <li>• 1 1/4" Full port bronze ball valve</li> <li>• 1 1/4" Brass close nipple</li> <li>• 1 1/4" x 1" Brass reducing bushing</li> <li>• SS saddle with 1 1/4" outlet</li> </ul>                 | NSF certified for use in domestic water applications. |

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## INSTALLATION HARDWARE SELECTION GUIDE

### HOT TAP INSTALLATION HARDWARE



For F-1000 Series, FSM-3 and F-3500 Flow Meters

| Pipe Material          | Model Number  | Description                                                                                                                                                                  | Items Included with Kit                                                                                                                                                                                                    | Notes                                                                                                                                              |
|------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stainless Steel</b> | INSTL0031-FMH | Insertion flow meter hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are SS, for use in SS piping systems.                                  | <ul style="list-style-type: none"> <li>• 1 1/4" Full port 316 SS ball valve</li> <li>• 1 1/4" 316 SS close nipple</li> <li>• 1 1/4" x 1" 316 SS reducing bushing</li> <li>• 1 1/4" Weld-on 316 SS branch outlet</li> </ul> | Designed for use in process water applications.                                                                                                    |
| <b>HDPE/PPR</b>        | INSTL5001-FMH | Insertion flow meter hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are bronze, brass and steel, for use in HDPE/PPR piping systems.       | <ul style="list-style-type: none"> <li>• 1 1/4" Full port bronze ball valve</li> <li>• 1 1/4" Brass close nipple</li> <li>• 1 1/4" x 1" Brass reducing bushing</li> </ul>                                                  | Designed for use in hydronic systems.<br>For use with customer supplied branch/fusion outlet, total stack height not to exceed 7".                 |
|                        | INSTL5002-FMH | Insertion flow meter hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are bronze, CU and low lead brass, for use in HDPE/PPR piping systems. | <ul style="list-style-type: none"> <li>• 1 1/4" Full port bronze ball valve</li> <li>• 1 1/4" SS close nipple</li> <li>• 1 1/4" x 1" SS reducing bushing</li> </ul>                                                        | NSF certified for use in domestic water applications.<br>For use with customer supplied branch/fusion outlet, total stack height not to exceed 7". |
|                        | INSTL5003-FMH | Insertion flow meter hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are SS, for use in HDPE/PPR piping systems.                            | <ul style="list-style-type: none"> <li>• 1 1/4" Full port SS ball valve</li> <li>• 1 1/4" SS close nipple</li> <li>• 1 1/4" x 1" SS reducing bushing</li> </ul>                                                            | Designed for use in hydronic systems.<br>For use with customer supplied branch/fusion outlet, total stack height not to exceed 7".                 |

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# INSTALLATION HARDWARE SELECTION GUIDE

## HOT TAP INSTALLATION HARDWARE



### For F-2000 Series and F-1500 Flow Meters

| Pipe Material                                         | Model Number  | Description                                                                                                                                                  | Items Included with Kit                                                                                                                                                                                           | Notes                              |
|-------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>Carbon Steel/<br/>Black Iron<br/>(welded pipe)</b> | INSTL201S-FMH | Insertion flow meter hot tap installation kit for pipe size range from 2" to 42". Wetted materials are carbon steel, for use in carbon steel piping systems. | <ul style="list-style-type: none"> <li>• 2" Full port carbon steel class 150 flanged gate valve</li> <li>• 2" Carbon steel class 150 weld-neck flange</li> <li>• 2" Weld-on carbon steel branch outlet</li> </ul> | Designed for use in steam systems. |
|                                                       | INSTL203S-FMH | Insertion flow meter hot tap installation kit for pipe size range from 2" to 42". Wetted materials are carbon steel, for use in carbon steel piping systems. | <ul style="list-style-type: none"> <li>• 2" Full port carbon steel class 300 flanged gate valve</li> <li>• 2" Carbon steel class 300 weld-neck flange</li> <li>• 2" Weld-on carbon steel branch outlet</li> </ul> | Designed for use in steam systems. |

### For F-5000 Series Flow Meters

| Model Number  | Description                                                                                                                             | Notes                                              |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| INSTL0094-FMH | Insertion flow meter hot tap installation kit, wetted materials are bronze, brass and steel, for use in carbon steel piping systems.    | Designed for use in NG and compressed air systems. |
| INSTL0096-FMH | Insertion flow meter hot tap installation kit, wetted materials are SS, brass and carbon steel, for use in carbon steel piping systems. | Designed for use in NG and compressed air systems. |

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## INSTALLATION HARDWARE SELECTION GUIDE

### HOT TAP INSTALLATION HARDWARE



For System-10, System-20 and F-4300 BTU Meters (For use with current (mA) based temperature sensors)

| Pipe Material                         | Model Number  | Description                                                                                                                                                                     | Items Included with Kit                                                                                                                                                                                                                                                                                                             | Notes                                 |
|---------------------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Steel Pipe Welded</b>              | INSTL0033-TSH | Temperature sensor (pair) hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are bronze, brass and steel, for use in carbon steel piping systems. | <ul style="list-style-type: none"> <li>• (2) SS hot tap thermowells with brass hot tap adapters</li> <li>• (2) Full port bronze ball valves, 1"</li> <li>• (2) Brass close nipples</li> <li>• (2) Weld-on carbon steel branch outlets, 1" NPT</li> <li>• (2) Transmitter enclosures</li> </ul>                                      | Designed for use in hydronic systems. |
|                                       | INSTL0035-TSH | Temperature sensor (pair) hot tap installation kit for pipe size range from 1.25" to 42". Wetted materials are SS, for use in carbon steel piping systems.                      | <ul style="list-style-type: none"> <li>• (2) SS hot tap thermowells with stainless steel hot tap adapters</li> <li>• (2) Full port stainless steel ball valves, 1"</li> <li>• (2) SS close nipples</li> <li>• (2) Weld-on carbon steel branch outlets, 1"</li> <li>• (2) Transmitter enclosures</li> </ul>                          | Designed for use in hydronic systems. |
| <b>Steel or PVC Pipe with Saddles</b> | INSTL0060-TSH | Temperature sensor (pair) hot tap installation kit for pipe size range from 1.5" to 6". Wetted materials are SS, for use in PVC piping systems.                                 | <ul style="list-style-type: none"> <li>• (2) SS hot tap thermowells with brass hot tap adapters</li> <li>• (2) Full port bronze ball valves, 1"</li> <li>• (2) Brass close nipples</li> <li>• (2) Ductile iron saddles for Sch std./40/80 pipe with 1" NPT outlets</li> <li>• (2) Transmitter enclosures</li> </ul>                 | Designed for use in hydronic systems. |
|                                       | INSTL060A-TSH | Temperature sensor (pair) hot tap installation kit for pipe size range from 8" to 14". Wetted materials are SS, for use in PVC piping systems.                                  | <ul style="list-style-type: none"> <li>• (2) SS hot tap thermowells with brass hot tap adapters</li> <li>• (2) Full port bronze ball valves, 1"</li> <li>• (2) Brass close nipples</li> <li>• (2) Ductile iron saddles for Sch std./40/80 pipe with 1" NPT outlets</li> <li>• (2) Transmitter enclosures</li> </ul>                 | Designed for use in hydronic systems. |
|                                       | INSTL0065-TSH | Temperature sensor (pair) hot tap installation kit for pipe size range from 6" to 12". Wetted materials are SS, for use in PVC piping systems.                                  | <ul style="list-style-type: none"> <li>• (2) SS hot tap thermowells with stainless steel hot tap adapters</li> <li>• (2) Full port stainless steel ball valves, 1"</li> <li>• (2) SS close nipples</li> <li>• (2) Ductile iron saddles for Sch std./40/80 pipe with 1" NPT outlets</li> <li>• (2) Transmitter enclosures</li> </ul> | Designed for use in hydronic systems. |

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## INSTALLATION HARDWARE SELECTION GUIDE

### HOT TAP AND CLAMP-ON INSTALLATION HARDWARE

For System-10, System-20 and F-4300 BTU Meters (For use with current (mA) based temperature sensors)

| Pipe Material                   | Model Number  | Description                                                                                                                                      | Items Included with Kit                                                                                                                                                                                                                                                                           | Notes                                 |
|---------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Copper Tube with Saddles</b> | INSTL0040-TSH | Temperature sensor (pair) hot tap installation kit for pipe size range from 2" to 6". Wetted materials are SS, for use in copper piping systems. | <ul style="list-style-type: none"> <li>• (2) SS hot tap thermowells with brass hot tap adapters</li> <li>• (2) Full port bronze ball valves, 1"</li> <li>• (2) Brass close nipples</li> <li>• (2) SS saddles for copper tube with 1" NPT outlets</li> <li>• (2) Transmitter enclosures</li> </ul> | Designed for use in hydronic systems. |
| <b>Steel Pipe</b>               | INSTL0501-TSC | Installation kit for pipe size range from 0.5" to 48". Materials consist of aluminum and brass, for use in carbon steel piping systems.          | <ul style="list-style-type: none"> <li>• (2) Clamp-on thermowell blocks w/clamps</li> <li>• (2) Transmitter enclosures</li> </ul>                                                                                                                                                                 | Designed for use in hydronic systems. |
| <b>Copper Tube</b>              | INSTL0502-TSC | Installation kit for pipe size range from 0.5" to 6". Materials consist of aluminum and brass, for use in copper piping systems.                 | <ul style="list-style-type: none"> <li>• (2) Clamp-on thermowell blocks w/clamps</li> <li>• (2) Transmitter enclosures</li> </ul>                                                                                                                                                                 | Designed for use in hydronic systems. |

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